

New Sites, New Services: Evidence from a Randomized Dialysis Payment Reform

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Abstract

I use the geographic randomization of Medicare’s End-Stage Renal Disease Treatment Choices (ETC) Model to study how payment incentives affect home dialysis supply. Linking facility entry histories to annual cost reports, I follow fixed entry cohorts and incumbent facilities through 2024. Statistically significant increases in home dialysis–providing facilities and Medicare home treatments are concentrated among early entrants. A similar pattern is not evident among incumbents or across the covered market. For the first payment-period cohort, an accounting decomposition assigns most of the supply increase to entrant numbers, with a smaller contribution from service per entrant. Higher reported training and administrative staffing accompany this expansion. Supplementary market estimates show lower long-term catheter use and higher fistula use; effects on mortality, hospitalization, and geographic access remain imprecisely estimated. These findings suggest that new facilities can be a margin of supply adjustment to payment reform, although broader welfare gains remain uncertain.

JEL codes: I11, I13, L11, L22, L50.

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1. Introduction

Payment reform can change the supply of care by influencing which facilities open and the services they subsequently provide. New facilities can incorporate a service into their staffing, training, and patient relationships as they establish operations; incumbents adapt an existing organization. These initial choices may shape supply for several years, even when entrants account for a small share of the market. Following facilities after entry can therefore reveal adjustment that is difficult to detect in aggregate outcomes. I examine this adjustment in home dialysis, asking how a randomized payment reform affects the subsequent service contribution of successive entry cohorts.

Medicare’s End-Stage Renal Disease Treatment Choices (ETC) Model provides the randomized comparison. Beginning in 2021, the model changed payments to encourage home dialysis and transplantation. It assigned 91 of 302 hospital referral regions (HRRs) to treatment and 211 to control, stratifying assignment across four regions (Ji et al. 2022). Home dialysis requires providers to train patients and caregivers, arrange supplies, and support care outside the facility. Entry thus offers an opportunity to organize these services from the outset. The reform combined a temporary home-dialysis payment bonus with continuing performance incentives, allowing cohort contributions to be followed as the payment schedule changed (Centers for Medicare & Medicaid Services 2020a).

The paper contributes evidence on facility entry as a margin of supply adjustment to payment reform. Huckfeldt et al. (2013) document responses in provider entry and exit to payment reform in post-acute care. I extend this entry perspective by tracing the services supplied by successive cohorts of new dialysis facilities after opening. A new location’s contribution depends on whether it provides home dialysis and how much care it subsequently delivers. Measuring both active service locations and treatment volume therefore connects entry to the supply of care over several years. The cohort totals incorporate changes in both entry and entrant composition, while the fixed original roster retains facilities that later stop providing home dialysis or exit.¹

Comparisons with incumbents and the covered market place these cohort contributions in the context of total regional supply. Large proportional responses within a small entrant cohort can accompany modest changes in aggregate supply, so identifying where service expands and assessing net market growth answer distinct economic questions. The decomposition quantifies the contributions of original entrant counts and average service per entrant to each cohort’s supply difference. I adapt the distinction between firm numbers and average cohort performance in Sedláček and Sterk (2017) to the service supplied by new health-care facilities. This accounting separates the contribution of a larger entrant roster from that of changes in activity across its original members. Training and staffing measures provide complementary evidence on the operating resources accompanying the service response.

¹Entry denotes a new physical site, including a location opened by an existing chain. The estimates concern recorded contributions in retained HRRs; sample selection and incomplete reports qualify their interpretation.

I combine annual dialysis facility cost reports from the Healthcare Cost Report Information System (HCRIS) with facility histories from the Provider of Services (POS) files. Cost reports measure treatment volume, training, staffing, and expenditure; the POS files record certification, location, and operating histories. Historical addresses and state and operator records help classify physical entry, while public dialysis facility reports, census data, and geographic crosswalks provide patient and market characteristics. The reporting population contains 6,971 facilities operating in 2019 and 400 classified eligible entrants in 2021–2024; the 2020 transition cohort is examined separately.² I hold each entry-year roster fixed and follow its recorded service through 2024 within a fixed HRR sample for that cohort. I compare cohort totals across treated and control HRRs within randomization regions, adjusting for 2019 market outcomes and predetermined characteristics. These outcomes reflect both who enters and what entrants subsequently supply, with companion analyses of incumbents and the covered market placing the cohort responses in the context of overall supply.

The positive supply response is concentrated among early entrants. Averaged over 2021–2024, the 2021 cohort supplies an additional 0.125 home dialysis–providing facilities and 183.01 Medicare home-treatment equivalents per HRR, with both estimates statistically significant. These increases are substantial relative to control means of 0.182 facilities and 158.44 treatment equivalents. The 2022 cohort also has a positive provision response, while the 2023 cohort has a significant decline in treatment volume and both 2024 estimates are imprecise.³ Incumbent and covered-market averages are statistically insignificant. The small scale of entrant cohorts allows sizeable proportional responses to coexist with an uncertain change in total regional supply.

A two-factor decomposition connects these service responses to original entrant counts and average service per entrant.⁴ For the 2021 cohort, the point allocation assigns most of the provider-count difference and approximately two-thirds of the treatment difference to entrant counts. The remaining treatment contribution reflects mean service per original entrant, retaining known nonproviders and later exits in the denominator. There is substantial sampling uncertainty around these accounting components. Reported training and administrative staffing also rise for this cohort, while most other input estimates are imprecise. These accompanying responses are consistent with resources devoted to establishing and supporting service at new locations; the separate roles of entrant composition and operating decisions remain unresolved. Robustness checks re-estimate the

²The population includes independent and chain-owned freestanding facilities eligible to file the Independent Renal Dialysis Facility Cost Report (Form 265). The cohort-specific main HRR samples retain 362 of the 400 entrants; Section 3 explains entry classification and reporting coverage.

³Cohort averages cover entry through 2024. Differences across cohorts combine entry-year conditions, facility age, and subsequent policy exposure.

⁴The decomposition uses standardized predictions from the linear cohort models. Causal attribution to entrant counts and average service requires additional assumptions.

main outcomes using Poisson pseudo-maximum likelihood (PPML), exclude the largest markets, restrict the sample to complete facility histories, and extend follow-up through 2025.

The analysis also relates to a broader literature on reimbursement and dialysis care. Studies of physician and hospital payment changes document adjustments in treatment supply, inputs, and technology adoption (Acemoglu and Finkelstein 2008; Clemens and Gottlieb 2014). In dialysis markets, Dai (2014) examines service choice under regulated prices. ETC evaluations study home dialysis use, transplantation, and related patient outcomes, with findings that vary across populations and follow-up periods (Centers for Medicare & Medicaid Services 2025c; Dixit et al. 2026; Ji et al. 2022; Johansen et al. 2023; Koukounas et al. 2024). The cohort approach complements these evaluations with evidence on how newly established locations contribute to recorded home dialysis supply.

Sections 2–4 describe the setting, data, and empirical design. Section 5 presents cohort supply, its decomposition, and incumbent and market comparisons. Section 6 examines training and operating inputs, and Section 7 reports robustness checks before the conclusion.

2. Institutional Setting

Home dialysis requires facilities to train patients or caregivers, arrange supplies, and provide continuing clinical support, even though treatment takes place outside the facility (Office of the Federal Register 2026). Medicare reimburses dialysis facilities through a prospective payment system that bundles routine services (Centers for Medicare & Medicaid Services 2023). The Centers for Medicare & Medicaid Services (CMS) introduced ETC to encourage home dialysis and transplantation by adjusting payments to eligible dialysis facilities and managing clinicians (Centers for Medicare & Medicaid Services 2020a).⁵ The model covers incumbent facilities and eligible new sites, including locations opened by existing chains. Entry in this paper therefore refers to the opening of a physical location.

Participation was mandatory for eligible providers in selected HRRs. Within the randomized study frame, 91 of 302 HRRs were assigned to ETC and 211 to control, with assignment stratified across four Census regions (Centers for Medicare & Medicaid Services 2020a; Ji et al. 2022).⁶ CMS proposed the model in July 2019, announced the final rule and geographic assignments in September 2020, and began payments in January 2021. The 2019 incumbent roster thus covers the proposal year and predates disclosure of geographic assignments.

ETC combines a temporary bonus on home-dialysis payments with performance-based adjustments to both home and in-center payments. The Home Dialysis Payment Adjustment (HDP)

⁵The measured facility response reflects exposure to both facility and clinician incentives.

⁶HRRs with at least 20 percent of postal codes in Maryland were assigned to ETC under a separate rule and are excluded from this frame (Centers for Medicare & Medicaid Services 2020b). Figure A1 in Appendix A maps the assignment; the opening paragraph of Appendix B.2 lists the regional treatment and control counts.

directly increased eligible home-dialysis payments, regardless of the provider's performance score. The bonus was 3 percent in 2021, 2 percent in 2022, and 1 percent in 2023, and ended on December 31, 2023 (Centers for Medicare & Medicaid Services 2020c, secs. 512.340–512.350). This temporary increase in revenue from eligible home care could help cover the training and continuing support required to expand provision.

The Performance Payment Adjustment (PPA) ties eligible home and in-center dialysis payments to performance, extending the incentive to a broader payment base. Its first payment period began in July 2022 and used performance measured during 2021; subsequent six-month payment periods likewise use earlier measurement windows. Home dialysis receives two-thirds of the combined score and transplant-related performance one-third. For each component, CMS uses the more favorable score from achievement relative to comparison benchmarks or improvement relative to the participant's own history (Centers for Medicare & Medicaid Services 2022).⁷ Maximum facility bonuses and penalties rose from 4 and 5 percent, respectively, in the first two payment periods to 7 and 9 percent in July–December 2025. Actual adjustments depend on performance within these limits (Centers for Medicare & Medicaid Services 2025b). Figure 1 summarizes the payment schedule.

CMS pools performance across facilities owned wholly or in part by the same legal entity within an HRR, linking a site's financial incentive to care provided elsewhere in its local ownership group (Centers for Medicare & Medicaid Services 2025b).⁸ Operators can expand training at existing home providers, introduce home services at previously in-center-only locations, or open new sites. A new facility can incorporate home care into its staffing and patient-support arrangements at opening, while incumbents adjust established operations. Pooling gives operators an incentive to consider these choices jointly across their local facilities. Following successive entry cohorts alongside the fixed incumbent roster traces where recorded supply responds; covered-market outcomes assess whether these contributions coincide with a detectable increase in total regional supply.

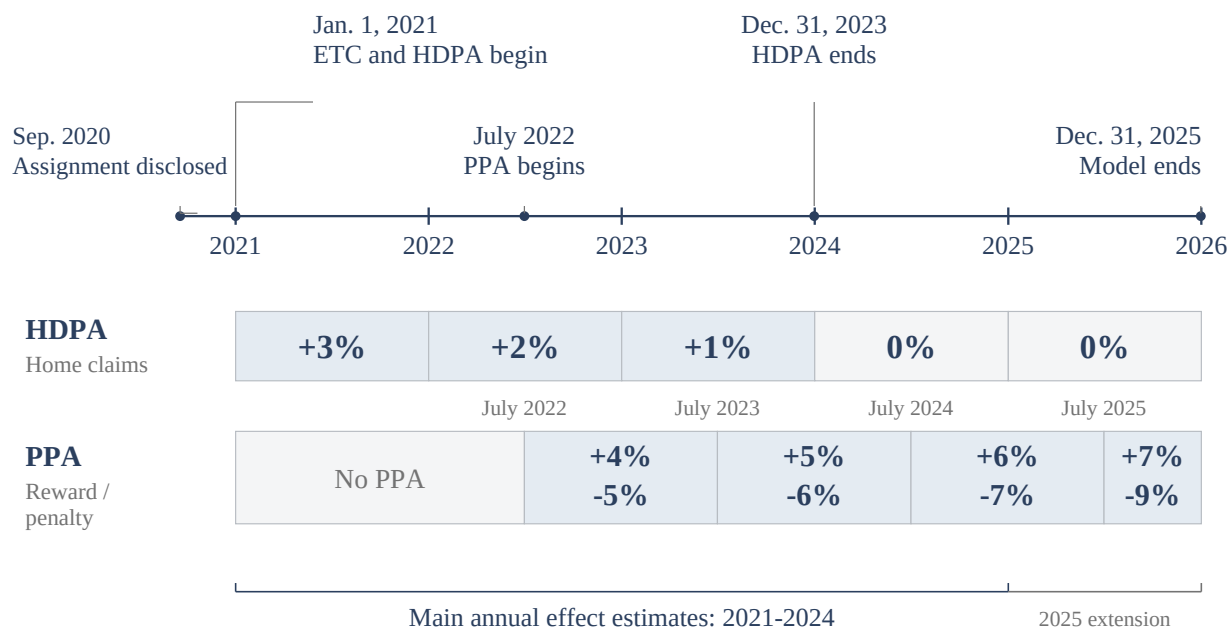
The payment schedule shapes the interpretation of the cohort estimates. Facilities entering in different years faced different initial bonuses and subsequent performance incentives, alongside differences in facility age and market conditions. The PPA continued during 2024–2025 after the HDEPA ended, so service supplied after 2023 reflects activity under ongoing performance incentives. In March 2025, CMS announced its intention to end ETC on December 31, 2025 (Centers for

⁷The home measure uses attributed beneficiary-months, with partial credit for in-center self-dialysis and, from measurement year 3, nocturnal in-center dialysis. The transplant component combines age-adjusted waitlisting and living-donor transplantation; later periods add health-equity features (Centers for Medicare & Medicaid Services 2025b). Relating facility treatment totals to these scores requires patient attribution, denominators, and performance benchmarks.

⁸The PPA exempts ownership groups and unaffiliated facilities with fewer than 11 attributed beneficiary-years (132 beneficiary-months) per measurement period. Eligibility also depends on presence in the benchmark and measurement years and on organizational stability. CMS ownership groups may differ from the economic-parent links in the research data.

Medicare & Medicaid Services 2025a). This announcement may have changed expectations about future payments during the supplementary follow-up. The estimates therefore capture responses to the combined ETC payment environment, which can affect both entry decisions and subsequent operations.⁹

Figure 1. ETC policy timing and payment phases



Notes: HDPa rates apply to eligible home-dialysis payments; PPA rates are maximum facility bonuses and penalties. Adjacent six-month periods with identical PPA limits are combined. Sources: CMS model rules and payment schedules.

3. Data and Sample Construction

I combine public facility histories and cost reports to measure home dialysis supply in the 302 randomized HRRs from 2019 through 2024, with supplementary follow-up in 2025. The core service data come from HCRIS, specifically Form CMS-265-11, because these reports record the treatment volumes and training activity needed to measure realized provision.¹⁰ I apply the form's reporting-category scope to entry counts, established facilities, market totals, and facility-level comparisons. Population membership follows reporting eligibility, with annual report availability assessed separately. Other public sources supplement Form 265 with certification histories, patient

⁹Separating the effects of HDPa, PPA, and investment at entry requires independent variation in these channels.

¹⁰Form CMS-265-11 is the annual Medicare cost report for freestanding dialysis facilities, recording treatments by modality, training, staffing, revenues, and costs (Centers for Medicare & Medicaid Services 2025d).

characteristics, and geographic measures. Table A1 in Appendix A summarizes the data sources, and Figure A1 maps the randomization geography.

3.1. Population and Facility Histories

Facilities eligible to file Form 265 account for approximately 95 percent of registry-active facilities in the randomized markets. This population includes both independently owned and chain-owned facilities and excludes hospital-based units outside the reporting category; throughout the paper, “market-wide” outcomes refer to this common scope within each HRR.¹¹ Within this reporting population, I define the baseline roster as the 6,971 eligible facilities operating in 2019, including 264 facilities that opened that year.¹² I hold baseline membership fixed throughout follow-up, assigning later openings to separate cohorts and retaining facilities in the roster regardless of subsequent exit. The 2019 reference period precedes the disclosure of treatment assignments, although the policy proposal was already public during that year.

I classify physical entry by reconciling CMS certification and POS histories with Dialysis Facility Compare (DFC), historical addresses, and available state and operator records. The classification combines individually corroborated openings with administrative histories screened for prior operations; a first certification date alone does not establish that a new location opened. The reviewed file contains 876 candidate physical entrants during 2019–2024, of which 853 fall within the Form 265 reporting scope.¹³ The 264 entrants in 2019 belong to the incumbent roster. The subsequent cohorts contain 189, 149, 108, 80, and 63 eligible locations in 2020–2024, respectively. These counts describe the classified entry roster; the available records cannot independently confirm the opening history of every location.

3.2. Outcomes and Observation Windows

The main outcomes measure the recorded contribution of a fixed facility population to home dialysis supply. Home dialysis–providing facilities, P_{hct} , count original cohort members with positive measured Medicare home activity in year t , and Medicare home treatments, Q_{hct} , sum their home treatment equivalents.¹⁴ Original eligible cohort size, N_{hc} , counts locations classified as physical entrants in year c and remains fixed throughout follow-up. Known nonproviders and documented

¹¹Eligibility follows certification categories and CMS reporting instructions (Centers for Medicare & Medicaid Services 2025d). Coverage measures the eligible share of registry-active facilities; treatment coverage and report completeness are assessed separately. The January 2026 register proxies year-end 2025 operating status.

¹²The roster includes facilities operating during 2019; registry coverage uses year-end counts.

¹³The review removes three New York facilities acquired at existing addresses and one Florida record with evidence of prior operation at the same site from the new-location cohorts. Their verified service remains in covered-market totals. Ownership-history uncertainty is recorded separately from evidence that a site already operated. The paragraph on the New York and Florida records in Appendix A gives the supporting evidence.

¹⁴Home treatment equivalents equal Worksheet D, column 4, lines 7 and 8, plus three times the weekly counts in lines 9 and 10. Patients can contribute multiple equivalents. Positive reported activity can establish provision when annual quantity is incomplete; home-service authorization records capability.

exits remain in this roster with zero activity; unresolved service remains unknown. The main incumbent and covered-market panels span 2019–2024, with average effects over 2021–2024. Each fixed entry cohort is followed from entry through 2024. The 2020 cohort provides a supplementary transition-period comparison because its formation overlaps assignment disclosure and pandemic disruption before payment implementation; its core service average includes 2020. Covered-market totals also retain verified activity from existing-site administrative successors and facilities whose physical entry year remains unresolved.

I aggregate verified facility components within each HRR and retain the previously selected HRRs over each population’s main observation window. A local population with some unresolved facilities contributes the sum of its known activity. An empty local entry cohort has a known zero contribution; a nonempty local population with no known contribution remains unavailable. The samples contain 301 HRRs for incumbents and the covered market, and 290, 290, 296, 293, and 295 HRRs for the 2020–2024 cohorts, respectively. They were selected by joint availability of the two main outcomes in every required year and are held fixed after the entry-classification review. The HRR set is constant within each population, while cohorts can cover different markets. Across the 2021–2024 cohorts, 362 of the 400 classified eligible entrants lie in their corresponding main HRR samples.¹⁵ The primary outcomes use observed records, documented identities and closures, and deterministic report reconciliations, excluding statistical donor-based completions. The resulting comparisons concern recorded supply in retained markets. Selection using follow-up availability and unreported activity remain identification concerns, discussed in Section 4.

Training treatments measure reported work supporting home care. Training outcomes use qualifying reports from the year after entry through 2024. A selected report must last 330–400 days, cover at least 90 percent of its calendar year, and begin no earlier than entry.¹⁶ The 2024 cohort consequently has no full-report outcome in the main window. Continuous provision requires positive home activity in every year from the first full year after entry through the observation year. The common-calendar comparison requires provision in both 2023 and 2024 and includes cohorts entering by 2023. A known zero resolves either history to zero; otherwise an unknown required year leaves the history missing. Initial configuration uses dated entry-period records, and operator composition uses documented ownership and pre-policy experience. Each additional HRR outcome retains a fixed subset of its approved cohort sample with an observed aggregate in every required year; conditional facility comparisons use the actual observed facilities within that geographic support.

¹⁵The paragraph on sample coverage in Appendix A details the retained geography and its facility records. Annual totals can use different observed facilities within that geography.

¹⁶Training counts use Worksheet D, column 1, lines 3–6 and can include repeated treatments for one person. Reports may cross calendar years; totals follow the qualifying reference year. Revisions are reconciled before selection, and nonoverlapping reports can recover calendar-year totals. Identified acute-kidney-injury activity is excluded.

Operating input comparisons follow incumbents and the covered market over 2019–2024 and each fixed entry cohort from entry through 2024, with cohort averages including the entry year. They use accepted annual input fields; an entry-year report can cover a partial operating year. Inputs comprise expenses, clinical and administrative full-time equivalents (FTEs), stations, direct-care compensation, and matched ratios per all-payer treatment or patient.¹⁷ Supplementary financial and patient-related panels end in 2024. Financial outcomes comprise Medicare payments and patient-service net income. Monetary amounts are in 2024 dollars. Clinical outcomes comprise standardized mortality, hospitalization and readmission ratios, dialysis adequacy, long-term catheter and fistula-use shares, and a vascular-access index.¹⁸ Geographic outcomes use fixed demand weights to measure distance to the nearest documented home-capable facility, nearby facility counts, and the population share without a documented provider within 30 miles. The market all-payer home-treatment share uses matched observed reports.¹⁹ These supplementary outcomes retain their own reporting requirements and use no statistical filling of unresolved values. Source reconciliation restores missing observations; reported FTEs, capacity records, and unresolved identity cases remain subject to source errors documented in the audit ledger.

I append available 2025 observations to the main panels for the extension in Section 7, retaining all earlier observations and the previously fixed HRR lists. Incumbent and entry-cohort service can be followed into that year. The market extension follows the verified 2024 facility roster into 2025.²⁰

3.3. Baseline Adjustment and Balance

Table 1 reports the principal baseline characteristics and outcome levels. It shows why both market size and pre-policy home dialysis supply are useful adjustment variables: facilities and activity are unevenly distributed across HRRs. Additional characteristics and the eight controls appear in Table A2 of Appendix A. Table A3 in the same appendix defines the study populations and variables used throughout the paper.²¹ The descriptive differences show the cross-market variation used for baseline adjustment.

Every regression includes the same eight predetermined market controls, $\mathbf{X}_h$: the frozen 2019 facility-count measure of market size; patient age and female, Black, and Hispanic shares; poverty and college-education shares; and coronavirus disease 2019 (COVID-19) mortality measured around

¹⁷Clinical staffing intensity is FTEs per 100 all-payer patients. Incumbent and market input totals retain 301 HRRs; matched ratios retain 300. Ratios require jointly observed components and positive denominators throughout follow-up, excluding empty local cohorts.

¹⁸Comparable catheter and vascular-access-index observations are unavailable in 2020.

¹⁹These supplementary market outcomes retain 299–301 HRRs. Geographic measures use documented positive capability; unknown capability remains unresolved.

²⁰In 2025, both service outcomes are unavailable in 2, 4, 4, and 5 main HRRs for the 2021–2024 cohorts, respectively. Incumbent and market totals retain 301 HRRs, summing verified facility contributions. Missing values remain unfilled. The 2025 entry roster is unavailable, so 2025 entrants are excluded.

²¹Tables A4 and A5 in Appendix A report outcome distributions and facility coverage, respectively.

Table 1. Pre-Policy Market Characteristics and Home Dialysis Supply

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
Eligible active facilities	23.37 [22.52]	22.80 [30.84]	0.52 (3.16)	302
Big Two facility share	75.45 [24.84]	76.25 [23.51]	−0.82 (3.05)	301
Other-chain facility share	16.19 [20.47]	14.89 [19.11]	1.32 (2.38)	301
Independent facility share	7.78 [17.00]	8.29 [17.70]	−0.50 (2.16)	301
Dialysis stations	349.72 [334.00]	272.97 [283.74]	72.27* (40.66)	266
Medicare treatments	120.71 [117.97]	90.84 [92.59]	28.56* (14.17)	266
Home-capable facilities	10.16 [9.62]	10.86 [14.93]	−0.71 (1.43)	302
Home dialysis—providing facilities	9.58 [9.08]	10.11 [14.03]	−0.55 (1.35)	302
Home-service share	42.33 [14.45]	45.71 [18.92]	−3.38* (1.99)	301
Home patients	178.54 [162.11]	148.21 [147.24]	29.62 (20.77)	266
Home treatments	17.28 [16.19]	16.25 [20.55]	1.00 (2.19)	302
Treatments per providing facility	2,103.43 [1,395.00]	1,854.12 [1,134.90]	253.70 (161.52)	296

Notes: HRR-level statistics refer to 2019; differences adjust for randomization regions. Shares are percentages; treatment totals are thousands of equivalents; other totals are counts. Eligible active facilities are year-end counts. Home provision and treatments use verified 2019-roster contributions, and the home-service share uses the original roster denominator. Brackets report standard deviations (SD). Parentheses report standard errors (SE) with the heteroskedasticity-consistent type 2 (HC2) adjustment. Stars denote the 10, 5, and 1 percent levels (*, **, ***), based on 10,000 region-stratified randomization-inference draws.

assignment disclosure. The market-size control comes from the earlier administrative baseline and totals 6,921 facilities across the 302 HRRs; it is retained as a predetermined covariate, distinct from the subsequently reconciled 6,971-facility incumbent roster.²² Patient, socioeconomic, and pandemic controls draw on the adjustment domains in Koukounas et al. (2024); I construct HRR-level measures from public data and add market size. Patient characteristics refer to 2019, socioeconomic measures use the 2015–2019 American Community Survey (ACS), and cumulative COVID-19 mortality ends on September 17, 2020.²³

²²The count regression in the decomposition additionally adjusts for the reconciled 2019 eligible roster. Table A2 labels the frozen market-size control separately; the accompanying paragraph in Appendix A records its source.

²³Patient counts weight demographic measures. Controls are centered and scaled within randomization regions; COVID-19 mortality uses the inverse hyperbolic sine transformation. The COVID-19 cutoff is the date on the archived assignment workbook and precedes the September 18 public fact sheet; the archive does not establish the workbook's exact public release time. Outcome baselines use verified 2019 service or input measures.

4. Empirical Strategy

I estimate two related responses to ETC assignment. The first is the change in supply from incumbents—the fixed roster of eligible facilities operating in 2019, including those that entered that year—or from the covered market. The second is the subsequent contribution of facilities entering in a particular year. These comparisons allow entry-cohort size and composition to respond to the reform while incumbent membership remains fixed.

4.1. Market and Cohort Specifications

For HRR h in calendar year t from 2019 through 2024, I estimate outcomes for incumbent facilities using the following specification:

$$Y_{ht} = \alpha_h + \lambda_{s(h),t} + \sum_{r=2020}^{2024} \mathbf{1}\{t = r\} [\tau_r T_h + \gamma'_r \mathbf{X}_h] + \varepsilon_{ht}. \quad (1)$$

The main outcomes Y_{ht} are the number of incumbent facilities providing home dialysis and the total Medicare home-treatment equivalents supplied by incumbents, both measured at the HRR–year level. T_h indicates ETC assignment, and $s(h)$ identifies its randomization region. The HRR fixed effect α_h absorbs time-invariant market characteristics. Region-by-year effects $\lambda_{s(h),t}$ allow each randomization region to have its own annual shocks and already include national year effects. The treatment coefficients τ_r compare treated and control paths relative to the omitted reference year of 2019, while the coefficients on $\mathbf{X}_h$ allow baseline characteristics to predict different annual paths. For market-wide outcomes, I apply the same specification to totals over all eligible facilities in each HRR, including later entrants. Supplementary incumbent and market operating inputs, as well as the other market outcomes, use this panel design.

I estimate the effect of ETC assignment on the total home dialysis service supplied by each fixed entry cohort using the following additive specification:

$$Y_{hct} = \lambda_{s(h),ct} + \tau_{ct} T_h + \rho_{ct} Y_{h,2019} + \gamma'_{ct} \mathbf{X}_h + \varepsilon_{hct}. \quad (2)$$

For the core service outcomes, $c \in \{2020, \dots, 2024\}$ denotes entry year and $t = c, \dots, 2024$ indexes calendar years of follow-up; Y_{hct} is the recorded service total supplied by that fixed cohort in HRR h . The stratum–cohort–year intercept $\lambda_{s(h),ct}$ compares treated and control HRRs within the randomized regions.²⁴ $Y_{h,2019}$ measures baseline market supply: home dialysis–providing facilities for provision and Medicare home treatments for treatment quantities and training.²⁵ Baseline slopes and the

²⁴HRR fixed effects would absorb assignment within a cohort–year comparison.

²⁵Baselines cover facilities operating in 2019. Input regressions use the corresponding 2019 market total or matched ratio.

eight control slopes vary by cohort and observation year. With assignment-invariant observation and sample membership, the geographic comparison identifies an intention-to-treat response that includes changes in entry, entrant composition, and subsequent activity. Facility-average and positive-denominator ratio estimates additionally condition on the realized entrant population; equations (A1) and (A2) in Appendix B.1 specify the facility comparisons.

4.2. Functional Form, Inference, and Reporting

I estimate equations (1) and (2) by ordinary least squares (OLS), using outcomes and their baseline measures in levels. The coefficients measure changes in facilities, treatment quantities, or the other stated outcome units. This additive specification connects the supply estimates to the accounting decomposition in Section 5.2. Probability and share effects are generally expressed in percentage points; the clinical adequacy index retains its source fraction scale. As a functional-form check, Section 7 re-estimates the main supply outcomes using PPML, which accommodates zero outcomes and a multiplicative conditional mean without requiring equality of the conditional mean and variance (Santos Silva and Tenreiro 2006, 2011).²⁶ Table A4 in Appendix A documents the outcome distributions.

I use randomization inference (RI) to assess the sharp null of no assignment effect using the known assignment mechanism (Athey and Imbens 2017). Sparse cohort outcomes and uneven treatment volumes make this a useful complement to conventional inference: concentrated leverage can impair large-sample approximations in randomized experiments (Young 2019). I compare a studentized efficient-score statistic with its values under 10,000 assignments preserving treatment totals within the four original regions. The null nuisance model is fitted once. Stars and significance statements refer to these individual sharp-null tests; for coefficient averages, the statistic targets the corresponding weighted average of annual scores.²⁷ Reported standard errors use the HC2 adjustment and are clustered at the HRR level, including for facility regressions. HC2 intervals summarize uncertainty in the estimated average and can differ from the RI significance assessment.

Three features qualify the causal interpretation. Chains spanning treated and control HRRs may change service in both groups. The fixed HRR samples are selected using follow-up availability, which can itself respond to ETC. Within retained HRRs, totals sum known facility contributions, so changes in reporting can affect measured supply even when the geography is constant. Random assignment alone does not remove the latter two concerns. Table A5 in Appendix A reports coverage and Section 7 compares alternative samples; high coverage and imprecise coverage differences

²⁶PPML uses $\log(1 + Y_{h,2019})$ for cohort baselines. HRR effects absorb time-invariant baselines in the incumbent and market panels.

²⁷The two-sided statistic targets a signed average and is calibrated under no effect in every included year. It does not by itself test the weaker hypothesis of a zero average effect with offsetting heterogeneous effects; see Wu and Ding (2021). Equation (A3) in Appendix B.2 defines the statistic.

cannot establish that missingness is ignorable. The maintained interpretation is therefore a response in recorded supply on the stated sample, with extrapolation to complete market supply requiring additional assumptions. Main tables average annual coefficients equally and retain their covariance within HRRs. Control means are observed means over the fitted control observations, including pooled available HRR-years in the 2025 extension; adjusted counterfactual means are labeled separately in the decomposition. PPML tables report log coefficients, converted in the text using $100[\exp(\hat{\tau}) - 1]$.

5. Main Results and Supply Decomposition

I first examine entry-cohort contributions, separate those contributions into entrant counts and average service, and then place them alongside incumbent and covered-market supply. Entrant outcomes combine the size and composition of newly formed cohorts with their subsequent activity. Incumbent comparisons follow a predetermined roster, while market totals establish whether the cohort contribution accompanies a detectable expansion of aggregate recorded home dialysis supply.

5.1. Entry Cohorts and Subsequent Contributions

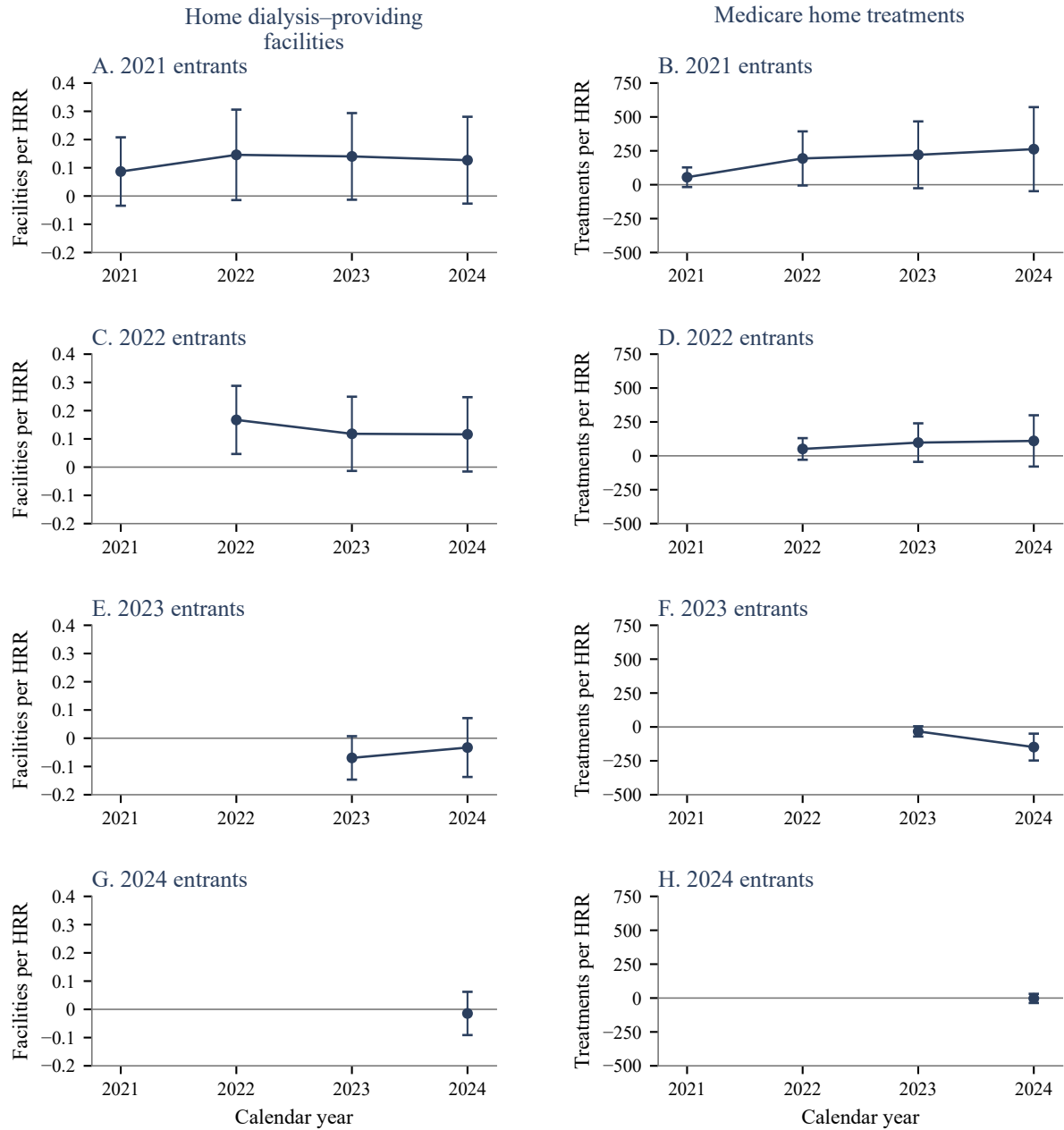
Positive home dialysis supply responses are concentrated among cohorts entering early in the payment period. Figure 2 traces the annual responses of the 2021–2024 entry cohorts.²⁸ The early cohorts generally contribute more home dialysis supply in treated markets, whereas later cohorts have weaker responses and, in some years, treatment declines. The 2021 cohort’s positive treatment contribution grows over follow-up and remains visible after the temporary HDPA ends. This pattern is consistent with new facilities responding to the incentives present when they established operations, followed by continued service as the PPA remains in force. Cohort age and calendar-year conditions also vary across these paths.

Table 2 reports Home dialysis–providing facilities and Medicare home treatments for each fixed 2021–2024 entry cohort.²⁹ For the 2021 cohort, the average estimates are 0.125 additional providing facilities and 183.01 additional Medicare home-treatment equivalents per HRR; both are statistically significant. The 2022 estimates are 0.134 facilities and 86.12 treatments, with stronger evidence for provision, which is statistically significant, than for treatment volume. The 2023 estimates are –0.051 facilities and –90.74 treatments; the treatment decline is statistically significant, while the provision response is imprecise. The 2024 estimates, –0.015 facilities and –2.71 treatments, provide little evidence of a first-year response. Positive average supply responses are therefore concentrated among the first two payment-period cohorts, with evidence for both outcomes in 2021.

²⁸The 2020 cohort spans assignment disclosure and precedes payment implementation. Table A6 in Appendix C.1 collects its supplementary outcomes.

²⁹Averages run from entry through 2024. Table A7 in Appendix C.2 reports the annual coefficients.

Figure 2. Home dialysis supply from fixed entrant cohorts



Notes: Annual linear estimates follow equation (2), using fixed facility rosters and cohort-specific HRRs from entry through 2024. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table 2. Home Dialysis Supply from Fixed Entry Cohorts

Outcome	2021	2022	2023	2024
Home dialysis—providing facilities				
ETC effect	0.125* (0.073)	0.134** (0.063)	−0.051 (0.045)	−0.015 (0.039)
Control mean	0.18	0.15	0.14	0.09
HRR-years	1,160	888	586	295
HRRs	290	296	293	295
Medicare home treatments				
ETC effect	183.01* (102.86)	86.12 (66.15)	−90.74** (33.49)	−2.71 (17.23)
Control mean	158.44	87.14	93.27	32.06
HRR-years	1,160	888	586	295
HRRs	290	296	293	295

Notes: Estimates follow equation (2) on fixed cohort HRR samples. Averages weight annual coefficients equally from entry through 2024. Outcomes are providing facilities and Medicare home-treatment equivalents per HRR, summing verified contributions and known empty-cohort zeros. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. HRR-years count entry-through-2024 observations. The retained cohorts contain 135, 101, 70, and 56 facilities.

The two outcomes describe complementary dimensions of local supply. A larger number of providing facilities expands the set of entrant locations actively supporting home care; a higher treatment total records more care delivered by that cohort. For the 2021 cohort, both responses are substantial relative to control means of 0.182 providing facilities and 158.44 treatment equivalents per HRR. The positive contribution continuing through the main follow-up is consistent with new sites establishing an ongoing home-service activity.³⁰ The cohort’s contribution to aggregate supply also depends on activity elsewhere in the market. Section 7 examines model choice, influential markets, reporting restrictions, and the supplementary 2025 extension.

5.2. Entrant Counts and Average Service

I separate the cohort’s supply contribution into the number of original entrants and the service supplied per entrant. For HRR h , entry cohort $c \in \{2021, \dots, 2024\}$, and follow-up year $t = c, \dots, 2024$, providing facilities satisfy $P_{hct} = N_{hc}p_{hct}$ and home treatments satisfy $Q_{hct} = N_{hc}\bar{q}_{hct}$ when $N_{hc} > 0$. Here N_{hc} is the original eligible cohort size, p_{hct} is the recorded proportion providing home dialysis, and $\bar{q}_{hct}$ is recorded Medicare home treatments per original entrant. Both averages retain known nonproviders and later exits in the denominator.³¹ They therefore describe the subsequent activity of the original cohort, including changes in its composition of active providers.

³⁰Linked patient histories would distinguish additional home dialysis from transfers between providers and clarify effects on patient choice and welfare.

³¹Empty cohorts have zero totals and undefined local ratios. Numerators use verified service; denominators include every original entrant, including facilities with unresolved activity.

The decomposition uses the fitted values from equation (2) underlying the main cohort results. I retain those regressions for providing facilities, P , and treatments, Q , and apply the same specification to the original entrant count, N . Each count regression uses the same observed HRRs and equal weights as its paired service regression, including markets with zero entrants.³² I then average predictions under each assignment over this common sample, following the regression-standardization approach described by Bartlett (2018). For outcome $K \in \{N, P, Q\}$ and assignment $d \in \{0, 1\}$, the adjusted mean is

$$\mu_{K,d,ct} = \frac{1}{|\mathcal{S}_{ct}|} \sum_{h \in \mathcal{S}_{ct}} \widehat{K}_{hct}(T_h = d). \quad (3)$$

Here $\mathcal{S}_{ct} = \mathcal{S}_c$ is the cohort's fixed main HRR sample, and $\widehat{K}_{hct}(T_h = d)$ is the fitted outcome from equation (2) with assignment set to d . Baseline outcomes and controls retain their observed values in both predictions. Averaging over the same markets places the two assignment states on a common distribution of market characteristics. These adjusted levels supply the counts and service rates needed for the decomposition: mean treatments per original entrant are $\bar{q}_{d,ct} = \mu_{Q,d,ct} / \mu_{N,d,ct}$.³³

I apply the Shapley decomposition framework of Shorrocks (2013) to entrant counts and average service. It averages the contribution of each factor over the two possible orders of replacing control counts and service rates with their treatment counterparts. Suppressing cohort and year subscripts, the components are

$$\phi_N = (\mu_{N,1} - \mu_{N,0}) \frac{\bar{q}_1 + \bar{q}_0}{2}, \quad \phi_q = (\bar{q}_1 - \bar{q}_0) \frac{\mu_{N,1} + \mu_{N,0}}{2}. \quad (4)$$

Here ϕ_N is the treatment-volume difference assigned to original entrant counts. It multiplies the adjusted difference in cohort size, $\mu_{N,1} - \mu_{N,0}$, by the mean of the two assignment-specific service rates, $(\bar{q}_1 + \bar{q}_0)/2$. Each additional entrant is thus valued at a common treatment volume. The mean-service component, ϕ_q , multiplies the difference in treatments per original entrant by the average cohort size, $(\mu_{N,1} + \mu_{N,0})/2$. Its denominator includes all original entrants, so it incorporates known nonproviders and later exits. Both components are treatment equivalents per HRR: a positive value contributes to higher recorded treatment volume under ETC, and opposite signs indicate offsetting contributions. They sum to the adjusted treatment difference, $\phi_N + \phi_q = \mu_{Q,1} - \mu_{Q,0}$, assigning half the interaction between count and service changes to each factor.

³²The 2019 baselines are eligible facilities for N , providing facilities for P , and Medicare home treatments for Q . All regressions include region effects and $\mathbf{X}_h$, with slopes varying by cohort and year.

³³Taking ratios of adjusted means retains HRRs with no entrants. In the linear model, $\mu_{K,1,ct} - \mu_{K,0,ct} = \widehat{\tau}_{ct}^K$, the assignment coefficient for outcome K .

For providing facilities, define the adjusted provision proportion as $p_d = \mu_{P,d}/\mu_{N,d}$. The corresponding count and provision-rate components are

$$\phi_N^P = (\mu_{N,1} - \mu_{N,0}) \frac{p_1 + p_0}{2}, \quad \phi_p^P = (p_1 - p_0) \frac{\mu_{N,1} + \mu_{N,0}}{2}. \quad (5)$$

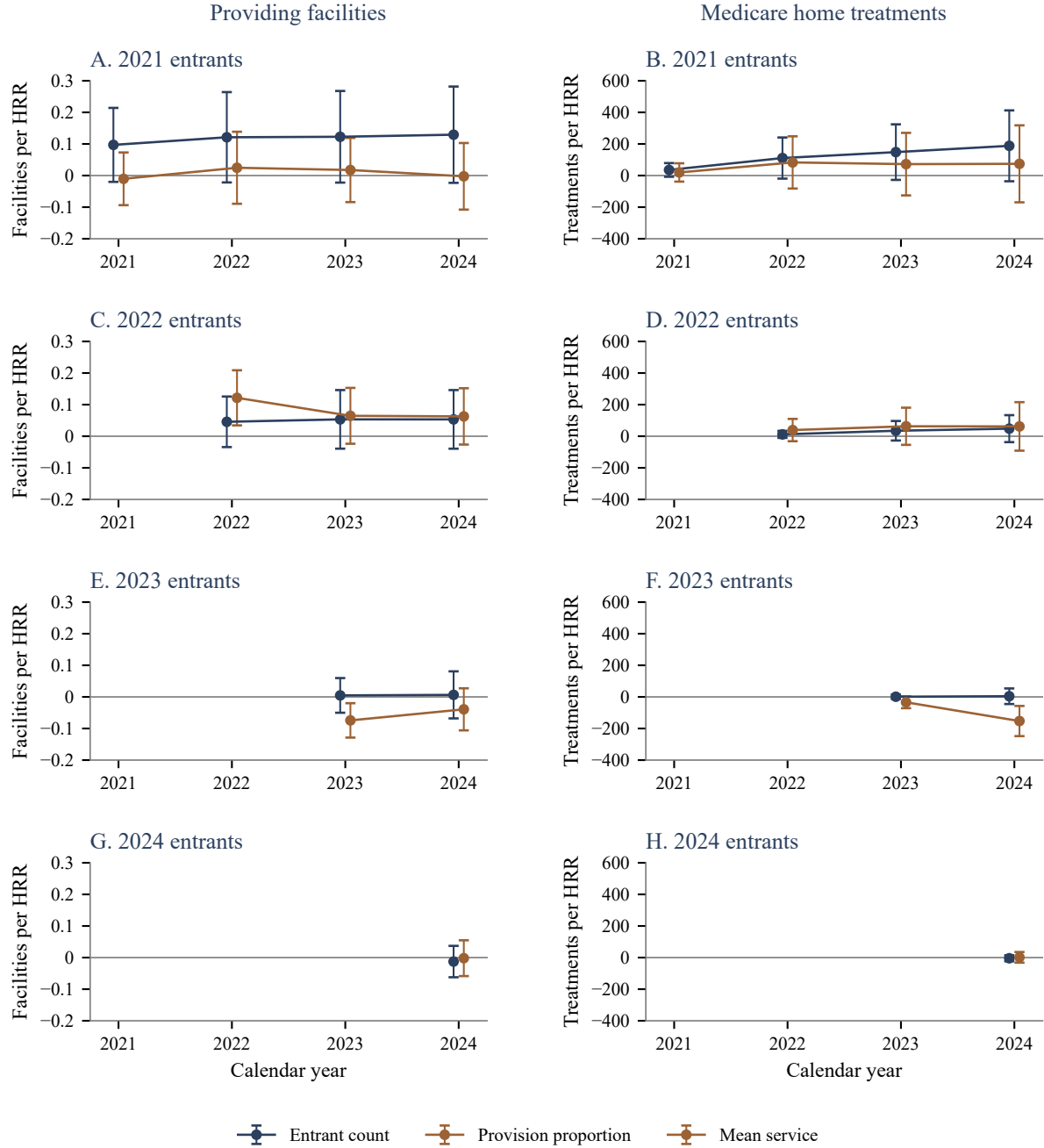
The superscript P identifies the providing-facility outcome. Its count component, ϕ_N^P , values the difference in original entrant counts at the average provision proportion, $(p_1 + p_0)/2$. The provision-rate component, ϕ_p^P , values the difference in the share of original entrants providing home dialysis at the average cohort size. Both are measured in providing facilities per HRR and sum to $\mu_{P,1} - \mu_{P,0}$. A larger cohort can therefore contribute more providing facilities even with an unchanged provision proportion; a lower proportion can offset part of that count contribution. Known nonproviders and exits remain in the original denominator.³⁴

Figure 3 shows the two annual components for both supply outcomes. Their sum equals the corresponding cohort and year effect in Figure 2. The 2021 cohort's positive providing-facility response is largely assigned to original entrant counts, with smaller changes in the provision proportion. Its treatment response has positive count and mean-service components throughout follow-up, and the count component rises over time. The 2022 provision response places more weight on the provision proportion, whereas the 2023 treatment decline is concentrated in mean service. Later cohorts have shorter observed histories, and the intervals leave the relative component sizes uncertain.

Table 3 quantifies these allocations. For the 2021 cohort, Panel A assigns 0.118 of the total 0.125 additional providing facilities per HRR to original entrant counts and 0.007 to the provision proportion, approximately 94 percent and 6 percent of the point allocation. Panel B assigns 120.7 of the 183.0 additional treatment equivalents to counts and 62.3 to mean service, approximately 66 percent and 34 percent. Both treatment components have substantial sampling uncertainty. The provision and treatment allocations thus describe different aspects of the same cohort: its additional provider locations are mainly associated with cohort size, while average service makes a larger contribution to the treatment difference. For 2022, the provision-proportion component is 0.083 of a total 0.134 facilities; its two treatment components are positive. The 2023 treatment decline is almost entirely assigned to mean service, and the 2024 differences are small. Panel C reports the adjusted counts, service totals, and rates entering the calculation. All 2,000 bootstrap draws for the 2021 treatment allocation satisfy the economic bounds, while 781 draws for 2023 contain a negative fitted quantity. This limits a literal service-rate interpretation of the latter allocation. The opening paragraph of Appendix G.4 specifies the resampling procedure and economic admissibility criteria.

³⁴Annual components are averaged over each cohort's follow-up. Standard errors use 2,000 HRR bootstrap draws, stratified by region and assignment and shared across outcomes and years.

Figure 3. Entrant counts and average service: annual accounting components



Notes: Components from equations (4)–(5) sum to the annual cohort effects in Figure 2. Bars are estimates plus or minus 1.96 bootstrap standard errors, using 2,000 HRR draws stratified by region and assignment and shared across outcomes and years.

Table 3. Entrant Counts and Average Service: Two-Factor Allocations

Component	2021	2022	2023	2024
Panel A. Home dialysis–providing facilities				
Entrant count	0.118 (0.071)	0.051 (0.045)	0.006 (0.033)	−0.013 (0.025)
Provision proportion	0.007 (0.050)	0.083 (0.042)	−0.057 (0.029)	−0.002 (0.029)
Total difference	0.125 (0.071)	0.134 (0.062)	−0.051 (0.043)	−0.015 (0.037)
HRRs per year	290	296	293	295
HRR-years	1,160	888	586	295
Panel B. Medicare home treatments				
Entrant count	120.7 (72.9)	31.4 (28.6)	2.7 (16.1)	−4.3 (9.7)
Mean service per original entrant	62.3 (81.8)	54.7 (54.7)	−93.4 (32.3)	1.6 (17.2)
Total difference	183.0 (99.9)	86.1 (65.1)	−90.7 (33.2)	−2.7 (17.0)
HRRs per year	290	296	293	295
HRR-years	1,160	888	586	295
Panel C. Adjusted levels (control / ETC)				
Original entrants per HRR	0.391 / 0.644	0.314 / 0.403	0.235 / 0.247	0.198 / 0.171
Providing facilities per HRR	0.179 / 0.304	0.143 / 0.277	0.138 / 0.087	0.096 / 0.081
Treatments per HRR	163.0 / 346.0	87.0 / 173.1	98.3 / 7.6	31.8 / 29.1
Provision proportion	0.457 / 0.471	0.455 / 0.686	0.588 / 0.352	0.485 / 0.474
Treatments per original entrant	416.6 / 536.9	277.0 / 429.5	417.7 / 30.6	160.8 / 169.7

Notes: Annual Shapley allocations use equations (3)–(5) and are averaged equally from entry through 2024. Components sum to the corresponding effects in Table 2. Service measures retain verified contributions; entrant counts retain the original roster. Parentheses report standard errors from 2,000 HRR bootstrap draws, stratified by region and assignment and shared across outcomes and years. These components describe accounting allocations. Panel C gives adjusted control and ETC levels from equation (3), averaged over follow-up. These differ from the observed control means in Table 2.

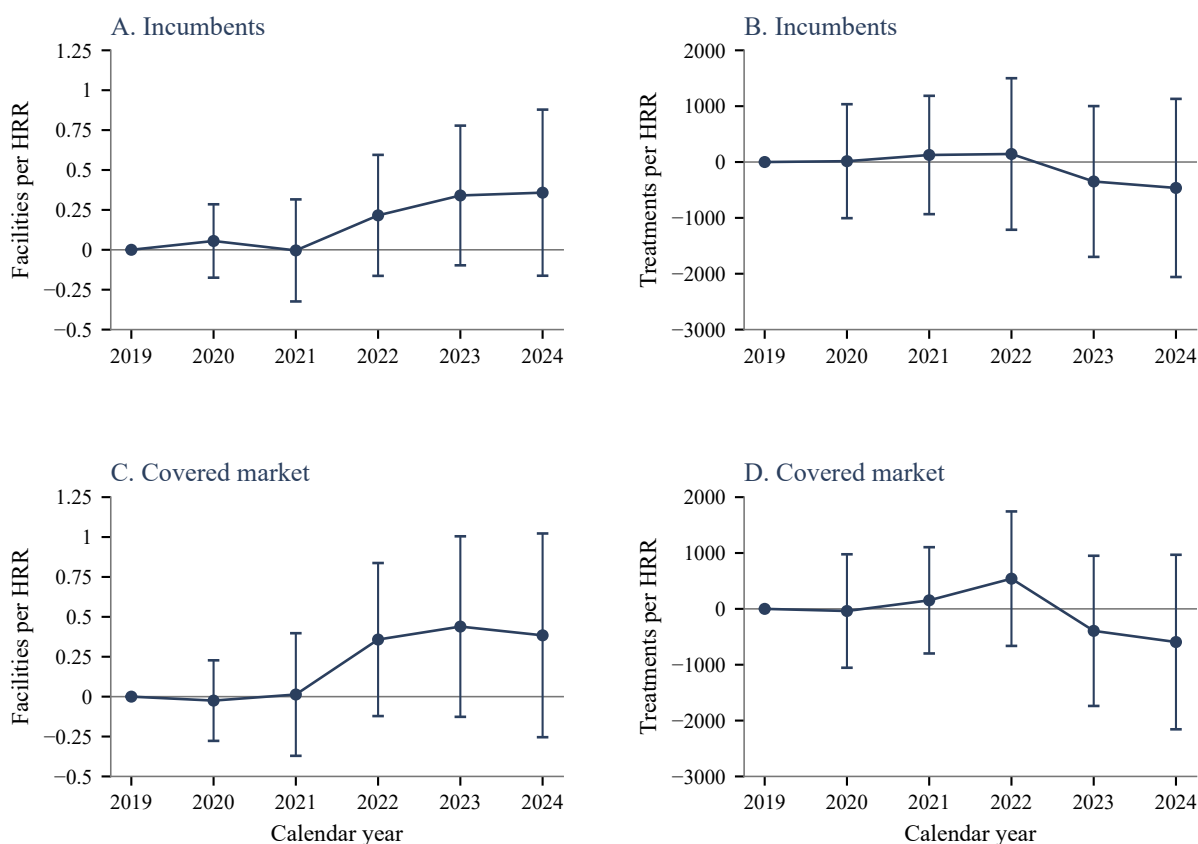
These are accounting allocations. A provision proportion can change through entrant composition, subsequent adoption, or cessation of home service; mean treatments can also reflect patient mix and treatment intensity.³⁵ The results therefore leave causal mediation through entry and operating decisions unresolved. This distinction parallels the separation of firm numbers and average firm size in Sedláček and Sterk (2017); their analysis of unobserved growth potential additionally relies on a structural model.

³⁵Figure A2 and Table A8 in Appendix D.1 report continued provision; Table A9 in Appendix D.2 reports initial configuration and operator composition.

5.3. Incumbent and Market-Wide Home Dialysis Supply

Figure 4 places the entrant response alongside incumbent and covered-market supply.³⁶ Both populations show modest positive provider-count estimates in the later years, while treatment-volume estimates turn negative toward the end of the main window. Wide intervals surround these paths. They provide limited evidence of sustained expansion across the broader market, despite the positive contribution from early entrants.

Figure 4. Incumbent and covered-market home dialysis supply



Notes: Annual linear estimates follow equation (1), with 2019 as the reference year. Both populations retain 301 HRRs during 2019–2024 and sum verified facility contributions. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table 4 summarizes the average responses. Panel A follows the original incumbent roster and reports 0.228 additional providing facilities and 135.62 fewer Medicare home-treatment equivalents per HRR. Panel B includes the broader eligible reporting population, with 0.299 additional providing facilities and 73.52 fewer treatments. All four averages are statistically insignificant. Their uncertainty

³⁶Both panels retain 301 HRRs; the facility records contributing to annual totals can change. Table A10 in Appendix E reports annual estimates.

leaves both modest increases and decreases in aggregate supply plausible. The early-cohort estimates consequently establish where a positive recorded contribution appears, while the net response across established facilities and the covered market remains unresolved.³⁷

Table 4. Incumbent and Covered-Market Home Dialysis Supply

Outcome	Home dialysis–providing facilities	Medicare home treatments
Panel A. Fixed 2019 incumbent roster		
ETC effect	0.228 (0.192)	–135.623 (640.823)
Control mean	10.01	13,336.65
HRR-years	1,204	1,204
HRRs	301	301
Panel B. Covered market		
ETC effect	0.299 (0.243)	–73.524 (579.568)
Control mean	10.82	14,122.39
HRR-years	1,204	1,204
HRRs	301	301

Notes: Estimates follow equation (1), relative to 2019, on 301 fixed HRRs. Averages weight the 2021–2024 coefficients equally. Outcomes are providing facilities and Medicare home-treatment equivalents per HRR, summing verified facility contributions. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Each equation uses 1,806 HRR-years over 2019–2024; the average-window count is 1,204.

The scale of the entry cohorts helps explain this contrast. In the common accounting samples, the 2021–2024 cohorts account for 4.0 percent of providing facilities and 2.2 percent of Medicare home treatments in control markets.³⁸ Substantial changes within these small cohorts can therefore make a modest contribution to total regional supply. The decomposition in Section 5.2 characterizes the entry and average-service margins behind the cohort differences. Table A11 in Appendix E places the market components in one common linear design. The residual category includes administrative successors and facilities without an assigned physical opening year; it accounts for much of the negative treatment point estimate in that common sample. The accounting consequently leaves both patient reallocation and the role of unresolved entry histories open. Section 6 examines accompanying training and operating inputs among entrants. The results locate a positive recorded contribution among early entrants, with insufficient evidence to establish a net market expansion.

³⁷Cohort and incumbent models use different baseline adjustments. Table A11 in Appendix E reports additive contributions under a common linear specification and sample.

³⁸Shares pool matched control HRR-years in 2021–2024: 271 HRRs for provision and 260 for treatments. The 2021 cohort accounts for 2.0 and 1.2 percent, respectively.

6. Additional Outcomes and Economic Implications

I report additional outcomes for the 2021–2024 entrant cohorts covering training activity, operating expenses, dialysis stations, clinical and administrative staffing, direct-care compensation, and cost and staffing intensity. Supplementary comparisons cover incumbent and market-wide inputs, financial and clinical outcomes, geographic availability, and the all-payer home-treatment share.

Table 5. Additional outcomes in fixed entry cohorts: average responses

Outcome / statistic	2021	2022	2023	2024
Panel A. Reported training activity				
Training treatments per HRR				
ETC effect	25.99* (14.00)	29.51 (20.22)	−28.58** (12.09)	— —
Control mean	20.97	16.93	28.81	—
HRR-years	864	590	292	—
HRRs	288	295	292	—
Panel B. Operating-input totals				
Operating expenses (million dollars per HRR)				
ETC effect	0.462 (0.329)	0.075 (0.116)	−0.092 (0.133)	0.007 (0.057)
Control mean	0.816	0.496	0.344	0.161
HRR-years	1,140	879	572	290
HRRs	285	293	286	290
Dialysis stations per HRR				
ETC effect	0.691 (0.912)	0.130 (0.535)	−0.229 (0.472)	−0.508 (0.392)
Control mean	2.233	1.461	0.742	0.791
HRR-years	1,140	873	568	289
HRRs	285	291	284	289
Clinical FTEs per HRR				
ETC effect	1.500 (0.980)	0.214 (0.330)	−0.189 (0.301)	−0.392 (0.361)
Control mean	2.370	1.343	0.809	0.689
HRR-years	1,140	879	572	290
HRRs	285	293	286	290
Administrative and management FTEs per HRR				
ETC effect	0.378* (0.220)	0.055 (0.104)	−0.008 (0.096)	−0.114* (0.059)
Control mean	0.558	0.379	0.240	0.173
HRR-years	1,140	879	572	290
HRRs	285	293	286	290

Continued on next page

Table 5. Additional outcomes in fixed entry cohorts: average responses (continued)

Outcome / statistic	2021	2022	2023	2024
Direct-care compensation (million dollars per HRR)				
ETC effect	0.127 (0.089)	0.036 (0.034)	−0.023 (0.031)	0.000 (0.014)
Control mean	0.232	0.124	0.092	0.040
HRR-years	1,140	876	572	290
HRRs	285	292	286	290
Panel C. Matched input ratios				
Expenses per treatment (dollars)				
ETC effect	5.26 (106.45)	−200.78* (119.32)	−35.05 (302.93)	−554.08 (662.36)
Control mean	569.45	762.01	1,028.77	1,737.51
HRR-years	272	189	76	42
HRRs	68	63	38	42
Clinical FTEs per 100 patients				
ETC effect	1.52 (1.31)	−0.43 (1.55)	1.88 (4.26)	−5.90 (4.03)
Control mean	10.96	11.59	11.71	16.03
HRR-years	272	183	74	40
HRRs	68	61	37	40
Treatments per clinical FTE				
ETC effect	−523.40 (406.16)	−173.06 (167.80)	−482.36* (291.24)	66.92 (137.70)
Control mean	1,085.16	955.39	1,120.57	481.06
HRR-years	272	189	76	40
HRRs	68	63	38	40

Notes: Equal-weight cohort averages use equation (2) on fixed outcome-specific HRRs. All-payer training uses qualifying full reports after entry through 2024; inputs cover entry through 2024, allowing partial entry-year reports. Baselines are 2019 home treatments for training and the corresponding input otherwise. Totals retain partial facility coverage; ratios require matched components and positive denominators. Monetary totals are millions of 2024 dollars; per-treatment expenses are 2024 dollars. Panel C uses all-payer denominators. FTE denotes full-time equivalent; administrative staffing includes management. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes indicate unavailable outcomes.

6.1. Reported Training Activity

Reported training rises for the 2021 cohort and falls for the 2023 cohort, paralleling their home-treatment responses (Panel A of Table 5). For the 2021 cohort, the average response is 25.99 additional training treatments per HRR, equivalent to approximately 124 percent of the control mean of 20.97 and statistically significant. The 2022 estimate is also positive, at 29.51 treatments, but is imprecise, whereas the 2023 cohort supplies 28.58 fewer training treatments per HRR relative to a

control mean of 28.81, a statistically significant decline.³⁹ These treatments record work preparing patients or caregivers to perform dialysis at home. The positive 2021 response is therefore consistent with greater preparation accompanying that cohort's larger home-service contribution, while the 2023 decline accompanies lower recorded home-treatment volume. Because regional training totals combine the number of entrants with the number of people trained per facility and the training received per person, the response can arise through both cohort size and activity within facilities.⁴⁰ Figure A3 in Appendix F.1 displays the annual HRR training paths.⁴¹

6.2. Operating Inputs

The 2021 cohort also has positive estimated responses in operating expenditure, physical capacity, and employment (Panel B of Table 5).⁴² The average estimates amount to an additional \$0.462 million in operating expenses, 0.691 dialysis stations, 1.500 clinical full-time equivalents (FTEs), 0.378 administrative and management FTEs, and \$0.127 million in direct-care compensation per HRR. The administrative and management response is statistically significant; the other four averages remain imprecisely estimated. Their positive direction is consistent with greater operating resources accompanying the cohort's larger service contribution, although the estimates leave considerable uncertainty about the amounts. These HRR totals combine the number and composition of entrants with resources used per facility, so the comparison concerns the aggregate operating scale of the cohort.

The administrative staffing increase amounts to approximately 68 percent of the control mean of 0.558 FTEs, suggesting that the 2021 cohort's expansion is accompanied by additional labor devoted to organizing and managing its facilities. A larger number of entrant locations can itself increase these labor requirements, alongside changes in staffing within each facility.⁴³ For the 2024 cohort, administrative and management staffing falls by 0.114 FTEs, approximately 66 percent of the control mean of 0.173, with the decline also statistically significant. All remaining input-total averages for the 2022–2024 cohorts are statistically insignificant. The administrative estimates thus have opposite signs for the earliest and latest cohorts, while broader changes in capacity, clinical employment, and expenditure remain less precisely determined. Annual expense and station paths appear in Figure A4, staffing in Figure A5, and compensation in Figure A6, all in Appendix F.2.

³⁹Training follow-up begins the year after entry (Section 3.2), leaving no main-window observation for 2024 entrants.

⁴⁰Unique trainees, completed courses, and linked patient histories are unobserved, limiting conclusions about recruitment and successful home-dialysis adoption.

⁴¹Table A12 in Appendix F.1 reports HRR and facility estimates, including the 2020 cohort; Table A13 reports coverage. Facility comparisons condition on realized entry and report availability.

⁴²Each input uses a fixed subset of the cohort's main HRRs and its corresponding 2019 market baseline in equation (2). Totals retain verified partial contributions; Section 3.2 specifies reporting rules.

⁴³The estimates leave the role of administrative requirements in entry decisions unresolved. Resources shared across dialysis modalities also limit conclusions about home-dialysis expenditure and efficiency.

6.3. Cost and Staffing Intensity

Panel C of Table 5 compares cost and staffing ratios in nonempty local cohorts with jointly observed components and positive denominators throughout follow-up. These conditional samples are substantially narrower than the samples for totals. Operating expenses per treatment fall for the represented 2022 cohort, while most other ratios are imprecisely estimated. The estimated average reduction is \$200.78 per treatment, approximately 26 percent of the control mean of \$762.01, and is statistically significant; the corresponding averages for the other three cohorts are statistically insignificant. This ratio measures operating expenditure relative to all-payer treatment volume, linking the resources used by entrants to the amount of care they supply. A decline can arise as treatment volume expands relative to operating outlays or as the composition of entrants and services changes. The result therefore provides suggestive evidence of lower recorded expenditure per unit of care among the represented 2022 entrants.⁴⁴ The contrast conditions on realized entry and reporting under each assignment.⁴⁵

The labor ratios relate clinical staffing to patient numbers and treatment volume. Average responses in clinical FTEs per 100 patients are statistically insignificant in all four cohorts; for 2021 entrants, the estimate is an additional 1.52 FTEs per 100 patients relative to a control mean of 10.96. Treatments per clinical FTE fall by 482.36 for the 2023 cohort, approximately 43 percent of the control mean of 1,120.57, with the decline statistically significant. The 2021 estimate is also negative, at 523.40 fewer treatments per clinical FTE, but is imprecise, as are the 2022 and 2024 averages. The 2023 response indicates lower recorded treatment volume per unit of clinical labor, which can reflect changes in service volume, care requirements, or entrant composition. Its interpretation as clinical productivity, and the implications of staffing intensity for workload and quality, depend on patient needs and the services delivered. Figure A6 reports the annual cost-per-treatment paths, and Figure A7 reports staffing intensity, both in Appendix F.2. Supplementary estimates cover incumbent and market operating inputs and broader covered-market outcomes.⁴⁶

7. Robustness Checks

I examine sensitivity to functional form, influential markets, and facility reporting, followed by the supplementary 2025 extension. The 2021 contribution remains positive across these comparisons, while its precision depends on the sample.

⁴⁴Assessing profitability or patient financial gains also requires facility revenues and patient payments.

⁴⁵The 2021 totals use 285 HRRs, compared with 68 for matched ratios; later-cohort ratios use 37–63. Ratios require jointly observed components and positive denominators, excluding empty cohorts. Differences in samples and reporting windows limit cross-outcome comparisons.

⁴⁶Tables A14, A15, and A16 in Appendix F.2 report market, incumbent, and cohort inputs, respectively; Table A6 in Appendix C.1 covers 2020 entrants. Table A17 in Appendix F.3 reports broader market outcomes.

7.1. Functional Form

Figures A8 and A9 in Appendix G.1 show the annual PPML responses for cohorts and for incumbent and covered-market supply. The early cohorts generally have positive estimates, the 2023 treatment path is negative, and aggregate responses remain uncertain. PPML allows assignment to change the conditional mean proportionally.⁴⁷ The 2021 comparison retains the same 290 HRRs under both functional forms. For 2023, separation reduces fitted support from 293 to 211 HRRs, so its comparison combines functional form, baseline transformation, and sample composition.

Table A18 in Appendix G.1 reports the annual and average estimates. The 2021 provider and treatment coefficients are 0.644 and 0.781, corresponding to increases of 90.4 and 118.4 percent; both are statistically significant.⁴⁸ The 2022 provider increase and 2023 treatment decline are statistically significant. Other cohort averages and all incumbent and market averages are imprecise. This agreement supports the direction of the early-cohort response under a multiplicative mean, while differences in scales and fitted support limit direct comparisons of magnitudes.

7.2. Influential Markets and Facility Reporting

Figures A10 and A11 in Appendix G.2 compare the main averages with estimates excluding large markets or requiring complete facility histories. The 2021 cohort's provider and treatment estimates remain positive, the 2022 provider response remains positive, and the 2023 treatment response remains negative. Incumbent and market estimates remain much less precise. The figures retain the original windows and linear specifications; each restriction changes the represented markets.

Table A19 in Appendix G.2 excludes the largest baseline market or the three largest markets within each population's sample, ranked by the reconciled 2019 eligible-facility count. Excluding one market leaves the 2021 estimates at 0.127 additional providing facilities and 181.16 additional Medicare home-treatment equivalents per HRR, both statistically significant. Excluding three reduces them to 0.117 facilities and 160.12 treatments, with neither statistically significant. The 2022 provider increase and 2023 treatment decline remain significant under both exclusions. Thus the positive 2021 point estimates persist beyond the largest markets, but their statistical precision is sensitive to those observations.

Table A20 in Appendix G.2 requires both main outcomes to be resolved for every original facility in every required year.⁴⁹ The 2021 cohort retains 281 of 290 HRRs, with estimates of 0.106 providing facilities and 172.54 treatment equivalents. The treatment increase is statistically significant; the provider estimate is imprecise. The 2022 provider increase and the 2023 treatment

⁴⁷The specification starts from the main panels, retains assignment and the eight controls, and transforms cohort baselines as described in Section 4.

⁴⁸Percentage responses apply $100[\exp(\hat{\tau}) - 1]$ after averaging annual log coefficients. The supplementary 2020 core-service average covers 2020–2024 in both functional forms.

⁴⁹Documented physical exits contribute zeros; unresolved reports remain missing.

decline are also statistically significant. Incumbent and market samples fall from 301 to 97 and 86 HRRs, respectively, with their averages remaining imprecise. These comparisons assess sensitivity to complete reporting in substantially selected samples. They cannot establish that missingness is random or identify the unreported contribution in excluded markets.

7.3. Supplementary Follow-Up in 2025

Figures A12 and A13 in Appendix G.3 extend the paths in Figures 2 and 4. The 2021 cohort retains positive provider and treatment point estimates, the 2023 treatment decline persists, and the aggregate paths remain uncertain. I append available 2025 observations and re-estimate the full annual specifications, retaining all earlier observations, main HRR lists, and baseline adjustments.⁵⁰ This extra year includes the announcement of early termination and therefore reflects a changing expected policy horizon as well as further cohort aging.

Table A21 in Appendix G.3 reports extended 2021 averages of 0.131 additional providing facilities and 210.63 additional Medicare home-treatment equivalents per HRR, both statistically significant. Several annual estimates remain imprecise, leaving a distinct effect in every year unresolved. The 2022 provider average is statistically significant, while its treatment response is imprecise; the 2023 treatment average is -123.04 , statistically significant. Table A22 in the same appendix reports incumbent averages of 0.248 facilities and -181.19 treatments, and market averages of 0.315 facilities and -167.42 treatments. All four are statistically insignificant. Continued recorded activity among early entrants consequently leaves net regional expansion and patient reallocation unresolved.

8. Conclusion

Following fixed entry cohorts reveals a continuing supply response to ETC among facilities that opened early in the payment period. The 2021 cohort supplies an estimated 0.125 additional home dialysis-providing facilities and 183.01 additional Medicare home-treatment equivalents per HRR on average through 2024, with both responses statistically significant. The 2022 provision response is also positive, while the 2023 treatment response is negative. Incumbent and covered-market estimates remain imprecise, leaving net regional expansion uncertain. PPML and supplementary 2025 follow-up preserve this pattern, with 2021 precision sensitive to excluding the three largest baseline markets.

The decomposition assigns most of the 2021 provider-count difference and approximately two-thirds of its treatment difference to original entrant counts. Higher reported training and administrative staffing accompany the cohort's larger service contribution. These findings are consistent with new locations building the organization needed to support home dialysis. Uncertainty

⁵⁰The market extension follows the verified 2024 roster. Section 3.2 specifies the 2025 gaps.

around the accounting components and limited evidence on changes per entrant leave the roles of composition and operating behavior unresolved. Separating incentives at entry from subsequent exposure and facility age requires additional variation.

Evaluations of payment reform should therefore follow entrants' subsequent service alongside incumbent and market-wide supply. The evidence concerns recorded supply within the freestanding reporting population through 2024. Selection of regions using follow-up availability, incomplete facility reports, and remaining uncertainty about physical entry histories constrain its interpretation. More complete reports and linked patient histories would help determine whether entrants bring additional patients into home dialysis or primarily serve patients transferring from other providers. Assessing gains in access and welfare also requires evidence on resource costs and patient outcomes.

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A. Data, Definitions and Coverage

Table A1 identifies the sources used to construct facility populations and outcomes, and Figure A1 displays the randomization geography. The provenance archive records report identifiers, worksheet cells, dates, and acceptance decisions, distinguishing outcome measurements from records used only to corroborate a facility's identity.

Table A1. Data Sources and Measurement Roles

Measure or source	Definition and use
ETC assignment and rules	CMS HRR assignment workbook dated September 17, 2020; final rule and subsequent model documents. The frozen assignment universe contains 302 HRRs in four Census regions. CMS ETC model materials .
HRR geography	Dartmouth Atlas 2019 crosswalk from Zone Improvement Plan (ZIP) codes to hospital service areas (HSAs) and HRRs (ZipHsaHrr19.csv.zip). The same geography is retained across years. Dartmouth supplemental files .
Facility registry and identity	Historical POS snapshots from the Quality Improvement and Evaluation System (QIES), supplemented by the Internet Quality Improvement and Evaluation System (iQIES) for the 2026 transition. Fields include the CMS Certification Number (CCN), certification and termination dates, address, facility category, services, staffing, and stations. Snapshot and download dates are recorded separately. CMS Provider of Services .
Service reports	HCRIS Form CMS-265-11, all-years renal facility release: report headers, numeric cells and alphanumeric cells. Calendar outcomes use report identifiers and audited period mappings, with separate missingness flags. CMS renal cost reports; all-years archive .
Reporting instructions	Provider Reimbursement Manual, Part 2, Chapter 42; the May 13, 2025 instructions are Transmittal 11. Each observation follows the reporting instructions applicable to its report period. 2025 Form 265 instructions .
Public facility reports	DFC historical releases and public Dialysis Facility Reports (DFRs), with release-specific dictionaries. Facility addresses, services, stations, patient characteristics, and clinical measures are assigned to their measurement years. The fiscal-year 2022 DFR contains the 2019 demographic fields used in the controls. CMS dialysis-facility data .
Community controls	2015–2019 ACS five-year files: B17001 for poverty and B15003 for educational attainment. Population numerators and denominators are aggregated through the fixed geographic crosswalk. Census sequence-based summary files .
COVID-19 around disclosure	Dartmouth HRR COVID-19 series, cumulative deaths per 100,000 through September 17, 2020. The predetermined series is transformed by the inverse hyperbolic sine before the frozen covariate standardization. Dartmouth COVID-19 resources .

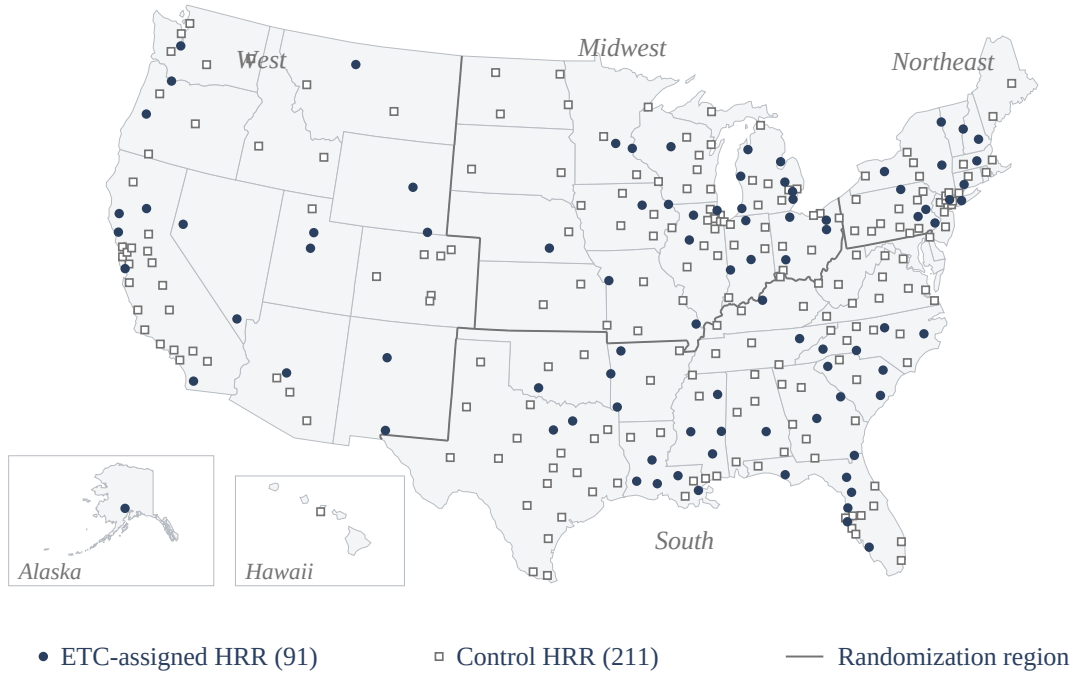
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Table A1. Data Sources and Measurement Roles (continued)

Measure or source	Definition and use
Geographic access	Census Geocoder and ZIP Code Tabulation Area (ZCTA) Gazetteer coordinates; Dartmouth crosswalk; fixed population and rural classifications; chronic kidney disease (CKD) estimates from the Centers for Disease Control and Prevention (CDC) Population Level Analysis and Community Estimates (PLACES), 2022 ZCTA release, based on the 2020 Behavioral Risk Factor Surveillance System (BRFSS). This demand weight partly overlaps the 2020 transition year. CDC PLACES ; Census Gazetteer .
Prices	Bureau of Labor Statistics (BLS) annual-average Consumer Price Index for All Urban Consumers (CPI-U), U.S. city average, all items. Monetary outcomes are expressed in 2024 dollars. BLS price data .
Entry and operator verification	National Plan and Provider Enumeration System (NPPEs) records and the National Provider Identifier (NPI) Registry; Medicare enrollment records; state licensing and certificate-of-need records; Securities and Exchange Commission (SEC) Electronic Data Gathering, Analysis, and Retrieval (EDGAR) filings; and dated operator announcements. These distinguish physical openings from moves, renumbering, and ownership changes; ownership at entry requires dated supporting evidence.
Dated configuration supplements	North Carolina State Medical Facilities Plans and Illinois annual end-stage renal disease (ESRD) facility profiles provide explicitly dated station counts. California's System for Integrated Electronic Reporting and Auditing (SIERA) reports and early HCRIS reports supply supplemental period measures; opening configuration requires a dated observation. Facility-specific provenance preserves the document, page or field, observation date, and acceptance decision.

Notes: Public records are linked using documented facility identities, report periods, and worksheet definitions. The source archive distinguishes accepted measurements from corroborating evidence. The archived assignment workbook is dated September 17, 2020; the public CMS fact sheet is dated September 18.

Figure A1. Geographic assignment of the ETC model



Notes: Points locate the 302 randomized HRRs: 91 ETC and 211 control. Dark boundaries delineate randomization regions; light boundaries delineate states. Alaska and Hawaii are inset.

Table A2 reports additional baseline characteristics and the eight controls. Retrospective 2019 patient characteristics from the fiscal-year 2022 public dialysis report are weighted by patient counts within HRR; community measures use aggregated ACS numerators and denominators. The COVID-19 cutoff is September 17, 2020. The frozen market-size covariate comes from the earlier Form 265 administrative baseline, totaling 6,921 facilities, compared with 6,971 in the reconciled incumbent roster; the latter count also supplies the outcome baseline for entrant-count regressions. The measures have different data vintages and remain distinct regressors. The source archive retains their HRR-level reconciliation. Table A3 defines the populations and outcomes.

Table A2. Additional Baseline Characteristics and Predetermined Controls

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
COVID-19 deaths by Sept. 17, 2020	56.16 [44.57]	52.01 [49.35]	3.99 (5.25)	302
All dialysis patients	2.13 [2.12]	1.69 [1.75]	0.43 (0.26)	266
All dialysis treatments	216.23 [219.46]	171.66 [176.47]	42.92 (26.77)	266

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Table A2. Additional Baseline Characteristics and Predetermined Controls (continued)

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
In-center patients	1.97 [1.97]	1.55 [1.62]	0.41* (0.24)	263
In-center treatments	188.00 [194.88]	148.21 [155.29]	38.24 (23.64)	266
All-payer home-treatment share	13.79 [4.90]	14.51 [6.85]	-0.67 (0.73)	265
Clinical FTE	236.61 [245.35]	187.37 [196.26]	47.47 (29.87)	266
Administrative FTE	46.53 [47.24]	37.01 [37.10]	9.07 (5.69)	266
Clinical FTE per 100 patients	11.090 [1.882]	11.088 [2.147]	0.022 (0.255)	265
Treatments per clinical FTE	925.95 [115.16]	932.30 [122.95]	-7.11 (15.40)	265
Total expense	69.17 [70.67]	55.12 [57.29]	13.59 (8.60)	266
Expense per treatment	325.44 [37.82]	322.13 [36.90]	3.99 (4.53)	265
Total Medicare payment	46.45 [46.33]	35.37 [35.97]	10.67* (5.57)	266
Medicare payment per treatment	385.06 [38.86]	388.25 [36.63]	-2.04 (4.58)	265
Net income from patient services	19.23 [21.57]	13.99 [14.52]	5.25** (2.55)	266
Net income per treatment	97.74 [134.73]	85.48 [51.64]	12.90 (14.77)	265
Distance to nearest home-capable facility	13.38 [20.45]	13.06 [19.70]	0.36 (2.54)	302
Home-capable facilities within 15 miles	5.09 [7.59]	5.93 [9.29]	-0.83 (0.99)	302
CKD risk beyond 30 miles	8.29 [13.84]	9.39 [19.20]	-1.05 (1.97)	302
Vascular-access index	0.117 [0.807]	-0.051 [0.829]	0.166* (0.100)	301
Long-term catheter share	12.18 [3.71]	12.14 [3.59]	0.04 (0.45)	301
Fistula-use share	62.93 [6.91]	65.21 [6.99]	-2.26*** (0.82)	301
Mortality standardized ratio	0.987 [0.115]	0.996 [0.142]	-0.009 (0.014)	301
Hospital-admission standardized ratio	0.974 [0.121]	0.970 [0.125]	0.004 (0.015)	301
Readmission standardized ratio	0.996 [0.101]	0.998 [0.139]	-0.003 (0.014)	300
Dialysis adequacy share	0.969 [0.011]	0.969 [0.017]	0.000 (0.002)	301

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Table A2. Additional Baseline Characteristics and Predetermined Controls (continued)

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
Patient mean age	62.71 [1.81]	63.13 [1.75]	−0.41** (0.19)	302
Female patients	42.55 [3.36]	42.14 [2.98]	0.40 (0.36)	302
Black patients	29.49 [23.78]	25.14 [21.49]	4.22* (2.26)	302
Hispanic patients	10.40 [13.13]	14.97 [18.05]	−4.58*** (1.68)	302
Community poverty rate	14.31 [3.80]	14.11 [4.05]	0.18 (0.41)	302
Community college-or-higher share	28.39 [7.58]	28.69 [8.42]	−0.28 (0.91)	302
Frozen 2019 facility count (market-size control)	23.34 [22.43]	22.73 [30.72]	0.56 (3.15)	302

Notes: HRR-level differences adjust for randomization regions. Baselines are from 2019, except COVID-19 mortality through September 17, 2020. Monetary totals are millions of 2024 dollars; monetary ratios are dollars per treatment. Patient and treatment totals are thousands. Shares are percentages except adequacy, a fraction. Distances are miles, age is years, and COVID-19 mortality is deaths per 100,000. FTE denotes full-time equivalent. Parentheses report HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) from 10,000 region-stratified studentized reassignments. The frozen 2019 market-size control totals 6,921 facilities; the reconciled incumbent roster contains 6,971.

Table A3. Definitions of Study Populations and Outcomes

Study measure	Definition
Form 265 eligible facility	A nonhospital renal facility whose CMS Certification Number (CCN) ends in 2500–2899 or 2900–2999, the relevant Form 265 reporting categories. Eligibility is assessed separately from ownership and report availability and defines the common scope of all facility populations.
Incumbent facilities	The eligible facilities operating in 2019, including that year’s entrants. Membership remains fixed throughout follow-up. Physical exit requires corroboration beyond administrative identifier termination.
Original eligible cohort size; N_{hc}	Number of eligible locations classified as physical entrants in HRR h during entry year c . The original roster is held fixed throughout follow-up, retaining known nonproviders and subsequent exits.
Covered-market outcomes	Outcomes aggregated over the eligible facility population in an HRR in each year, including later openings and verified activity from administrative successors or locations with unresolved physical entry dates. The supplementary 2025 extension follows the verified 2024 roster.
Home dialysis–providing facilities; P_{hct}	Number of eligible cohort facilities with positive recorded Medicare home-dialysis activity in year t . Authorization records capability; provision is measured from treatment activity. A known positive component may establish provision when complete annual quantity remains unknown.

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Table A3. Definitions of Study Populations and Outcomes (continued)

Study measure	Definition
Providing home dialysis	Facility indicator equal to one for positive recorded Medicare home-dialysis activity, zero for a known nonprovider, and missing for unresolved status. Facility probability effects are expressed in percentage points.
Medicare home treatments; q_{it}, Q_{hct}	Facility quantity is Worksheet D, column 4, lines 7 + 8 + 3(9 + 10). Weekly peritoneal-dialysis entries are converted into three treatment equivalents per week. Q_{hct} sums q_{it} for the fixed eligible cohort. A patient may contribute multiple treatment equivalents.
Mean service per eligible entrant; $\bar{q}$	Total Medicare home-treatment equivalents divided by the original eligible entrant count, including known nonproviders. The decomposition uses a ratio of jointly standardized group quantities and counts. The denominator includes every original eligible entrant.
Baseline treatments per providing facility	Medicare home treatments divided by home dialysis–providing facilities in 2019, defined for a positive observed provider count. This ratio describes baseline treatment intensity in Table 1.
Home-service share; provision proportion p_{hct}	Recorded home dialysis–providing facilities divided by the original baseline or entry-cohort roster size. Empty cohorts have undefined local proportions. The decomposition uses the ratio of standardized provider and original-entrant counts.
Training treatments	Worksheet D, column 1, lines 3–6: reported all-payer self-care/home-dialysis training treatments. A trainee can contribute multiple treatments, and course completion remains unobserved. Primary annual measures require a 330–400-day report with at least 90 percent calendar-year coverage, starting no earlier than entry; follow-up begins in $c + 1$.
All-payer home and total treatments	Quantities from Worksheet D, column 1, underlying treatment ratios. Home treatments sum 7 + 8 + 3(9 + 10); total treatments sum lines 1–8 plus 3(9 + 10). Matched quantities exclude separately reported acute-kidney-injury activity.
All-payer home-treatment share	All-payer home treatments (Worksheet D, column 1, corresponding home rows) divided by all-payer total treatment equivalents from the same report and facility scope. The denominator sums lines 1–8 plus three times lines 9–10. Separate acute-kidney-injury rows are excluded.
Continuous provision from first full year	Positive provision in every required year from $c + 1$ through t . Any known zero makes the indicator zero; otherwise any unresolved required year leaves it missing. The first observation measures one-year provision status; later observations require progressively longer histories.
Provision in both 2023 and 2024	Positive home-dialysis activity in both calendar years on the original cohort roster. A known zero resolves the indicator to zero; otherwise an unresolved required year leaves it missing. For 2023 entrants the window includes their potentially partial entry year.

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Table A3. Definitions of Study Populations and Outcomes (continued)

Study measure	Definition
Count and service components	The symmetric allocations in equations (4) and (5). Count and mean-treatment components are treatment equivalents per HRR; count and provision-proportion components are providing facilities per HRR. Each pair sums to its standardized linear contrast.
Home dialysis training capability	Separate POS indicators for home hemodialysis and home peritoneal-dialysis training and support, using the first valid dated observation within 180 days of entry. These indicators record authorized capability.
Dialysis service capability	Separate POS indicators for hemodialysis and peritoneal-dialysis services under the same 180-day rule. Near-universal indicators can have insufficient variation for a finite comparison.
Registered nurses	Registered-nurse full-time equivalents in the accepted early POS record. Implausible all-zero personnel blocks remain missing.
Licensed practical nurses	Staff count in the accepted early POS record, retaining the source unit.
Dialysis / technical staff	Separate early full-time-equivalent fields for dialysis technicians and water-treatment/machine technical staff, using the validated POS dictionary and dated observation.
Dietitians / social workers / other personnel	Three separate early full-time-equivalent fields from the accepted POS personnel block. Missing fields and invalid personnel blocks retain their missing status.
Dialysis stations	Number of stations in a dated, address-verified DFC record; otherwise an accepted positive POS count or dated state capacity evidence. Recorded stations measure authorized capacity at the source observation date; opening-day equipment use remains unobserved.
Big Two / other-chain / independent entrants	Entry-time operator classification: DaVita or Fresenius, another identified chain, or an independent operator. Unknown identity is retained separately. Results may count each type per HRR or report its share among entrants with the stated denominator.
Documented operator experience	Linked pre-policy operator facility counts, home-service shares, and presence in the entrant's HRR. Experience averages retain documented histories; unlinked operator experience remains unknown.
Total operating expenses	HCRIS reported operating expenses, aggregated in the eligible HRR population and expressed in 2024 dollars. Missing or inconsistent accounting fields remain missing.
Expenses per treatment	Corresponding reported expenses divided by the matched treatment denominator in the validated cost-report construction; expressed in 2024 dollars per treatment.
Clinical / administrative staffing	HCRIS occupation-specific employee and contract full-time equivalents summed according to the frozen staffing dictionary. Unresolved components retain their missing status. These report-period totals differ from early POS snapshots.

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Table A3. Definitions of Study Populations and Outcomes (continued)

Study measure	Definition
Clinical FTE per 100 patients; treatments per FTE	The first ratio is 100 times clinical full-time equivalents divided by all-payer patients. The second divides all-payer treatment equivalents by clinical full-time equivalents. Both require matched observations and a positive denominator.
Direct-care compensation	HCRIS compensation assigned to direct-care labor under the validated accounting mapping; aggregated in 2024 dollars. The measure aggregates labor compensation across direct-care workers.
Medicare payments; payments per treatment	Reported Medicare payment totals and the corresponding matched Medicare per-treatment ratio, in 2024 dollars. They measure facility accounting receipts; total public spending per patient requires additional expenditure records.
Net patient-service income	Signed patient-service accounting income from the validated HCRIS mapping, reported as an HRR total or per matched treatment in 2024 dollars. Consolidated company profits and economic profits remain unobserved.
Mortality / hospitalization / readmission ratios	Three source-defined standardized clinical ratios from public dialysis reports, aggregated using the documented source weights. Each ratio retains its own event definition and adjustment benchmark.
Long-term catheter / fistula share	Separate public-report vascular-access shares within the source-defined patient populations. Effects are percentage points. Comparable 2020 catheter observations are unavailable.
Vascular-access index	$\frac{1}{2}(z_{\text{catheter}} - z_{\text{fistula}})$ on joint support. Component means and standard deviations are fixed from eligible 2019 HRR measures. The index is recomputed from verified eligible-facility observations; lower values are favorable. It summarizes the evidence in its catheter and fistula components.
Dialysis adequacy	The public-report fraction meeting the source-defined adequacy criterion, retained on its native fractional scale and aggregated with its specified weights. Adequacy retains its source fraction scale.
Nearest home-capable facility	Weighted distance in miles from fixed demand locations to the nearest eligible facility with administrative home-service capability. The national facility search can cross the demand HRR boundary. Capability differs from positive measured treatment activity.
Home-capable facilities within 15 miles	Number of eligible administratively home-capable facilities within 15 miles of each fixed demand location, using fixed demand weights. The national search includes facilities outside the demand HRR.
Home-service desert within 30 miles	Share of the fixed weighted demand population with no eligible administratively home-capable facility within 30 miles, allowing a national search across HRR boundaries. Effects are percentage-point changes in documented geographic availability.

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Table A3. Definitions of Study Populations and Outcomes (continued)

Study measure	Definition
Cost-report fields for operating inputs	Operating expenses: Worksheet C, line 18, column 2. All-payer treatment denominator: C, line 19, column 1, with the documented quantity reconciliation. Clinical FTEs: S-1, lines 23–26, column 3; administrative FTEs: S-1, lines 29–30, column 3. Direct-care compensation: A, lines 7 and 8, column 8. Stations and patient counts retain their dated registry/report fields.
Cost-report fields for financial outcomes	Medicare payments and treatment equivalents: Worksheet D, line 11, columns 8 and 4, respectively. Net patient-service revenue and net income: F-1, lines 3 and 18, column 2. Ratios use matched facility records with a positive aggregate denominator; monetary amounts are deflated to 2024 dollars.

Notes: Populations follow Form 265 eligibility. Outcomes use verified records without statistical imputation, distinguishing known zeros from missing activity. Section 3.2 defines windows and samples.

Table A4 describes the dispersion and prevalence of zeros in the main outcomes. Table A5 reports coverage using all required facility records within the fixed main HRRs as the denominator. Of the 149, 108, 80, and 63 classified entrants in 2021–2024 across the 302-HRR frame, the respective main samples retain 135, 101, 70, and 56. Thus the 400-location eligible roster yields 362 locations in the main cohort panels. The incumbent sample retains 6,970 of 6,971 locations. Availability of observed components determines the retained HRRs; completeness of every facility report is a separate criterion used in the robustness check. Empty local cohorts contribute zero, partially observed populations contribute known activity, and entirely unresolved nonempty populations remain unavailable.

The entry review distinguishes corroborated new openings, administratively classified openings without identified prior operations, and records with evidence of an existing site. New York’s January 2022 licensing review describes the acquisition of existing Faxton, Herkimer, and St. Luke’s clinics at the addresses later associated with CCNs 332852, 332854, and 332857. They are excluded from the 2023 new-location cohort. In Florida, CCN 682742 shares an address and suite with the earlier CCN 682686; original reports and enrollment records support exclusion from the 2022 new-location cohort but leave the intervening continuity of service unresolved.⁵¹ Their observed service remains in the covered-market category outside the classified new-location cohorts. Remaining alias and same-site ambiguities limit interpretation of administrative facility counts as distinct physical capacity. Previously excluded relocation and pre-existing-site cases remain excluded; exit zeros require evidence of physical closure.

⁵¹The New York evidence is in the January 27, 2022 Public Health and Health Planning Council exhibits, PDF pages 82–83 and 86. The Florida evidence includes original cost reports and NPPES records 1184141350 and 1558095513. These records do not resolve the August–September 2022 interval, so no continuous facility history or additional treatment quantity is imputed.

Source selection preserves report identity and period before aggregating values. For a given reporting interval, accepted revisions supersede the earlier filing. Calendar-year service totals use documented report intervals and exact reconciliations of nonoverlapping periods where available; an unresolved interval remains missing. The training outcome instead uses the qualifying report total attributed to its reference year under the 330–400-day and 90-percent-coverage rule. Input fields retain their accepted annual report mappings, including partial entry-year periods. Recoveries require compatible identities, explicit worksheet evidence, and the field’s period rule. A blank cell is treated as a source-coded zero only when companion cells and worksheet arithmetic establish that coding; report absence is never sufficient. The archive retains source-quality flags for unresolved staffing units, period conflicts, and identities, without inferring decimal corrections from neighboring facilities. These conventions improve documented coverage but cannot eliminate source errors or assignment-related reporting.

Table A4. Distributions of Main Supply Outcomes

Population / outcome	HRRs	Mean range	Zero share range	Maximum
2020 entrants: Home dialysis–providing facilities	290	0.17–0.26	81.4–86.6	7.0
2020 entrants: Medicare home treatments	290	46.73–317.16	81.7–87.2	12,024.0
2021 entrants: Home dialysis–providing facilities	290	0.18–0.24	83.4–85.9	4.0
2021 entrants: Medicare home treatments	290	63.76–339.68	83.4–86.2	7,812.0
2022 entrants: Home dialysis–providing facilities	296	0.16–0.20	86.5–87.8	4.0
2022 entrants: Medicare home treatments	296	39.59–175.63	86.8–87.8	5,889.0
2023 entrants: Home dialysis–providing facilities	293	0.11–0.14	90.4–91.5	4.0
2023 entrants: Medicare home treatments	293	29.65–112.46	90.4–91.5	4,044.0
2024 entrants: Home dialysis–providing facilities	295	0.09–0.09	91.2–91.2	2.0
2024 entrants: Medicare home treatments	295	30.98–30.98	91.2–91.2	2,325.0
Incumbents: Home dialysis–providing facilities	301	9.55–10.21	1.7–2.3	120.0
Incumbents: Medicare home treatments	301	12,117.09–17,683.17	1.7–3.0	164,118.0

Continued on next page

Table A4. Distributions of Main Supply Outcomes (continued)

Population / outcome	HRRs	Mean range	Zero share range	Maximum
Covered market: Home dialysis–providing facilities	301	9.99–10.78	0.7–1.7	123.0
Covered market: Medicare home treatments	301	13,311.79–17,766.90	1.0–2.0	164,229.0

Notes: Ranges summarize annual cross-sections on fixed HRRs from 2019 or cohort entry through 2024. Zero shares are percentages, including empty cohorts; means and maxima retain the stated outcome units.

Table A5. Facility Reporting Coverage within Fixed Main HRRs

Population / measure	2019	2020	2021	2022	2023	2024
Incumbents						
Fixed HRRs	301	301	301	301	301	301
Required facility records	6,970	6,970	6,970	6,970	6,970	6,970
Provision status known (percent)	96.9	97.0	97.0	97.0	96.9	96.9
Treatment quantity known (percent)	95.7	95.7	95.2	95.9	95.8	96.1
Covered market						
Fixed HRRs	301	301	301	301	301	301
Required facility records	6,970	7,180	7,348	7,481	7,570	7,648
Provision status known (percent)	96.9	96.8	96.8	96.9	96.7	96.8
Treatment quantity known (percent)	95.7	95.5	95.1	95.7	95.5	95.8
2020 entrants						
Fixed HRRs	—	290	290	290	290	290
Required facility records	—	177	177	177	177	177
Provision status known (percent)	—	93.8	97.7	98.9	98.3	98.3
Treatment quantity known (percent)	—	91.5	95.5	97.2	95.5	97.2
2021 entrants						
Fixed HRRs	—	—	290	290	290	290
Required facility records	—	—	135	135	135	135
Provision status known (percent)	—	—	96.3	98.5	99.3	99.3
Treatment quantity known (percent)	—	—	94.8	96.3	99.3	97.8
2022 entrants						
Fixed HRRs	—	—	—	296	296	296
Required facility records	—	—	—	101	101	101
Provision status known (percent)	—	—	—	94.1	97.0	98.0
Treatment quantity known (percent)	—	—	—	94.1	94.1	95.0
2023 entrants						
Fixed HRRs	—	—	—	—	293	293
Required facility records	—	—	—	—	70	70
Provision status known (percent)	—	—	—	—	97.1	98.6
Treatment quantity known (percent)	—	—	—	—	97.1	97.1

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Table A5. Facility Reporting Coverage within Fixed Main HRRs (continued)

Population / measure	2019	2020	2021	2022	2023	2024
2024 entrants						
Fixed HRRs	—	—	—	—	—	295
Required facility records	—	—	—	—	—	56
Provision status known (percent)	—	—	—	—	—	96.4
Treatment quantity known (percent)	—	—	—	—	—	96.4

Notes: Coverage is the share of required facility records with known outcomes within each fixed HRR sample. Documented exits and nonproviders count as observed zeros. Partially observed markets retain verified contributions; unreported activity remains unknown. Form 265 eligibility uses a separate denominator.

B. Supplementary Estimation and Inference

B.1. Facility Comparisons and Initial Configuration

For facility i in cohort c , the training and provision-history comparisons use

$$y_{ict} = \lambda_{s(h(i)),ct} + \beta_{ct}T_{h(i)} + \rho_{ct}Y_{h(i),2019} + \gamma'_{ct}\mathbf{X}_{h(i)} + \epsilon_{ict}. \quad (\text{A1})$$

Training uses the 2019 market home-treatment baseline, and provision histories use the home-provider baseline. The eight controls and assignment are measured at the facility's HRR. Comparisons weight observed facilities equally within the corresponding fixed aggregate geography, clustering standard errors and reassigning treatment at the HRR level. Because entry and reporting can respond to assignment, these coefficients describe the facilities observed under each assignment, including differences in entrant composition.

Initial configuration uses the first accepted observation z_{ic} within 180 days of entry:

$$z_{ic} = \lambda_{s(h(i)),c} + \delta_cT_{h(i)} + \rho_cY_{h(i),2019} + \gamma'_c\mathbf{X}_{h(i)} + v_{ic}. \quad (\text{A2})$$

Home-training capabilities use the 2019 home-provider baseline; other configuration measures use the 2019 eligible-facility count. The regressions retain dated records within the approved cohort HRRs. Selection into entry and the availability of early records also qualify these conditional comparisons.

B.2. Randomization and Coefficient Averages

Randomization inference preserves the original regional treatment totals over all 302 HRRs, even when a regression uses a subset. The Midwest contains 25 treated and 59 control HRRs, the Northeast 13 and 30, the South 35 and 80, and the West 18 and 42. The observed outcomes, covariates, and sample masks are held fixed. Validity for this selected sample therefore requires its

membership and measurement to be invariant to reassignment under the null; fixing the observed mask in the computation does not establish that condition.

For each specification, I first fit the nuisance model imposing zero treatment coefficients. Let $V(z)$ contain the assignment regressors under reassignment z , residualized against the nuisance design, and let r denote the null residual vector. In OLS, residualization is unweighted. In PPML, the design is weighted by the square root of the null fitted mean and r is the corresponding Pearson residual; HRR effects are included in the projection where applicable. The score-based coefficient vector is $\tilde{\tau}(z) = [V(z)'V(z)]^+ V(z)'r$, where $+$ denotes the generalized inverse. Its HRR influence vector is $\psi_h(z) = [V(z)'V(z)]^+ \sum_{i \in h} V_i(z)r_i / \sqrt{1 - \ell_i}$, with ℓ_i the null-model leverage, including absorbed HRR effects. For contrast weights w , the statistic is

$$S_w(z) = \frac{w'\tilde{\tau}(z)}{\{\sum_h [w'\psi_h(z)]^2\}^{1/2}}. \quad (\text{A3})$$

Annual contrasts select one coefficient; average contrasts weight the included annual coefficients equally. Summing the HRR influence vectors before squaring retains cross-year covariance. The two-sided comparison ranks $|S_w|$ under the observed assignment against 10,000 reassignments, with the observed assignment included in the Monte Carlo correction. The tabled point estimates and HC2 standard errors come from the unrestricted regressions, while the score statistic uses the restricted nuisance fit. Undefined statistics are reported as unavailable, without treating them as evidence of insignificance.

The reference null is no assignment effect on any represented unit and year. An average-score statistic is sensitive to a signed average departure, rather than being an omnibus test for every possible annual pattern. Rejection supplies evidence against this sharp null; validity for a weak zero-average null with heterogeneous effects requires additional arguments (Wu and Ding 2021). Ownership and operator-composition comparisons use fully re-estimated, studentized linear statistics under the same assignments. Significance refers to individual comparisons without a simultaneous adjustment across the reported outcomes. Conventional HC2 intervals characterize uncertainty in coefficient averages and can differ from the RI assessment.

C. Annual Cohort Results

C.1. The 2020 Transition Cohort

Table A6 collects the 2020 cohort's supply, training, continued provision, operating inputs, initial configuration, and operator characteristics. Its original roster contains 189 eligible facilities, and its core panel retains 290 HRRs. Core service and input averages include entry; training and provision histories follow the windows in Section 3.2.

Table A6. The 2020 Transition Cohort: Supplementary Outcomes

Outcome / unit	ETC effect	HRRs
Panel A. Home service, training, and continued provision		
Home dialysis–providing facilities per HRR	−0.048 (0.059)	290
Medicare home treatments per HRR	−86.13 (82.25)	290
Training treatments per HRR	−33.03 (28.70)	287
Training treatments per observed facility	−41.33 (45.67)	93
Provision in every full follow-up year (facilities per HRR)	−0.042 (0.058)	290
Provision in every full follow-up year (pp)	0.93 (8.35)	96
Provision in both 2023 and 2024 (facilities per HRR)	−0.062 (0.069)	290
Provision in both 2023 and 2024 (pp)	−0.18 (8.44)	96
Panel B. Operating inputs		
Operating expenses (million 2024 dollars per HRR)	−0.207 (0.241)	278
Dialysis stations per HRR	0.075 (1.192)	279
Clinical FTEs per HRR	−0.652 (0.737)	276
Administrative FTEs per HRR	−0.126 (0.175)	276
Direct-care compensation (million 2024 dollars per HRR)	−0.061 (0.069)	278
Expenses per treatment (2024 dollars)	35.05 (123.11)	81
Clinical FTEs per 100 patients	0.08 (0.88)	79
Treatments per clinical FTE	−141.78 (96.53)	80
Panel C. Initial facility configuration		
Home hemodialysis training and support (pp)	1.71 (6.16)	88
Home peritoneal dialysis training and support (pp)	−10.83 (7.74)	88
Registered nurses (FTEs)	0.54 (0.70)	88

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Table A6. The 2020 Transition Cohort: Supplementary Outcomes (continued)

Outcome / unit	ETC effect	HRRs
Dialysis stations	-0.85 (1.28)	68
Hemodialysis service (pp)	-4.75 (8.52)	88
Peritoneal-dialysis service (pp)	4.87 (4.55)	88
Licensed practical nurses (staff count)	-0.01 (0.06)	88
Dialysis technicians (FTEs)	0.79 (0.62)	88
Water/machine technicians (FTEs)	0.09 (0.09)	88
Dietitians (FTEs)	-0.21 (0.17)	88
Social workers (FTEs)	-0.17 (0.16)	88
Other staff (FTEs)	-0.27 (0.20)	88
Panel D. Entrant ownership		
DaVita and Fresenius (facilities per HRR)	-0.128 (0.103)	290
Other chains (facilities per HRR)	0.000 (0.048)	290
Independent facilities (facilities per HRR)	-0.033 (0.035)	290
Unknown ownership (facilities per HRR)	0.001 (0.020)	290
DaVita and Fresenius (pp)	-4.667 (9.804)	290
Other chains (pp)	5.649 (8.118)	290
Independent facilities (pp)	-1.860 (5.808)	290
Unknown ownership (pp)	0.878 (3.662)	290

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Table A6. The 2020 Transition Cohort: Supplementary Outcomes (continued)

Outcome / unit	ETC effect	HRRs
Panel E. Documented operator experience		
Operator facilities in 2018	−202.805 (208.757)	290
Operator home hemodialysis share in 2018 (pp)	0.859 (1.918)	290
Operator home peritoneal dialysis share in 2018 (pp)	2.069 (1.384)	290
Local operator facilities in 2019	−13.820* (7.908)	290

Notes: Core service and input averages cover 2020–2024; training and continuous-provision averages cover 2021–2024. The common-window outcome requires provision in 2023 and 2024. Annual coefficients receive equal weights. Panel C uses accepted records within 180 days of entry; facility estimates condition on entry and reporting. Panels D–E report standardized ownership and experience contrasts. HRRs are outcome-specific; pp denotes percentage points and FTE denotes full-time equivalent. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) from 10,000 region-stratified reassignments, using efficient scores in Panels A–C and re-estimated studentized statistics in Panels D–E.

C.2. Annual Service Estimates

Figure 2 displays the annual cohort paths; Table A7 reports the underlying coefficients and the averages summarized in Table 2, together with the 2020 transition cohort. Provision and treatments share the same fixed HRRs within each cohort. Each average runs from entry through 2024 and retains covariance across annual coefficients.

Table A7. Annual Home Dialysis Supply by Entry Cohort

Outcome / population	Average	2020	2021	2022	2023	2024
2020 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	−0.048 (0.059)	−0.050 (0.056)	−0.047 (0.060)	−0.069 (0.063)	−0.024 (0.068)	−0.049 (0.071)
Control mean	0.25	0.19	0.25	0.27	0.27	0.27
HRR-years	1,450	290	290	290	290	290
HRRs	290	290	290	290	290	290
2020 entrants: Medicare home treatments (HRR)						
ETC effect	−86.13 (82.25)	4.28 (32.86)	−27.05 (70.30)	−66.80 (93.63)	−162.53 (117.92)	−178.55 (134.98)
Control mean	240.20	48.30	177.87	271.71	339.19	363.93
HRR-years	1,450	290	290	290	290	290
HRRs	290	290	290	290	290	290

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Table A7. Annual Home Dialysis Supply by Entry Cohort (continued)

Outcome / population	Average	2020	2021	2022	2023	2024
2021 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	0.125*	—	0.087	0.146*	0.140*	0.127
	(0.073)	—	(0.062)	(0.082)	(0.078)	(0.078)
Control mean	0.18	—	0.16	0.18	0.19	0.20
HRR-years	1,160	—	290	290	290	290
HRRs	290	—	290	290	290	290
2021 entrants: Medicare home treatments (HRR)						
ETC effect	183.01*	—	55.23	193.86*	220.57*	262.37
	(102.86)	—	(36.90)	(101.82)	(125.72)	(158.16)
Control mean	158.44	—	46.16	134.62	195.34	257.63
HRR-years	1,160	—	290	290	290	290
HRRs	290	—	290	290	290	290
2022 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	0.134**	—	—	0.167***	0.118*	0.116*
	(0.063)	—	—	(0.062)	(0.067)	(0.067)
Control mean	0.15	—	—	0.11	0.17	0.17
HRR-years	888	—	—	296	296	296
HRRs	296	—	—	296	296	296
2022 entrants: Medicare home treatments (HRR)						
ETC effect	86.12	—	—	50.73	97.60	110.03
	(66.15)	—	—	(40.66)	(72.52)	(96.49)
Control mean	87.14	—	—	24.15	93.66	143.62
HRR-years	888	—	—	296	296	296
HRRs	296	—	—	296	296	296
2023 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	−0.051	—	—	—	−0.070*	−0.033
	(0.045)	—	—	—	(0.039)	(0.053)
Control mean	0.14	—	—	—	0.13	0.15
HRR-years	586	—	—	—	293	293
HRRs	293	—	—	—	293	293
2023 entrants: Medicare home treatments (HRR)						
ETC effect	−90.74**	—	—	—	−32.87	−148.60***
	(33.49)	—	—	—	(19.28)	(50.54)
Control mean	93.27	—	—	—	37.12	149.42
HRR-years	586	—	—	—	293	293
HRRs	293	—	—	—	293	293

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Table A7. Annual Home Dialysis Supply by Entry Cohort (continued)

Outcome / population	Average	2020	2021	2022	2023	2024
2024 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	–0.015 (0.039)	—	—	—	—	–0.015 (0.039)
Control mean	0.09	—	—	—	—	0.09
HRR-years	295	—	—	—	—	295
HRRs	295	—	—	—	—	295
2024 entrants: Medicare home treatments (HRR)						
ETC effect	–2.71 (17.23)	—	—	—	—	–2.71 (17.23)
Control mean	32.06	—	—	—	—	32.06
HRR-years	295	—	—	—	—	295
HRRs	295	—	—	—	—	295

Notes: Annual linear estimates follow equation (2) on fixed cohort HRRs. Averages weight coefficients equally from entry through 2024. Outcomes are providing facilities and Medicare home-treatment equivalents per HRR. Totals retain verified contributions and known empty-cohort zeros. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes indicate years before entry.

D. Supplementary Evidence on Entrant Responses

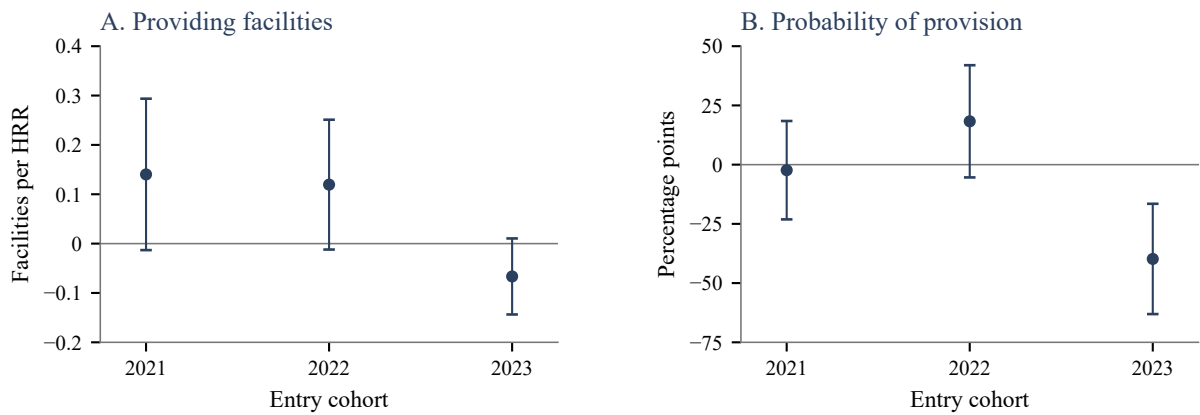
D.1. Continued Provision

Figure A2 compares provision in both 2023 and 2024 at the HRR and facility levels. The 2021 and 2022 cohorts have positive provider-count responses, while their conditional facility probabilities remain imprecise; the 2023 responses are negative. Table A8 reports these common-window comparisons and average responses for provision in every year from the first full follow-up year onward.⁵² Both measures retain the original roster. A known zero resolves a history to zero; otherwise an unresolved required year leaves it missing.

The 2021 and 2022 cohorts contribute an additional 0.140 and 0.120 facilities per HRR providing in both years, respectively; both increases are statistically significant. Their growing-history count responses are similar, with weaker evidence of a change in service continuity per observed entrant. The 2023 common-window estimates are –0.067 facilities per HRR and –39.80 percentage points at the facility level. The regional count is imprecise, while the conditional facility decline is statistically significant. These comparisons associate the early cohorts’ continuing supply contribution with a larger set of active locations, while leaving individual facilities’ persistence and selection into entry difficult to separate.

⁵²For 2023 entrants, the common window includes entry, and the first-full-year measure covers 2024 alone. The 2024 cohort has no corresponding follow-up.

Figure A2. Providing home dialysis in both 2023 and 2024



Notes: The outcome requires positive home provision in both 2023 and 2024, including the entry year for the 2023 cohort. Linear estimates follow equations (2) and (A1); facility comparisons condition on observed entry. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A8. Continued Home Dialysis Provision

Outcome / level	2021	2022	2023	2024
Provision in every full follow-up year / HRR count				
ETC effect	0.138*	0.119*	-0.033	—
	(0.080)	(0.067)	(0.053)	—
Control mean	0.18	0.16	0.15	—
HRR-years	870	592	293	—
HRRs	290	296	293	—
Provision in both 2023 and 2024 / HRR count				
ETC effect	0.140*	0.120*	-0.067	—
	(0.078)	(0.067)	(0.039)	—
Control mean	0.19	0.16	0.13	—
HRR-years	290	296	293	—
HRRs	290	296	293	—
Provision in every full follow-up year / facility probability				
ETC effect	-3.47	17.55	-19.30	—
	(11.13)	(12.16)	(15.59)	—
Control mean	45.31	51.94	65.96	—
Facility-years	401	197	69	—
HRRs	73	66	44	—
Provision in both 2023 and 2024 / facility probability				
ETC effect	-2.34	18.29	-39.80***	—
	(10.59)	(12.08)	(11.88)	—
Control mean	46.34	50.77	54.17	—
Facility-years	134	99	70	—
HRRs	73	66	44	—

Notes: Linear estimates follow equations (2) and (A1). Continuous provision requires positive activity from the first full year after entry through each follow-up year; annual effects are averaged equally through 2024. The common-window outcome requires provision in 2023 and 2024. A known zero resolves a history to zero; otherwise incomplete histories remain missing. Facility effects are percentage points, conditional on observed entrants; control means are percentages. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes denote unavailable outcomes.

D.2. Initial Configuration and Entrant Composition

Table A9 describes the initial configuration and operator composition of the 2021–2024 cohorts.⁵³ Configuration uses dated first-180-day records and equation (A2). Ownership counts, ownership shares, and documented operator experience use standardized HRR totals or ratios with region effects and the eight controls, without a separate outcome baseline. These comparisons describe who enters and the resources recorded near entry; latent capabilities and subsequent investment remain unobserved. The 2020 comparisons are collected in Table A6.

Table A9. Initial Configuration and Observable Entrant Composition

Outcome	2021	2022	2023	2024
Panel A. Dated facility configuration within 180 days of entry				
Home hemodialysis training and support (pp)				
ETC effect	−3.35 (8.99)	2.81 (9.16)	−10.94 (16.97)	5.21 (16.45)
Control mean	12.50	15.00	25.00	25.71
Facility-years	118	90	52	49
HRRs	65	61	35	40
Home peritoneal dialysis training and support (pp)				
ETC effect	−3.64 (10.01)	−1.21 (13.26)	−2.36 (21.19)	−3.06 (19.72)
Control mean	31.94	50.00	47.22	37.14
Facility-years	118	90	52	49
HRRs	65	61	35	40
Registered nurses (FTEs)				
ETC effect	1.59 (0.84)	0.11 (0.61)	−1.44 (1.48)	−0.13 (0.30)
Control mean	2.32	2.11	2.17	1.92
Facility-years	115	89	52	49
HRRs	64	60	35	40
Dialysis stations				
ETC effect	4.33 (3.46)	−5.84** (2.34)	— —	3.23 (3.45)
Control mean	13.11	13.78	12.83	12.41
Facility-years	41	40	6	33
HRRs	30	33	3	28

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⁵³The 2023 station sample is insufficient for the specified regression. Unknown operator histories remain missing.

Table A9. Initial Configuration and Observable Entrant Composition (continued)

Outcome	2021	2022	2023	2024
Hemodialysis service (pp)				
ETC effect	1.73 (8.42)	-4.12 (11.20)	12.46 (15.87)	-9.16 (11.56)
Control mean	76.39	71.67	72.22	82.86
Facility-years	118	90	52	49
HRRs	65	61	35	40
Peritoneal-dialysis service (pp)				
ETC effect	-1.29 (3.88)	3.03 (11.76)	-25.08** (9.54)	-14.14 (9.96)
Control mean	4.17	10.00	11.11	11.43
Facility-years	118	90	52	49
HRRs	65	61	35	40
Licensed practical nurses (staff count)				
ETC effect	0.11 (0.08)	-0.44* (0.34)	0.00 (0.06)	0.11 (0.10)
Control mean	0.15	0.27	0.06	0.13
Facility-years	118	90	52	49
HRRs	65	61	35	40
Dialysis technicians (FTEs)				
ETC effect	0.51 (0.39)	-0.41 (0.43)	-0.12 (0.31)	0.06 (0.40)
Control mean	1.00	1.36	1.04	1.26
Facility-years	118	90	52	49
HRRs	65	61	35	40
Water/machine technicians (FTEs)				
ETC effect	-0.12 (0.12)	0.17 (0.19)	0.06 (0.29)	0.06 (0.19)
Control mean	0.66	0.56	0.92	0.59
Facility-years	118	90	52	49
HRRs	65	61	35	40
Dietitians (FTEs)				
ETC effect	-0.14 (0.10)	0.01 (0.10)	0.13 (0.22)	0.13 (0.15)
Control mean	0.81	0.58	0.87	0.59
Facility-years	118	90	51	49
HRRs	65	61	35	40

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Table A9. Initial Configuration and Observable Entrant Composition (continued)

Outcome	2021	2022	2023	2024
Social workers (FTEs)				
ETC effect	-0.23 (0.13)	-0.55 (0.41)	0.12 (0.22)	0.14 (0.14)
Control mean	0.84	0.91	0.87	0.58
Facility-years	118	90	52	49
HRRs	65	61	35	40
Other staff (FTEs)				
ETC effect	0.27 (0.19)	0.60 (0.57)	1.06* (0.48)	-0.33 (0.23)
Control mean	0.26	0.43	0.54	0.42
Facility-years	118	90	52	49
HRRs	65	61	35	40
Panel B. Entrant counts per HRR				
DaVita and Fresenius				
ETC effect	0.066 (0.089)	0.044 (0.058)	-0.067 (0.044)	0.054 (0.039)
Control mean	0.20	0.18	0.13	0.07
HRR-years	135	101	70	56
HRRs	290	296	293	295
Other chains				
ETC effect	0.203 (0.122)	0.032 (0.037)	-0.003 (0.033)	-0.030 (0.020)
Control mean	0.09	0.06	0.05	0.04
HRR-years	135	101	70	56
HRRs	290	296	293	295
Independent facilities				
ETC effect	-0.008 (0.034)	0.013 (0.035)	0.062 (0.044)	-0.041 (0.028)
Control mean	0.08	0.07	0.05	0.08
HRR-years	135	101	70	56
HRRs	290	296	293	295
Unknown ownership				
ETC effect	-0.010 (0.007)	-0.007 (0.008)	0.014 (0.016)	-0.010 (0.008)
Control mean	0.01	0.01	0.01	0.01
HRR-years	135	101	70	56
HRRs	290	296	293	295

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Table A9. Initial Configuration and Observable Entrant Composition (continued)

Outcome	2021	2022	2023	2024
Panel C. Ownership shares (percentage points)				
DaVita and Fresenius				
ETC effect	-10.157 (12.263)	-0.229 (9.832)	-29.081** (12.757)	36.739** (16.245)
Control mean	52.26	55.53	56.03	34.76
HRR-years	135	101	70	56
HRRs	290	296	293	295
Other chains				
ETC effect	22.409 (12.719)	3.783 (8.879)	-1.877 (11.842)	-14.468 (9.655)
Control mean	23.52	20.44	20.57	18.18
HRR-years	135	101	70	56
HRRs	290	296	293	295
Independent facilities				
ETC effect	-9.295 (6.403)	-1.142 (7.849)	25.105* (13.835)	-17.185 (13.266)
Control mean	20.80	21.20	19.47	42.12
HRR-years	135	101	70	56
HRRs	290	296	293	295
Unknown ownership				
ETC effect	-2.958 (1.761)	-2.411 (2.519)	5.852 (6.682)	-5.086 (4.324)
Control mean	3.42	2.83	3.93	4.94
HRR-years	135	101	70	56
HRRs	290	296	293	295
Panel D. Documented operator experience				
Operator facilities in 2018				
ETC effect	-398.330 (311.075)	-262.509 (211.903)	-363.180 (357.127)	788.285*** (242.567)
Control mean	1,686.57	1,941.92	1,835.07	1,558.22
HRR-years	135	101	70	56
HRRs	290	296	293	295
Operator home hemodialysis share in 2018 (pp)				
ETC effect	-1.516 (2.058)	-2.208 (1.777)	10.749 (10.159)	3.052 (3.627)
Control mean	20.58	20.90	21.45	22.57
HRR-years	135	101	70	56
HRRs	290	296	293	295

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Table A9. Initial Configuration and Observable Entrant Composition (continued)

Outcome	2021	2022	2023	2024
Operator home peritoneal dialysis share in 2018 (pp)				
ETC effect	2.921* (1.741)	−0.210 (2.471)	7.095 (6.194)	−6.279*** (2.245)
Control mean	43.90	43.01	43.92	47.90
HRR-years	135	101	70	56
HRRs	290	296	293	295
Local operator facilities in 2019				
ETC effect	−5.588 (4.070)	−9.328* (5.062)	−13.561 (11.907)	8.604 (5.133)
Control mean	18.04	24.01	22.34	10.48
HRR-years	135	101	70	56
HRRs	290	296	293	295

Notes: Panel A uses equation (A2) and records within 180 days of entry; binary effects are percentage points. Panels B–D standardize HRR counts or ratios with region effects and eight controls; control means are adjusted and experience measures require documented values. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) from 10,000 region-stratified reassignments, using efficient scores in Panel A and re-estimated studentized statistics otherwise. Dashes denote unavailable estimates.

E. Incumbent Supply and Market Accounting

Figure 4 displays the annual incumbent and covered-market paths; Table A10 reports the underlying coefficients and the averages summarized in Table 4, including the 2020 comparison relative to 2019. Both panels retain 301 HRRs. Incumbent membership includes the 2019 entrants and remains fixed; annual totals retain verified facility contributions within the covered reporting population.

Table A10. Annual Incumbent and Covered-Market Supply

Outcome / population	Average	2020	2021	2022	2023	2024
Incumbents: Home dialysis–providing facilities (HRR)						
ETC effect	0.228 (0.192)	0.055 (0.117)	−0.004 (0.163)	0.216 (0.193)	0.340* (0.223)	0.358 (0.266)
Control mean	10.01	10.29	10.39	10.15	9.88	9.64
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Incumbents: Medicare home treatments (HRR)						
ETC effect	−135.62 (640.82)	14.95 (520.48)	126.07 (540.54)	143.90 (692.05)	−348.50 (688.72)	−463.96 (813.58)
Control mean	13,336.65	17,315.19	15,072.57	13,615.65	12,666.93	11,991.46
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301

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Table A10. Annual Incumbent and Covered-Market Supply (continued)

Outcome / population	Average	2020	2021	2022	2023	2024
Covered market: Home dialysis–providing facilities (HRR)						
ETC effect	0.299 (0.243)	−0.025 (0.129)	0.013 (0.196)	0.358 (0.245)	0.439 (0.289)	0.384 (0.326)
Control mean	10.82	10.51	10.88	10.86	10.81	10.75
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Covered market: Medicare home treatments (HRR)						
ETC effect	−73.52 (579.57)	−38.42 (518.17)	152.83 (485.37)	540.61 (614.19)	−393.67 (685.96)	−593.87 (797.12)
Control mean	14,122.39	17,416.62	15,447.29	14,277.34	13,559.32	13,205.62
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301

Notes: Annual linear estimates follow equation (1), relative to 2019, on 301 fixed HRRs. Averages weight the 2021–2024 coefficients equally. Outcomes are providing facilities and Medicare home-treatment equivalents per HRR, summing verified contributions. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null.

Table A11 connects the populations using a common linear design. The outcome identity is

$$Y_{ht}^{\text{market}} = Y_{ht}^{I2019} + \sum_{c=2020}^{2024} Y_{ht}^c + Y_{ht}^{\text{other}}. \quad (\text{A4})$$

Identical regressors, observations, and weights make the component coefficients add to the market coefficient. The shared samples retain 271 HRRs for providers and 260 for treatments. Pre-entry contributions are structural zeros. The residual contains verified activity from facilities outside the fixed incumbent and classified entry rosters, including the four existing-site records removed from entry. Across all 302 HRRs, the category has 21, 40, 65, 74, and 89 facility records in 2020–2024; the treatment-accounting sample contains 18, 34, 48, 54, and 66. Earlier certification dates in these histories do not by themselves resolve physical entry. On the common treatment sample, the residual contributes −249.2 of the market point estimate of −267.5 equivalents per HRR. Its size makes unresolved entry histories material to interpreting the market decomposition. The exercise establishes an additive identity; it cannot trace patient transfers or attribute the residual to a behavioral mechanism.

Table A11. Additive Market Accounting

Component	Average	2020	2021	2022	2023	2024
Home dialysis–providing facilities: Covered market						
ETC effect	0.238 (0.248)	−0.074 (0.133)	−0.115 (0.206)	0.198 (0.246)	0.441 (0.294)	0.427 (0.334)
Control mean	10.01	9.70	10.08	10.05	9.98	9.91
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Home dialysis–providing facilities: Incumbents						
ETC effect	0.225 (0.199)	0.007 (0.123)	−0.047 (0.172)	0.134 (0.198)	0.401* (0.227)	0.411 (0.280)
Control mean	9.23	9.48	9.58	9.36	9.09	8.87
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Home dialysis–providing facilities: 2020 entrants						
ETC effect	−0.069 (0.075)	−0.062 (0.064)	−0.067 (0.072)	−0.097 (0.078)	−0.036 (0.083)	−0.075 (0.084)
Control mean	0.27	0.19	0.26	0.27	0.27	0.27
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Home dialysis–providing facilities: 2021 entrants						
ETC effect	0.081 (0.085)	0.000 (0.000)	0.057 (0.072)	0.103 (0.094)	0.094 (0.090)	0.071 (0.091)
Control mean	0.20	0.00	0.17	0.19	0.20	0.22
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Home dialysis–providing facilities: 2022 entrants						
ETC effect	0.084 (0.057)	0.000 (0.000)	0.000 (0.000)	0.150** (0.074)	0.092 (0.081)	0.094 (0.081)
Control mean	0.11	0.00	0.00	0.11	0.17	0.16
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Home dialysis–providing facilities: 2023 entrants						
ETC effect	−0.026 (0.030)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	−0.068 (0.053)	−0.036 (0.070)
Control mean	0.07	0.00	0.00	0.00	0.14	0.16
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271

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Table A11. Additive Market Accounting (continued)

Component	Average	2020	2021	2022	2023	2024
Home dialysis–providing facilities: 2024 entrants						
ETC effect	0.005 (0.010)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.018 (0.042)
Control mean	0.02	0.00	0.00	0.00	0.00	0.08
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Home dialysis–providing facilities: Other						
ETC effect	−0.063* (0.038)	−0.019 (0.019)	−0.058** (0.030)	−0.092*** (0.038)	−0.043 (0.048)	−0.058 (0.050)
Control mean	0.11	0.03	0.07	0.11	0.12	0.15
HRR-years	1,084	271	271	271	271	271
HRRs	271	271	271	271	271	271
Medicare home treatments: Covered market						
ETC effect	−267.5 (643.3)	88.3 (397.7)	−42.0 (523.4)	−106.6 (691.7)	−417.7 (739.9)	−503.8 (813.5)
Control mean	13,294.79	16,115.67	14,417.94	13,511.09	12,769.18	12,480.96
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260
Medicare home treatments: Incumbents						
ETC effect	−91.1 (646.1)	139.0 (401.5)	96.0 (514.1)	14.7 (694.1)	−265.5 (735.6)	−209.6 (842.3)
Control mean	12,516.88	16,011.67	14,064.57	12,851.61	11,891.49	11,259.85
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260
Medicare home treatments: 2020 entrants						
ETC effect	−79.0 (113.4)	14.6 (40.9)	−13.7 (85.1)	−53.1 (110.6)	−113.3 (131.0)	−136.0 (148.7)
Control mean	283.31	45.37	178.17	279.55	324.40	351.11
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260
Medicare home treatments: 2021 entrants						
ETC effect	126.5 (111.6)	0.0 (0.0)	46.8 (40.9)	143.9 (109.2)	155.5 (136.7)	160.0 (173.6)
Control mean	164.54	0.00	45.90	138.05	200.46	273.74
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260

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Table A11. Additive Market Accounting (continued)

Component	Average	2020	2021	2022	2023	2024
Medicare home treatments: 2022 entrants						
ETC effect	69.3 (64.5)	0.0 (0.0)	0.0 (0.0)	54.6 (50.5)	101.4 (93.4)	121.3 (128.4)
Control mean	65.84	0.00	0.00	25.22	95.98	142.18
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260
Medicare home treatments: 2023 entrants						
ETC effect	-45.7** (21.3)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	-33.9 (25.3)	-148.7** (62.9)
Control mean	48.37	0.00	0.00	0.00	39.17	154.31
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260
Medicare home treatments: 2024 entrants						
ETC effect	1.6 (5.2)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	6.3 (20.9)
Control mean	7.18	0.00	0.00	0.00	0.00	28.73
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260
Medicare home treatments: Other						
ETC effect	-249.2*** (105.5)	-65.3 (54.1)	-171.0** (92.8)	-266.6** (117.7)	-262.0*** (115.5)	-297.1** (128.0)
Control mean	208.67	58.63	129.29	216.66	217.69	271.04
HRR-years	1,040	260	260	260	260	260
HRRs	260	260	260	260	260	260

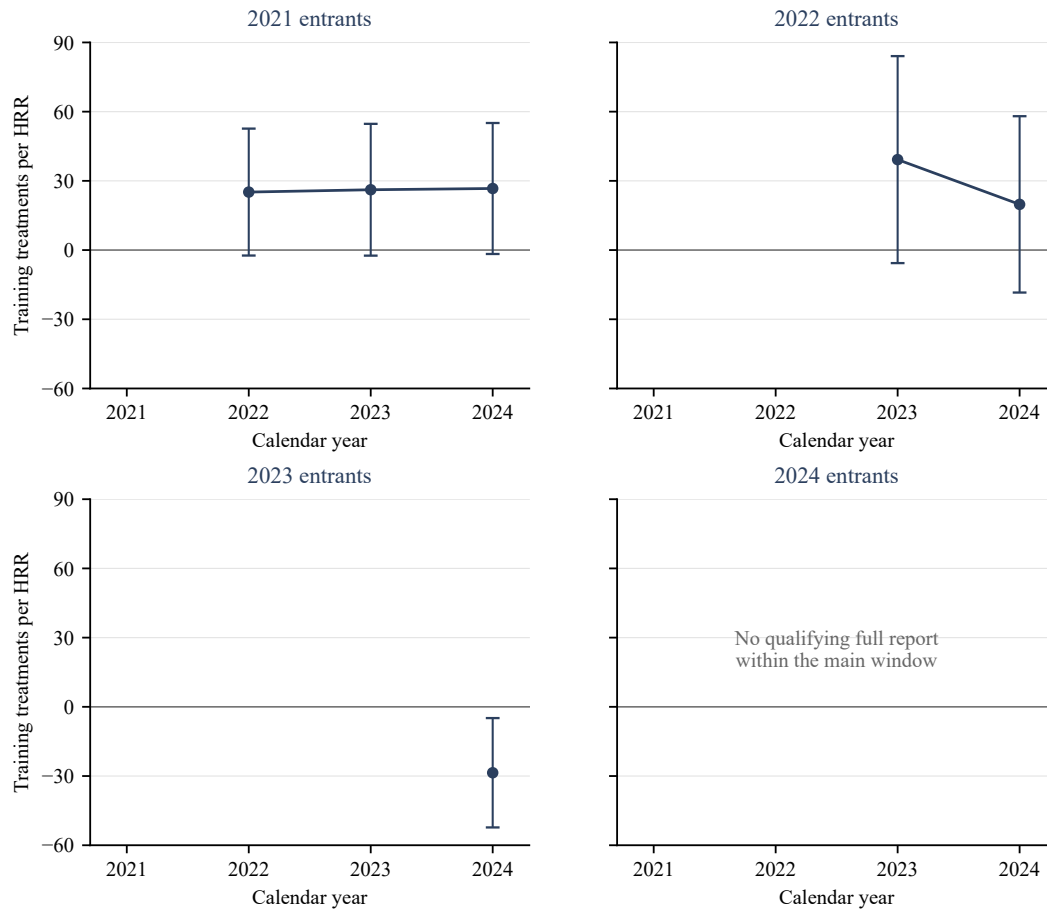
Notes: Equal-weight 2021–2024 averages follow equation (1). Components share observations, weights, and controls, so coefficients sum to the market effect. Pre-entry values are zero; the residual contains verified activity outside the incumbent and dated entry rosters. Provision and treatment samples contain 271 and 260 HRRs. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null.

F. Additional Activity, Inputs and Economic Outcomes

F.1. Training Activity and Reporting

Figure A3 displays the annual HRR training paths discussed in Section 6.1, with positive early-cohort responses and a negative 2023 response. Table A12 gives annual and average estimates at both the HRR and conditional facility levels, including the transition cohort. The 2021 HRR average is 25.99 additional training treatments and is statistically significant; its facility average is 21.28 and is imprecise. For 2023 entrants, the respective declines of 28.58 and 139.65 are statistically significant. The comparison therefore provides weaker evidence of more training per early entrant than of a larger regional training contribution.

Figure A3. Reported training activity by entry cohort



Notes: Annual all-payer training treatments per HRR follow equation (2), using the 2019 home-treatment baseline. Fixed cohorts and outcome-specific HRRs cover qualifying full reports after entry through 2024. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A12. Annual Reported Training Activity

Outcome / population	Average	2021	2022	2023	2024
2020 entrants / HRR					
ETC effect	-33.03 (28.70)	-41.42 (35.46)	-46.09 (36.82)	-41.86 (38.69)	-2.75 (13.70)
Control mean	53.63	55.06	60.31	63.70	35.45
HRR-years	1,148	287	287	287	287
HRRs	287	287	287	287	287

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Table A12. Annual Reported Training Activity (continued)

Outcome / population	Average	2021	2022	2023	2024
2020 entrants / facility					
ETC effect	-41.33 (45.67)	-52.75 (55.49)	-58.51 (58.13)	-54.95 (62.17)	0.88 (22.44)
Control mean	83.96	86.03	93.51	100.31	55.83
Facility-years	675	169	170	168	168
HRRs	93	93	93	93	93
2021 entrants / HRR					
ETC effect	25.99* (14.00)	—	25.14* (14.04)	26.14* (14.57)	26.69* (14.48)
Control mean	20.97	—	18.02	23.64	21.25
HRR-years	864	—	288	288	288
HRRs	288	—	288	288	288
2021 entrants / facility					
ETC effect	21.28 (26.81)	—	25.89 (32.02)	23.60 (25.30)	14.36 (25.22)
Control mean	53.43	—	46.30	59.98	53.91
Facility-years	395	—	131	132	132
HRRs	71	—	71	71	71
2022 entrants / HRR					
ETC effect	29.51 (20.22)	—	—	39.21* (22.89)	19.80 (19.50)
Control mean	16.93	—	—	12.95	20.92
HRR-years	590	—	—	295	295
HRRs	295	—	—	295	295
2022 entrants / facility					
ETC effect	74.52 (52.18)	—	—	102.55* (57.14)	46.50 (53.19)
Control mean	53.67	—	—	41.05	66.29
Facility-years	194	—	—	97	97
HRRs	65	—	—	65	65
2023 entrants / HRR					
ETC effect	-28.58** (12.09)	—	—	—	-28.58** (12.09)
Control mean	28.81	—	—	—	28.81
HRR-years	292	—	—	—	292
HRRs	292	—	—	—	292

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Table A12. Annual Reported Training Activity (continued)

Outcome / population	Average	2021	2022	2023	2024
2023 entrants / facility					
ETC effect	-139.65*** (58.90)	—	—	—	-139.65*** (58.90)
Control mean	131.24	—	—	—	131.24
Facility-years	66	—	—	—	66
HRRs	43	—	—	—	43

Notes: Linear estimates follow equations (2) and (A1), using the 2019 home-treatment baseline. Training counts cover all payers. Qualifying reports last 330–400 days and cover at least 90 percent of the calendar year; cross-year totals are permitted. Follow-up covers full years after entry through 2024, with equal weights in averages. HRR totals retain verified contributions and empty-cohort zeros; facility estimates condition on entry and reporting. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes indicate years outside the reporting window.

Table A13 records known training outcomes and original facility counts by assignment and year. The HRR regressions retain a fixed subset with observed aggregate training in every eligible year, while facility comparisons use the actual reports within that geography. The full-report rule begins after entry, leaving the 2024 cohort without a qualifying main-window observation. Coverage describes the represented activity and cannot establish that missingness is independent of assignment.

Table A13. Training-Report Coverage

Cohort / year	Control known	Control roster	ETC known	ETC roster
2020 / 2021	129	135	42	42
2020 / 2022	130	135	42	42
2020 / 2023	128	135	41	42
2020 / 2024	128	135	42	42
2021 / 2022	80	83	52	52
2021 / 2023	80	83	52	52
2021 / 2024	82	83	52	52
2022 / 2023	65	67	33	34
2022 / 2024	65	67	32	34
2023 / 2024	45	48	21	22

Notes: Coverage refers to original entrants within the main HRRs. Training regressions retain outcome-specific HRRs observed throughout follow-up. Unknown reports remain missing; coverage rates alone cannot establish random missingness.

F.2. Operating Inputs by Facility Population

Figures A4, A5, A6, and A7 display the eight entrant input paths underlying Table 5. They cover operating expenses and capacity, clinical and administrative staffing, compensation and expenses per treatment, and staffing intensity. Each outcome retains a fixed subset of the cohort’s main HRRs

and its corresponding observed 2019 input baseline. Totals include known empty cohorts; matched ratios require positive observed denominators and consequently represent narrower populations.

Figure A4. Entrant Operating Expenses and Dialysis Capacity

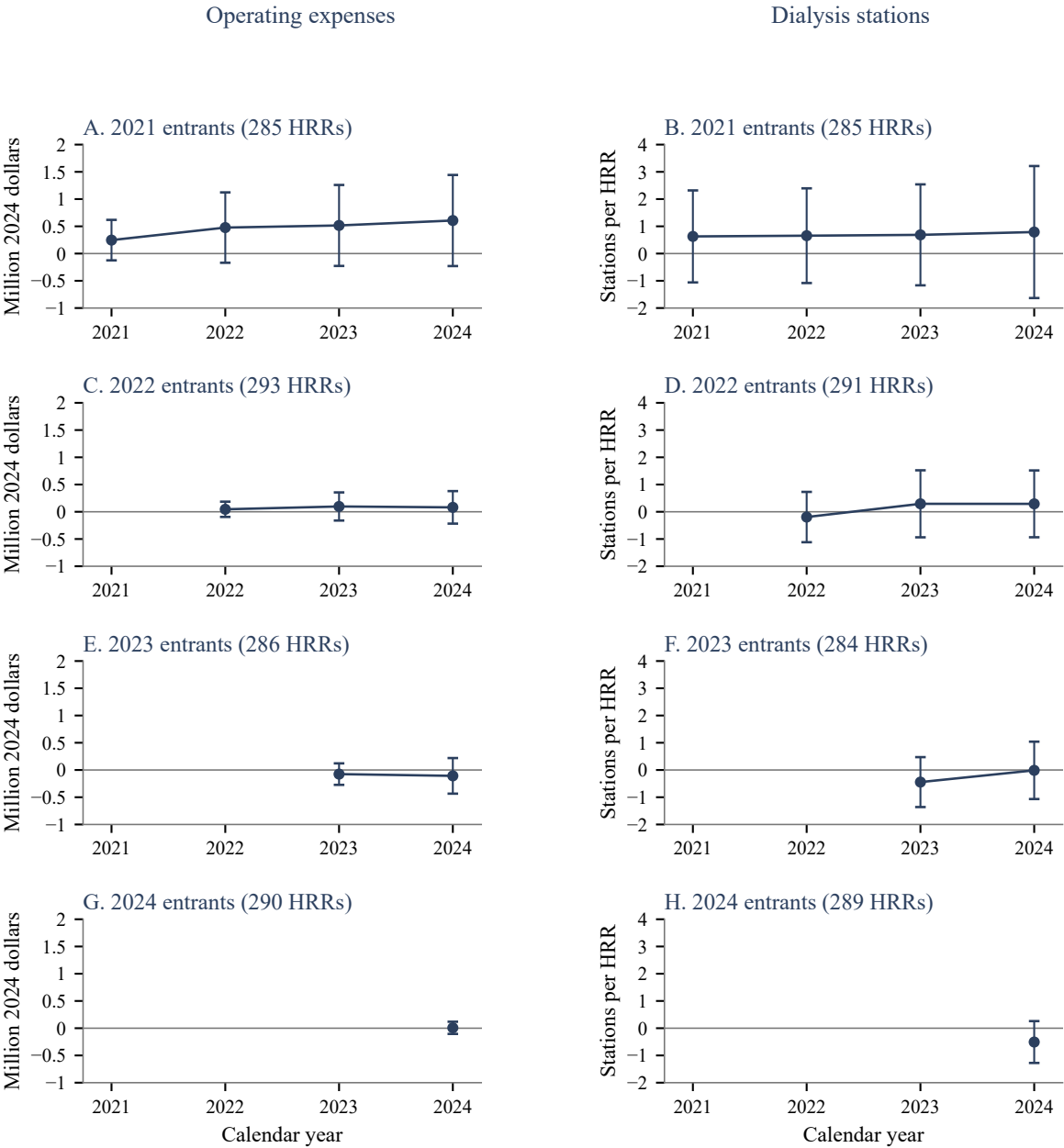
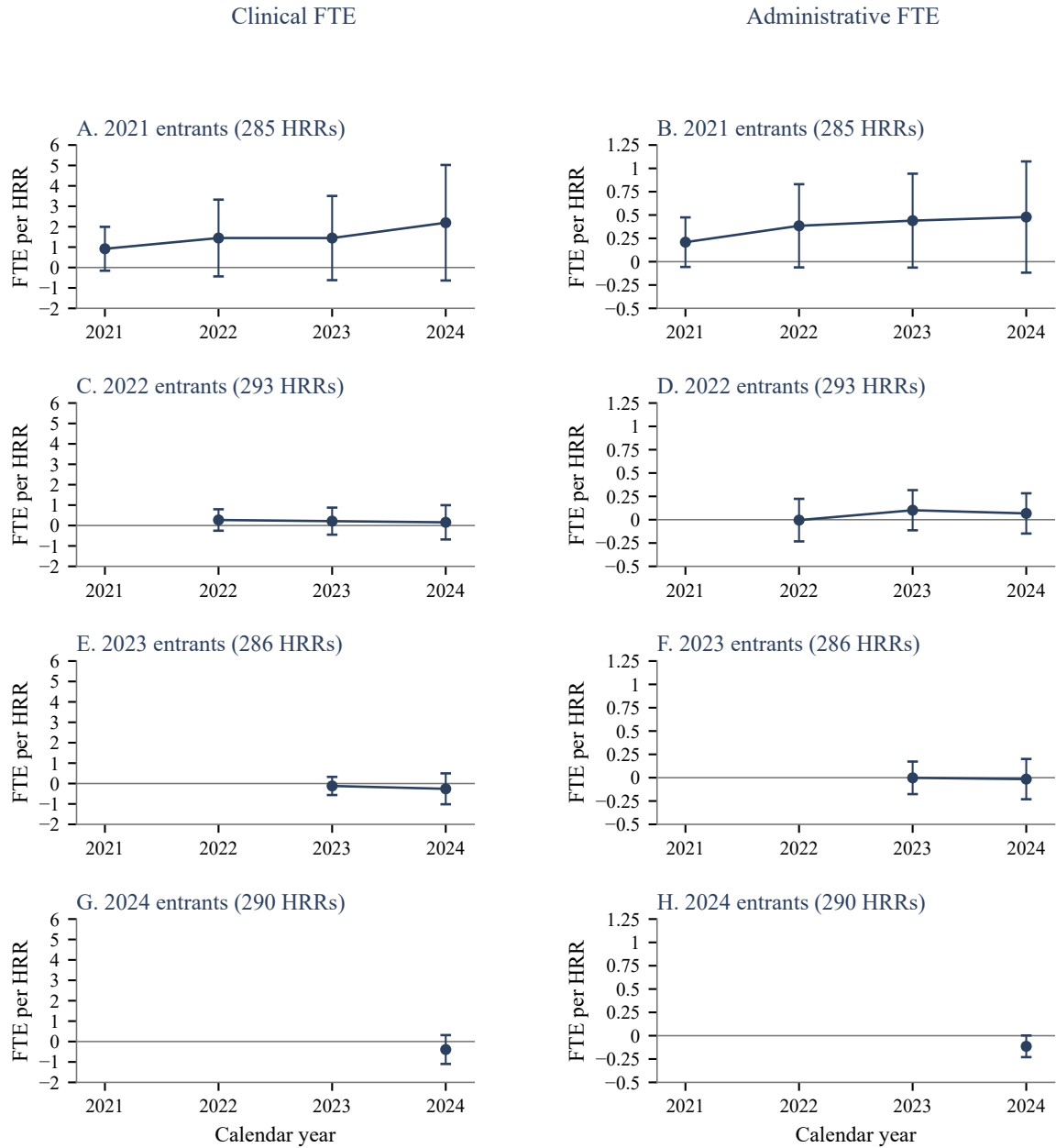
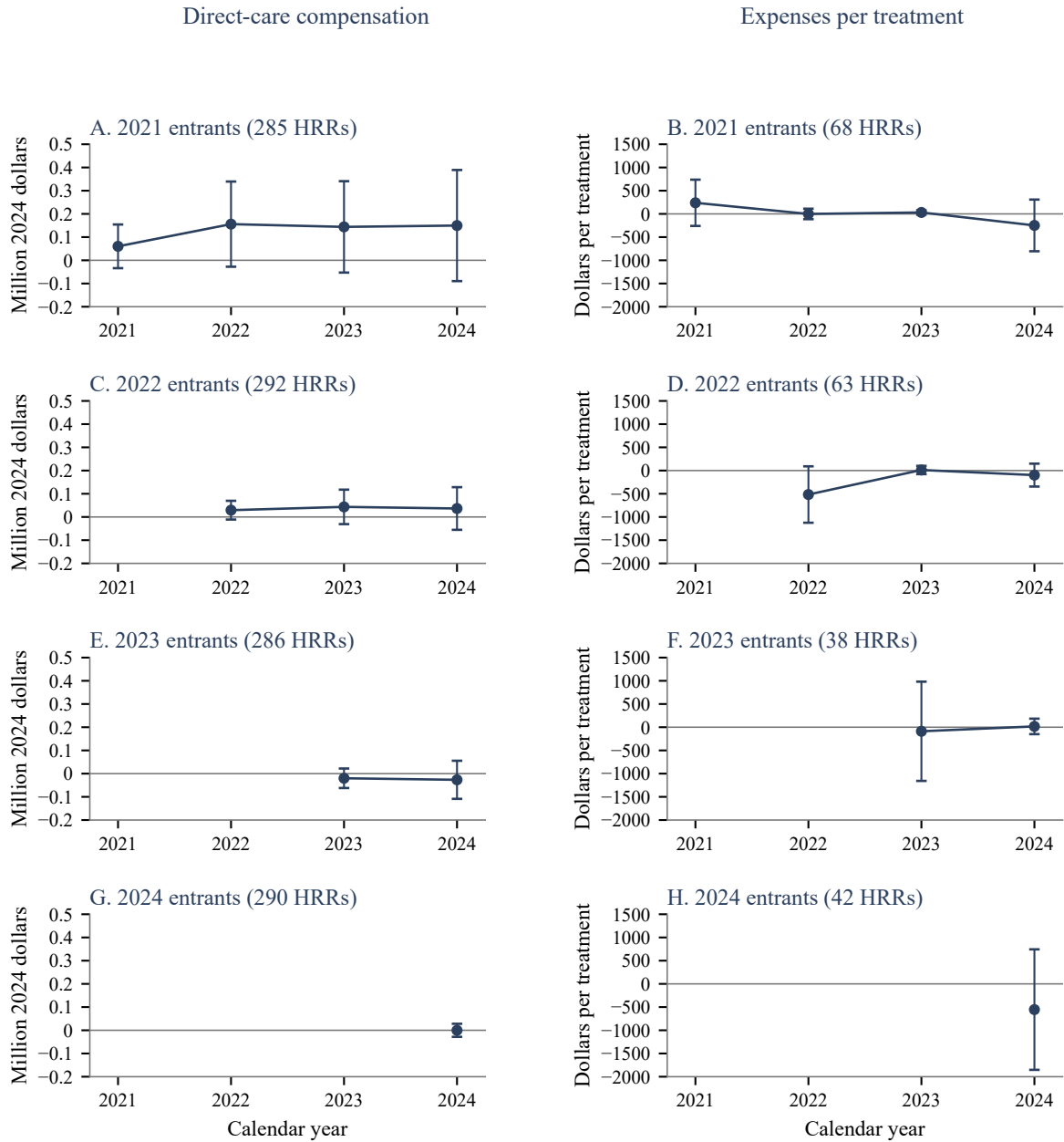


Figure A5. Entrant Clinical and Administrative Staffing



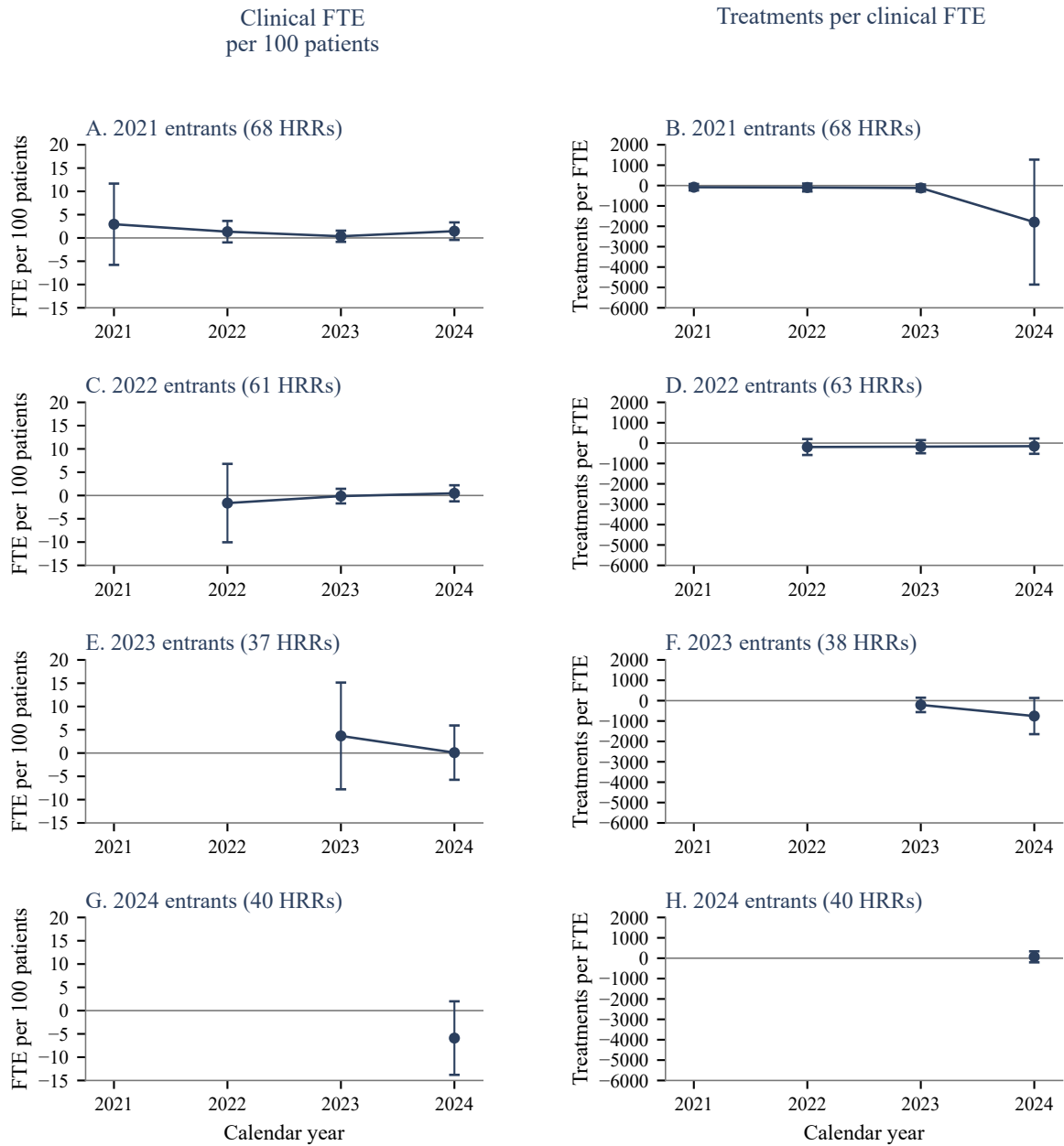
Notes: Annual staffing estimates follow equation (2) on fixed outcome-specific HRRs. Administrative staffing includes management. Follow-up covers entry through 2024; entry-year reports may be partial. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A6. Entrant Compensation and Expenses per Treatment



Notes: Annual estimates follow equation (2) on fixed outcome-specific HRRs. Compensation totals retain verified contributions; expenses per treatment use matched reports and positive all-payer treatment denominators. Monetary values are in 2024 dollars. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A7. Entrant Staffing Intensity and Treatments per Clinical FTE



Notes: Annual estimates follow equation (2) on fixed outcome-specific HRRs. Ratios require matched reports and positive denominators; patient and treatment counts cover all payers. Empty cohorts are excluded. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Tables A14 and A15 report annual and average responses for the covered market and incumbents. All eight averages are imprecise in each population, although station counts, direct-care compensation, and some expense measures show positive responses early in the payment period. Table A16 gives the corresponding 2021–2024 entrant estimates; Table A6 collects the transition cohort’s averages. Each table reports actual fitted observations and HRRs. The geography is fixed within each outcome, while verified facility contributions can remain incomplete. These inputs include resources shared across dialysis modalities, limiting their attribution to home-service expansion.

Table A14. Annual Covered-Market Operating Input Responses

Outcome / year	Average	2020	2021	2022	2023	2024
Operating expenses						
ETC effect	0.424 (0.803)	0.482 (0.570)	0.914* (0.584)	0.420 (0.869)	0.388 (1.036)	−0.024 (1.057)
Control mean	70.729	77.821	73.896	71.450	70.403	67.165
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Dialysis stations						
ETC effect	10.106 (6.804)	−0.180 (3.192)	12.164** (6.761)	12.153* (7.343)	11.106 (8.480)	5.000 (9.916)
Control mean	382.143	381.767	383.076	387.043	379.510	378.943
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Clinical FTE						
ETC effect	0.878 (4.185)	2.547 (4.129)	5.370 (4.042)	−0.805 (5.730)	−4.711 (5.998)	3.656 (9.205)
Control mean	248.727	257.669	249.891	231.730	224.538	288.749
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Administrative FTE						
ETC effect	−1.113 (1.269)	−2.233** (1.298)	−1.042 (1.461)	−1.244 (1.731)	0.009 (1.625)	−2.176 (2.439)
Control mean	50.067	51.250	53.359	51.117	49.719	46.071
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Direct-care compensation						
ETC effect	0.166 (0.268)	0.339* (0.216)	0.466** (0.206)	0.205 (0.328)	0.010 (0.332)	−0.017 (0.375)
Control mean	23.984	25.742	24.553	24.094	23.423	23.863
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301

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Table A14. Annual Covered-Market Operating Input Responses (continued)

Outcome / year	Average	2020	2021	2022	2023	2024
Expenses per treatment						
ETC effect	−2.23 (3.39)	−5.45* (3.54)	−0.53 (3.05)	−4.54 (4.14)	−2.80 (4.10)	−1.03 (4.29)
Control mean	308.35	327.54	317.70	313.41	308.81	293.50
HRR-years	1,200	300	300	300	300	300
HRRs	300	300	300	300	300	300
Clinical FTE per 100 patients						
ETC effect	−0.02 (0.17)	0.24 (0.17)	0.17 (0.16)	0.04 (0.22)	0.03 (0.23)	−0.33 (0.31)
Control mean	10.80	11.25	10.87	10.07	9.73	12.52
HRR-years	1,200	300	300	300	300	300
HRRs	300	300	300	300	300	300
Treatments per clinical FTE						
ETC effect	13.89 (15.81)	14.67 (18.03)	−10.21 (13.82)	3.99 (23.81)	3.26 (16.85)	58.54 (43.42)
Control mean	950.31	940.92	946.77	1,007.05	1,031.89	815.52
HRR-years	1,200	300	300	300	300	300
HRRs	300	300	300	300	300	300

Notes: Estimates follow equation (1) on fixed outcome-specific HRRs; averages weight 2021–2024 coefficients equally. Expenses and compensation are millions of 2024 dollars per HRR; stations and FTEs are HRR totals. Ratios require matched reports and positive denominators; patient and treatment counts cover all payers. Expenses per treatment are 2024 dollars. Totals retain partial facility coverage. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null.

Table A15. Annual Incumbent Operating Input Responses

Outcome / year	Average	2020	2021	2022	2023	2024
Operating expenses						
ETC effect	0.249 (0.872)	0.543 (0.530)	0.974* (0.566)	0.135 (0.882)	0.093 (1.097)	−0.206 (1.227)
Control mean	67.409	77.195	72.022	68.490	66.553	62.573
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Dialysis stations						
ETC effect	9.090 (6.660)	−0.835 (2.777)	11.310* (6.887)	11.222* (7.463)	9.907 (8.042)	3.920 (9.398)
Control mean	371.361	377.662	376.319	377.571	367.619	363.933
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301

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Table A15. Annual Incumbent Operating Input Responses (continued)

Outcome / year	Average	2020	2021	2022	2023	2024
Clinical FTE						
ETC effect	0.312 (4.162)	2.592 (4.085)	5.208 (3.688)	-2.092 (5.721)	-5.334 (6.343)	3.465 (8.997)
Control mean	239.275	255.888	244.866	223.898	214.249	274.086
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Administrative FTE						
ETC effect	-1.408 (1.230)	-2.167** (1.272)	-1.121 (1.380)	-1.616 (1.633)	-0.419 (1.639)	-2.478 (2.479)
Control mean	47.827	50.677	51.971	49.003	47.119	43.215
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Direct-care compensation						
ETC effect	0.139 (0.290)	0.350* (0.211)	0.484*** (0.184)	0.099 (0.329)	-0.046 (0.370)	0.019 (0.425)
Control mean	23.041	25.608	24.091	23.293	22.336	22.446
HRR-years	1,204	301	301	301	301	301
HRRs	301	301	301	301	301	301
Expenses per treatment						
ETC effect	-2.27 (3.28)	-5.40* (3.50)	0.08 (2.93)	-4.51 (4.01)	-2.71 (4.06)	-1.91 (4.16)
Control mean	305.35	326.31	314.71	309.83	305.77	291.10
HRR-years	1,200	300	300	300	300	300
HRRs	300	300	300	300	300	300
Clinical FTE per 100 patients						
ETC effect	-0.07 (0.18)	0.22 (0.17)	0.17 (0.16)	-0.02 (0.23)	-0.01 (0.24)	-0.43 (0.34)
Control mean	10.85	11.24	10.86	10.08	9.77	12.69
HRR-years	1,200	300	300	300	300	300
HRRs	300	300	300	300	300	300
Treatments per clinical FTE						
ETC effect	19.32 (16.18)	15.77 (18.36)	-10.07 (13.47)	12.84 (23.53)	6.54 (17.09)	67.96* (44.53)
Control mean	951.65	943.42	952.61	1,009.20	1,034.21	810.59
HRR-years	1,200	300	300	300	300	300
HRRs	300	300	300	300	300	300

Notes: Estimates follow equation (1) for the fixed 2019 roster; outcome-specific HRRs remain fixed. Averages weight 2021–2024 coefficients equally. Expenses and compensation are millions of 2024 dollars per HRR; stations and FTEs are HRR totals. Ratios require matched reports and positive denominators; patient and treatment counts cover all payers. Expenses per treatment are 2024 dollars. Totals retain partial facility coverage. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null.

Table A16. Annual Operating Input Responses of Fixed Entry Cohorts

Outcome / observation year	2021 entrants	2022 entrants	2023 entrants	2024 entrants
Operating expenses				
Average ETC effect	0.462 (0.329)	0.075 (0.116)	-0.092 (0.133)	0.007 (0.057)
2021	0.247 (0.190)	—	—	—
2022	0.477 (0.330)	0.046 (0.071)	—	—
2023	0.516 (0.379)	0.097 (0.132)	-0.076 (0.101)	—
2024	0.607 (0.427)	0.081 (0.152)	-0.108 (0.167)	0.007 (0.057)
Average control mean	0.816	0.496	0.344	0.161
HRR-years, average window	1,140	879	572	290
HRRs	285	293	286	290
Dialysis stations				
Average ETC effect	0.691 (0.912)	0.130 (0.535)	-0.229 (0.472)	-0.508 (0.392)
2021	0.629 (0.862)	—	—	—
2022	0.656 (0.887)	-0.194 (0.471)	—	—
2023	0.687 (0.945)	0.292 (0.628)	-0.444 (0.468)	—
2024	0.790 (1.236)	0.291 (0.626)	-0.013 (0.537)	-0.508 (0.392)
Average control mean	2.233	1.461	0.742	0.791
HRR-years, average window	1,140	873	568	289
HRRs	285	291	284	289
Clinical FTE				
Average ETC effect	1.500 (0.980)	0.214 (0.330)	-0.189 (0.301)	-0.392 (0.361)
2021	0.918 (0.548)	—	—	—
2022	1.445 (0.959)	0.271 (0.268)	—	—
2023	1.443 (1.052)	0.214 (0.338)	-0.119 (0.227)	—
2024	2.194 (1.444)	0.157 (0.429)	-0.259 (0.386)	-0.392 (0.361)
Average control mean	2.370	1.343	0.809	0.689
HRR-years, average window	1,140	879	572	290
HRRs	285	293	286	290

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Table A16. Annual Operating Input Responses of Fixed Entry Cohorts (continued)

Outcome / observation year	2021 entrants	2022 entrants	2023 entrants	2024 entrants
Administrative FTE				
Average ETC effect	0.378*	0.055	−0.008	−0.114*
	(0.220)	(0.104)	(0.096)	(0.059)
2021	0.209	—	—	—
	(0.136)	—	—	—
2022	0.384	−0.005	—	—
	(0.227)	(0.116)	—	—
2023	0.440*	0.101	−0.002	—
	(0.257)	(0.110)	(0.089)	—
2024	0.478	0.067	−0.015	−0.114*
	(0.304)	(0.110)	(0.110)	(0.059)
Average control mean	0.558	0.379	0.240	0.173
HRR-years, average window	1,140	879	572	290
HRRs	285	293	286	290
Direct-care compensation				
Average ETC effect	0.127	0.036	−0.023	0.000
	(0.089)	(0.034)	(0.031)	(0.014)
2021	0.060	—	—	—
	(0.048)	—	—	—
2022	0.156	0.029	—	—
	(0.094)	(0.021)	—	—
2023	0.144	0.043	−0.020	—
	(0.100)	(0.038)	(0.021)	—
2024	0.150	0.037	−0.027	0.000
	(0.122)	(0.047)	(0.042)	(0.014)
Average control mean	0.232	0.124	0.092	0.040
HRR-years, average window	1,140	876	572	290
HRRs	285	292	286	290
Expenses per treatment				
Average ETC effect	5.26	−200.78*	−35.05	−554.08
	(106.45)	(119.32)	(302.93)	(662.36)
2021	239.11	—	—	—
	(254.36)	—	—	—
2022	−1.06	−516.61*	—	—
	(57.51)	(310.01)	—	—
2023	31.09	12.21	−87.56	—
	(23.61)	(45.42)	(545.57)	—
2024	−248.11	−97.95	17.45	−554.08
	(284.02)	(125.84)	(85.08)	(662.36)
Average control mean	569.45	762.01	1,028.77	1,737.51
HRR-years, average window	272	189	76	42
HRRs	68	63	38	42

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Table A16. Annual Operating Input Responses of Fixed Entry Cohorts (continued)

Outcome / observation year	2021 entrants	2022 entrants	2023 entrants	2024 entrants
Clinical FTE per 100 patients				
Average ETC effect	1.52 (1.31)	−0.43 (1.55)	1.88 (4.26)	−5.90 (4.03)
2021	2.94 (4.45)	— —	— —	— —
2022	1.34 (1.18)	−1.63 (4.29)	— —	— —
2023	0.36 (0.61)	−0.13 (0.80)	3.68 (5.85)	— —
2024	1.46 (0.96)	0.47 (0.88)	0.09 (2.98)	−5.90 (4.03)
Average control mean	10.96	11.59	11.71	16.03
HRR-years, average window	272	183	74	40
HRRs	68	61	37	40
Treatments per clinical FTE				
Average ETC effect	−523.40 (406.16)	−173.06 (167.80)	−482.36* (291.24)	66.92 (137.70)
2021	−82.72 (80.20)	— —	— —	— —
2022	−95.92 (98.92)	−192.84 (199.64)	— —	— —
2023	−120.15 (87.30)	−175.77 (164.51)	−208.74 (180.41)	— —
2024	−1,794.82* (1,564.14)	−150.56 (191.63)	−755.98* (453.17)	66.92 (137.70)
Average control mean	1,085.16	955.39	1,120.57	481.06
HRR-years, average window	272	189	76	40
HRRs	68	63	38	40

Notes: Estimates follow equation (2) on fixed outcome-specific HRRs from entry through 2024; entry-year reports may be partial. Averages weight annual coefficients equally. Expenses and compensation are millions of 2024 dollars per HRR; stations and FTEs are HRR totals. Ratios require matched reports and positive denominators; patient and treatment counts cover all payers. Expenses per treatment are 2024 dollars. Totals include verified contributions and empty-cohort zeros. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes indicate years before entry.

F.3. Financial, Clinical and Geographic Outcomes

Table A17 consolidates the broader covered-market outcomes using equation (1) and fixed outcome-specific subsets of the main market sample. It retains every outcome in the four families, showing the 2021–2024 average alongside the 2020 coefficient. Patient-service net income falls by \$3.83 million per HRR on average, a statistically significant decline, but the negative \$3.02 million transition-year difference precedes payment implementation and complicates attribution to those incentives.

Long-term catheter use falls by 0.94 percentage points and fistula use rises by 1.04 percentage points, with both changes statistically significant; the vascular-access index summarizes these related improvements. Mortality, hospitalization, readmission, adequacy, geographic availability, and the all-payer home-treatment share have imprecise average responses.⁵⁴ Together, these comparisons leave the link between entrant supply, regional access, and patient welfare uncertain.

Table A17. Supplementary Covered-Market Outcomes

Outcome	2021–2024 average	2020	Control mean	HRR-years	HRRs
Panel A. Financial outcomes					
Medicare payments (million 2024 dollars)	–2.179 (1.572)	–0.250 (0.411)	27.238	1,196	299
Payments per treatment (2024 dollars)	0.757 (3.218)	–0.953 (1.702)	352.901	1,196	299
Patient-service net income (million 2024 dollars)	–3.834*** (1.525)	–3.024*** (1.319)	13.830	1,196	299
Net income per treatment (2024 dollars)	–15.923 (13.407)	–19.120* (12.313)	64.556	1,196	299
Panel B. Clinical indicators					
Standardized mortality ratio	0.002 (0.014)	0.014 (0.018)	0.975	1,200	300
Standardized hospitalization ratio	–0.008 (0.011)	–0.001 (0.011)	0.956	1,200	300
Standardized readmission ratio	0.001 (0.015)	0.000 (0.021)	0.997	1,196	299
Vascular-access index	–0.204*** (0.071)	— —	0.920	1,200	300
Dialysis adequacy fraction	–0.002 (0.002)	0.000 (0.001)	0.968	1,200	300
Long-term catheter share (percent)	–0.937*** (0.354)	— —	16.822	1,200	300
Fistula share (percent)	1.043** (0.435)	0.200 (0.303)	60.671	1,200	300
Panel C. Geographic availability					
Nearest documented home-capable facility (miles)	–0.393 (0.435)	–0.485 (0.388)	12.066	1,204	301

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⁵⁴Financial outcomes reflect accounting allocations; clinical outcomes depend on patient mix. Both span dialysis modalities and available reports. Geographic measures use documented capability nationwide and fixed demand weights partly based on 2020 health prevalence; actual travel, waiting times, and patient choices are unobserved.

Table A17. Supplementary Covered-Market Outcomes (continued)

Outcome	2021–2024 average	2020	Control mean	HRR-years	HRRs
Documented home-capable facilities within 15 miles	0.030 (0.116)	–0.025 (0.056)	6.302	1,204	301
No documented home-capable facility within 30 miles (percent)	0.058 (0.434)	0.025 (0.135)	8.377	1,204	301
Panel D. Home-treatment share					
All-payer home treatments as a share of all treatments (percent)	0.547 (0.543)	1.461** (0.787)	18.225	1,196	299

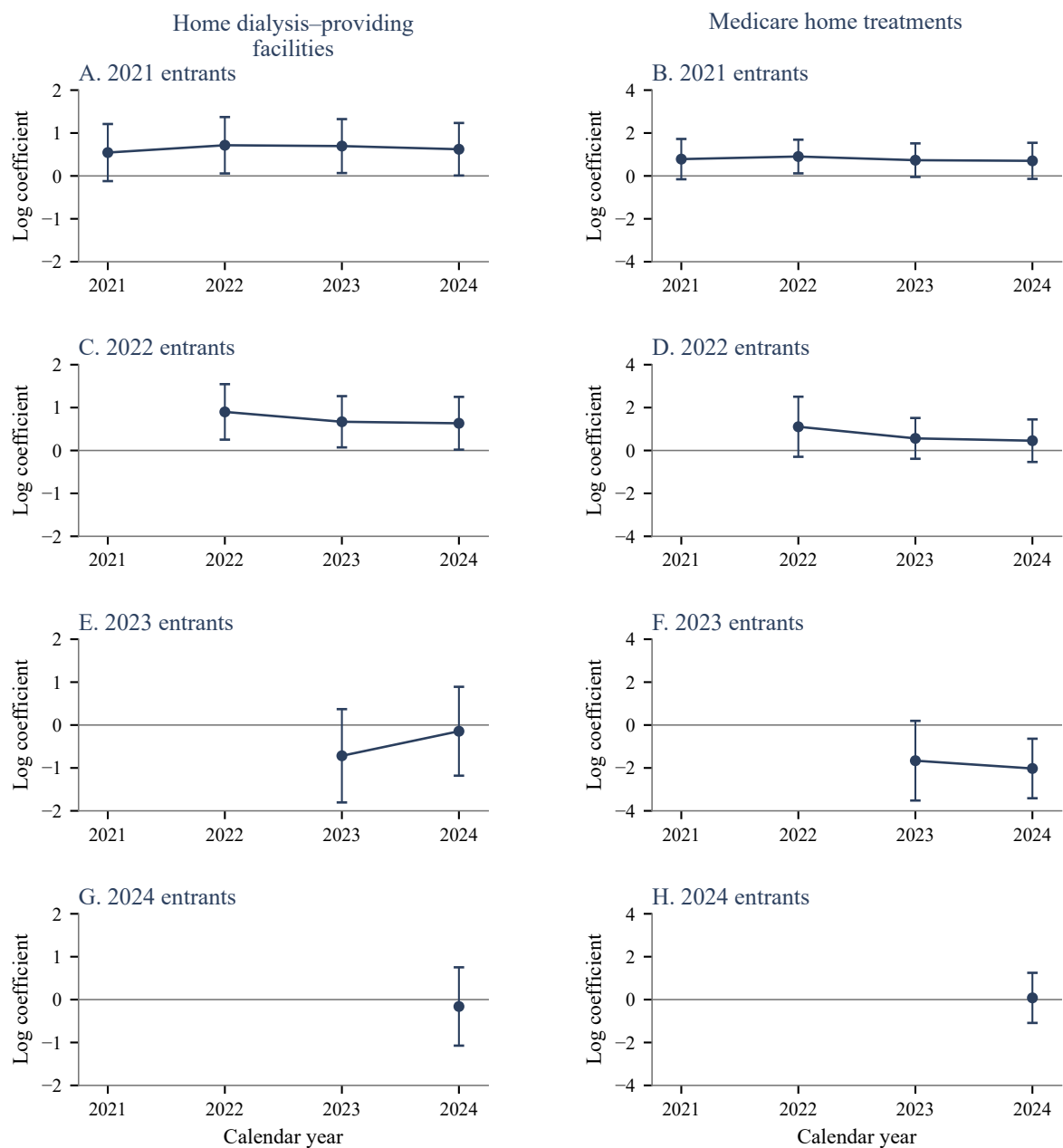
Notes: Estimates follow equation (1) on fixed outcome-specific HRRs. The first column averages 2021–2024 coefficients equally; the second reports 2020. Control means and sample counts refer to 2021–2024. Totals retain verified contributions; ratios require matched reports. Share effects are percentage points except adequacy, a fraction. Geographic measures use fixed demand weights and documented home-service capability. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes denote unavailable observations. Adequacy retains the source fraction scale; catheter and fistula effects are percentage points.

G. Robustness Checks

G.1. PPML Estimates of the Main Outcomes

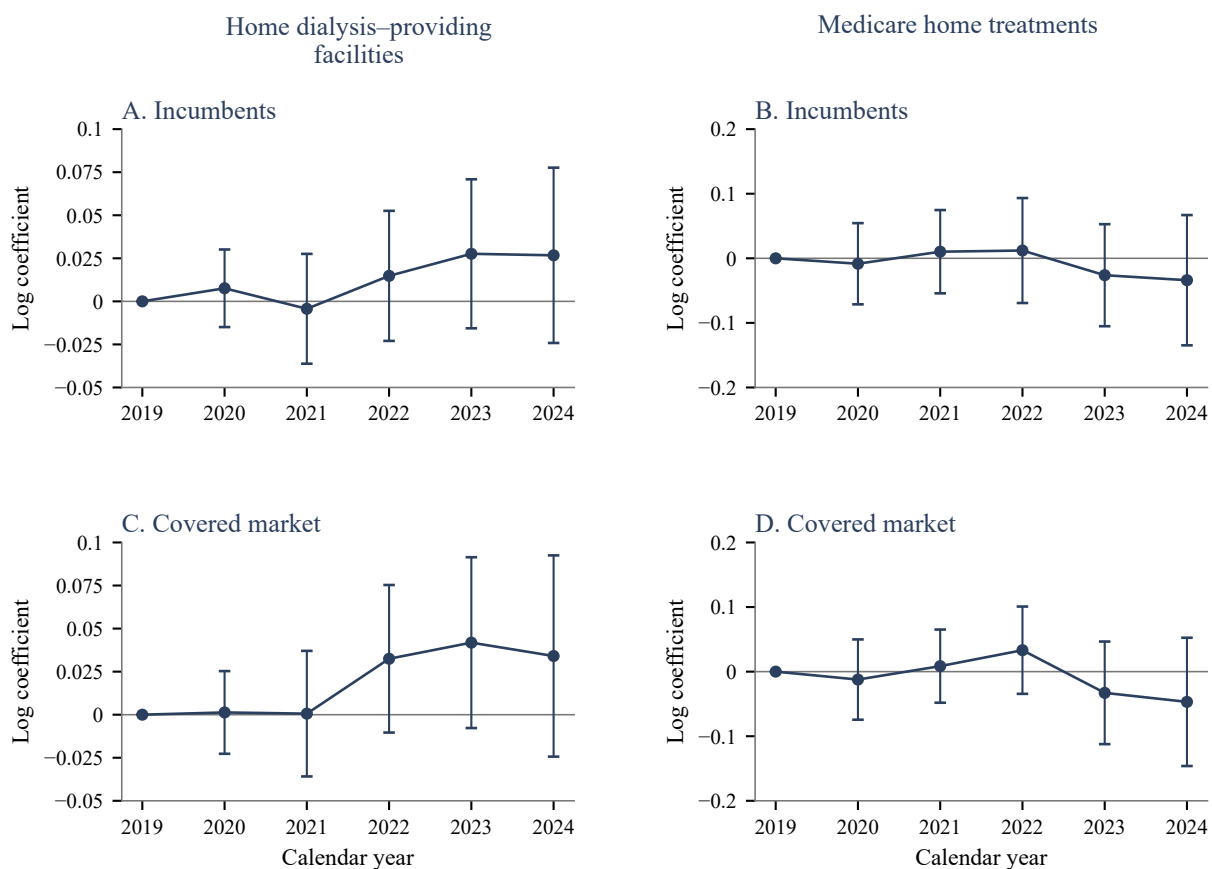
Figures A8 and A9 show the annual PPML responses for entry cohorts and for incumbents and the covered market. Table A18 reports the annual coefficients, equal-weight averages, and fitted support. Cohort baselines enter as $\log(1 + Y_{h,2019})$; the remaining regressor index follows the corresponding primary equation. Separation can reduce the represented observations. Coefficients are reported in log units, and significance follows individual randomization inference; plotted HC2 intervals use a normal approximation.

Figure A8. Annual PPML responses of fixed entry-cohort home dialysis supply



Notes: PPML estimates follow equation (2) with an exponential link and $\log(1 + Y_{h,2019})$ baselines. Coefficients are in log units; separation may reduce fitted support. Table A18 reports annual estimates and averages. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A9. Annual PPML responses of incumbent and covered-market home dialysis supply



Notes: PPML estimates follow equation (1) with an exponential link, relative to 2019. Coefficients are in log units; separation may reduce fitted support. Table A18 reports annual estimates and averages. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A18. Annual PPML Estimates of the Main Outcomes

Outcome / population	Average	2020	2021	2022	2023	2024
2020 entrants: Home dialysis-providing facilities (HRR)						
ETC effect	-0.126 (0.308)	-0.207 (0.397)	-0.143 (0.314)	-0.244 (0.321)	0.013 (0.303)	-0.050 (0.336)
Control mean	0.25	0.19	0.25	0.27	0.27	0.27
HRR-years	1,450	290	290	290	290	290
HRRs	290	290	290	290	290	290

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Table A18. Annual PPML Estimates of the Main Outcomes (continued)

Outcome / population	Average	2020	2021	2022	2023	2024
2020 entrants: Medicare home treatments (HRR)						
ETC effect	−0.27 (0.52)	0.12 (0.83)	−0.14 (0.52)	−0.20 (0.48)	−0.57 (0.57)	−0.58 (0.59)
Control mean	240.20	48.30	177.87	271.71	339.19	363.93
HRR-years	1,450	290	290	290	290	290
HRRs	290	290	290	290	290	290
2021 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	0.644* (0.321)	—	0.545 (0.340)	0.715* (0.335)	0.696* (0.321)	0.622* (0.312)
Control mean	0.18	—	0.16	0.18	0.19	0.20
HRR-years	1,160	—	290	290	290	290
HRRs	290	—	290	290	290	290
2021 entrants: Medicare home treatments (HRR)						
ETC effect	0.78* (0.41)	—	0.78 (0.48)	0.90* (0.40)	0.73* (0.40)	0.70 (0.43)
Control mean	158.44	—	46.16	134.62	195.34	257.63
HRR-years	1,160	—	290	290	290	290
HRRs	290	—	290	290	290	290
2022 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	0.734** (0.301)	—	—	0.899** (0.330)	0.669** (0.305)	0.634* (0.314)
Control mean	0.15	—	—	0.11	0.17	0.17
HRR-years	888	—	—	296	296	296
HRRs	296	—	—	296	296	296
2022 entrants: Medicare home treatments (HRR)						
ETC effect	0.71 (0.52)	—	—	1.11 (0.71)	0.56 (0.49)	0.46 (0.51)
Control mean	87.14	—	—	24.15	93.66	143.62
HRR-years	888	—	—	296	296	296
HRRs	296	—	—	296	296	296
2023 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	−0.431 (0.523)	—	—	—	−0.717 (0.555)	−0.145 (0.528)
Control mean	0.20	—	—	—	0.18	0.21
HRR-years	422	—	—	—	211	211
HRRs	211	—	—	—	211	211

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Table A18. Annual PPML Estimates of the Main Outcomes (continued)

Outcome / population	Average	2020	2021	2022	2023	2024
2023 entrants: Medicare home treatments (HRR)						
ETC effect	-1.84** (0.78)	—	—	—	-1.66 (0.95)	-2.03** (0.71)
Control mean	130.07	—	—	—	51.77	208.38
HRR-years	422	—	—	—	211	211
HRRs	211	—	—	—	211	211
2024 entrants: Home dialysis–providing facilities (HRR)						
ETC effect	-0.160 (0.466)	—	—	—	—	-0.160 (0.466)
Control mean	0.09	—	—	—	—	0.09
HRR-years	295	—	—	—	—	295
HRRs	295	—	—	—	—	295
2024 entrants: Medicare home treatments (HRR)						
ETC effect	0.08 (0.60)	—	—	—	—	0.08 (0.60)
Control mean	32.06	—	—	—	—	32.06
HRR-years	295	—	—	—	—	295
HRRs	295	—	—	—	—	295
Incumbents: Home dialysis–providing facilities (HRR)						
ETC effect	0.016 (0.019)	0.008 (0.011)	-0.004 (0.016)	0.015 (0.019)	0.028 (0.022)	0.027 (0.026)
Control mean	10.26	10.54	10.64	10.40	10.12	9.87
HRR-years	1,184	296	296	296	296	296
HRRs	296	296	296	296	296	296
Incumbents: Medicare home treatments (HRR)						
ETC effect	-0.01 (0.04)	-0.01 (0.03)	0.01 (0.03)	0.01 (0.04)	-0.03 (0.04)	-0.03 (0.05)
Control mean	13,661.93	17,737.51	15,440.19	13,947.74	12,975.88	12,283.93
HRR-years	1,184	296	296	296	296	296
HRRs	296	296	296	296	296	296
Covered market: Home dialysis–providing facilities (HRR)						
ETC effect	0.027 (0.022)	0.001 (0.012)	0.001 (0.019)	0.032 (0.022)	0.042* (0.025)	0.034 (0.030)
Control mean	10.93	10.61	10.99	10.97	10.91	10.85
HRR-years	1,196	299	299	299	299	299
HRRs	299	299	299	299	299	299

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Table A18. Annual PPML Estimates of the Main Outcomes (continued)

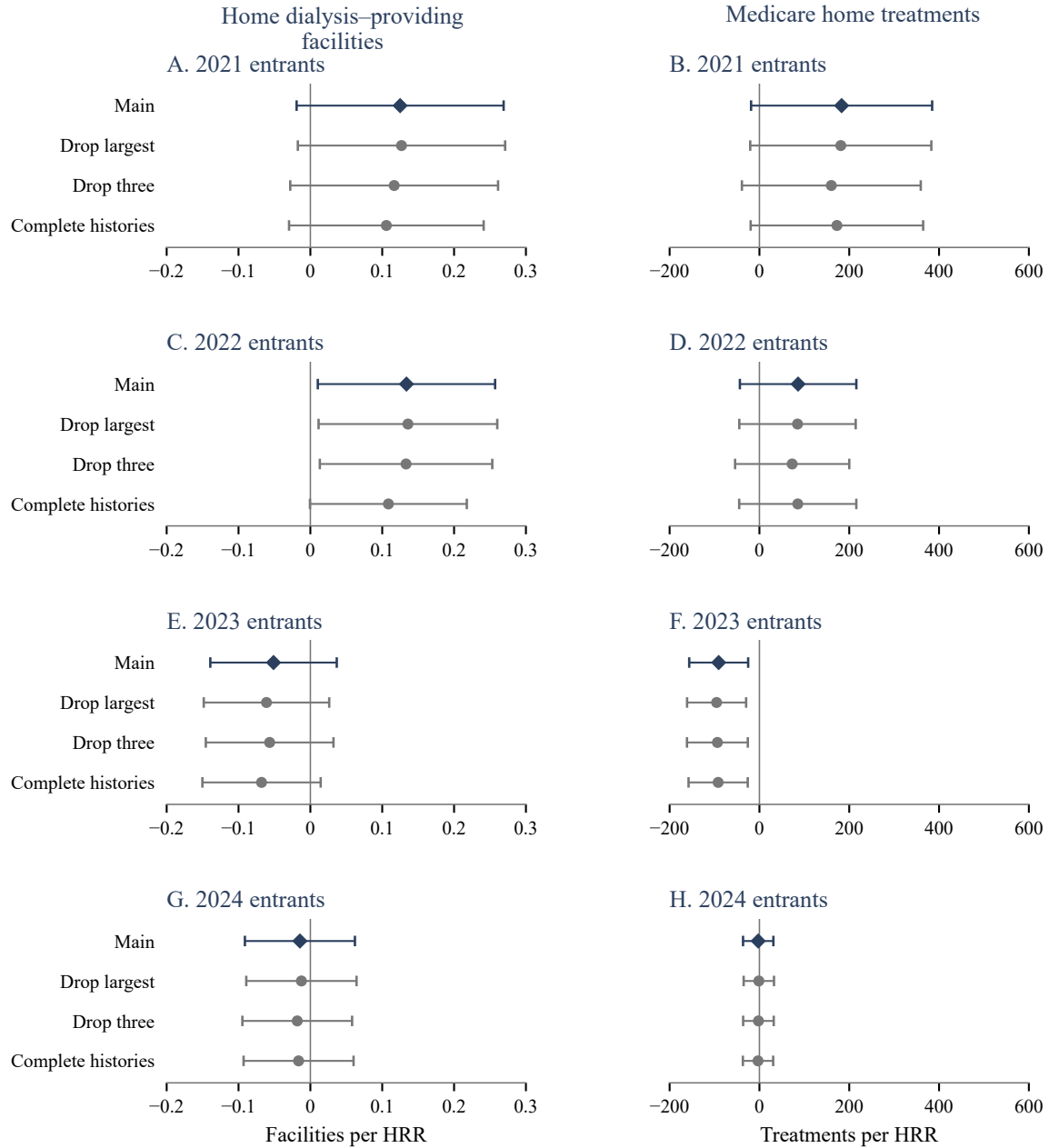
Outcome / population	Average	2020	2021	2022	2023	2024
Covered market: Medicare home treatments (HRR)						
ETC effect	−0.01 (0.03)	−0.01 (0.03)	0.01 (0.03)	0.03 (0.03)	−0.03 (0.04)	−0.05 (0.05)
Control mean	14,258.18	17,584.09	15,595.82	14,414.62	13,689.70	13,332.60
HRR-years	1,196	299	299	299	299	299
HRRs	299	299	299	299	299	299

Notes: Entries are PPML log coefficients. Specifications follow the main equations, with $\log(1 + Y_{h,2019})$ cohort baselines. Averages weight 2021–2024 coefficients equally, starting at entry for later cohorts. Separation may reduce fitted support; control means retain outcome units. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes denote unavailable estimates. The 2020 service average covers 2020–2024 in both functional forms.

G.2. Influential Markets and Complete Facility Histories

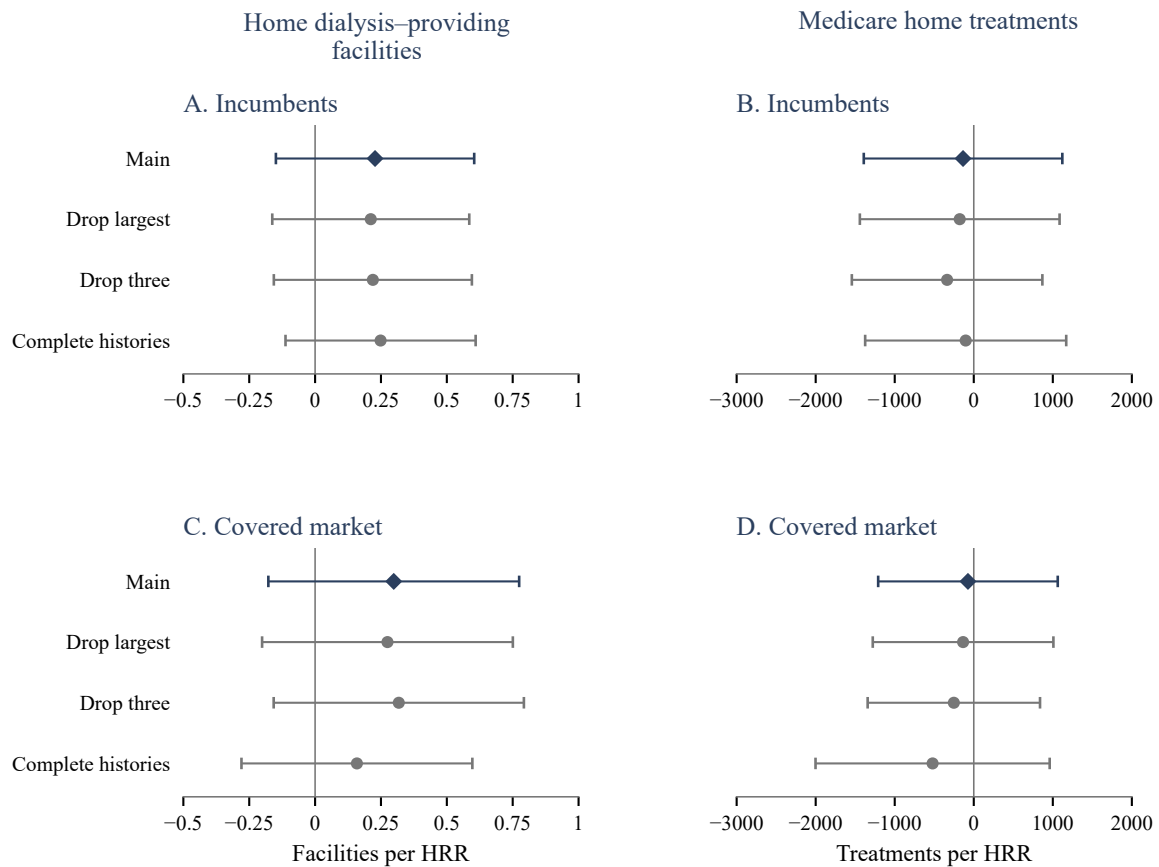
Figures A10 and A11 compare the main average responses with those under sample restrictions, retaining the original windows and linear specifications. Table A19 excludes the largest baseline market or the three largest markets within each population’s sample, ranked by 2019 eligible facility counts. Table A20 requires resolved provision and treatment quantities for every original facility in every required year. The latter restriction substantially narrows incumbent and market support. Both checks assess sensitivity to the represented markets; complete reporting leaves the process governing missing observations unresolved.

Figure A10. Entry-cohort average supply responses across sample restrictions



Notes: Points compare equal-weight cohort averages from entry through 2024. Exclusions remove the largest or three largest markets by 2019 facility count. Complete-history samples require both outcomes for every original facility-year. Specifications follow the main models; Tables A19 and A20 report sample sizes. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A11. Incumbent and covered-market average supply responses across sample restrictions



Notes: Points compare equal-weight 2021–2024 averages. Exclusions remove the largest or three largest markets by 2019 facility count. Complete-history samples require both outcomes for every required facility-year. Specifications follow the main models; Tables A19 and A20 report sample sizes. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A19. Influential-Market Sensitivity

Population / restriction	Providers	Treatments	HRRs
Exclude largest baseline market			
2020 entrants	–0.053 (0.063)	–118.821 (100.577)	289
2021 entrants	0.127* (0.074)	181.158* (102.896)	289
2022 entrants	0.136** (0.063)	84.772 (66.134)	295
2023 entrants	–0.061 (0.045)	–95.392*** (33.547)	292

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Table A19. Influential-Market Sensitivity (continued)

Population / restriction	Providers	Treatments	HRRs
2024 entrants	−0.012 (0.039)	−1.328 (17.234)	294
Incumbents	0.211 (0.191)	−177.836 (644.602)	300
Covered market	0.275 (0.243)	−135.255 (582.996)	300
Exclude three largest baseline markets			
2020 entrants	−0.023 (0.058)	−73.530 (96.167)	287
2021 entrants	0.117 (0.074)	160.124 (101.612)	287
2022 entrants	0.133** (0.061)	72.919 (64.863)	293
2023 entrants	−0.057 (0.045)	−93.602*** (34.558)	290
2024 entrants	−0.018 (0.039)	−2.197 (17.373)	292
Incumbents	0.219 (0.192)	−337.383 (614.945)	298
Covered market	0.318 (0.242)	−252.131 (556.301)	298

Notes: Equal-weight averages retain the main specifications and observation windows, excluding the largest or three largest markets within each main sample by eligible 2019 facility count. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null.

Table A20. Complete-Facility-Roster Sensitivity

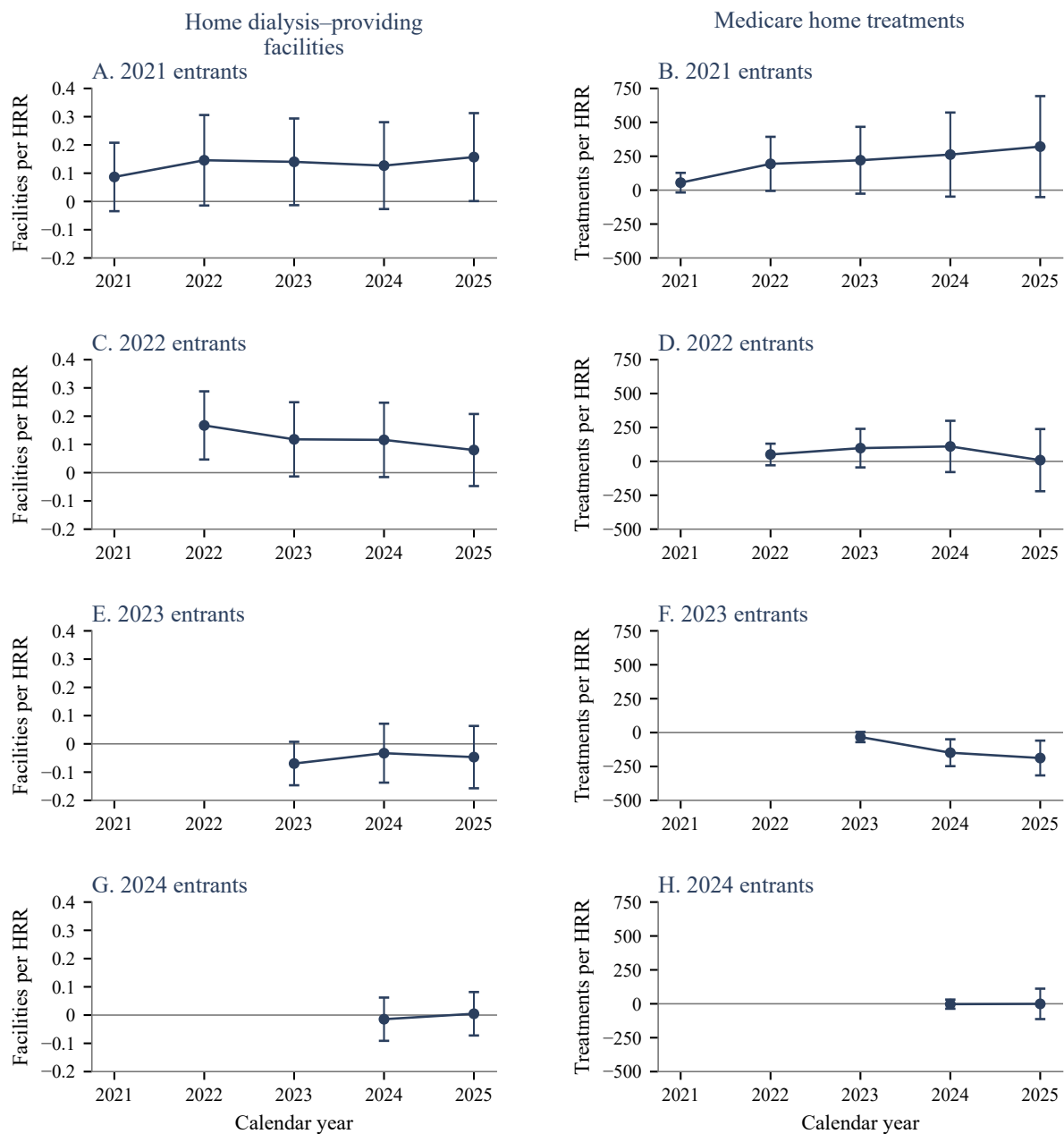
Population / restriction	Providers	Treatments	HRRs
Every original facility has both outcomes in every required year			
2020 entrants	−0.001 (0.053)	−102.924 (94.279)	272
2021 entrants	0.106 (0.069)	172.535* (97.995)	281
2022 entrants	0.109* (0.056)	85.343 (66.540)	288
2023 entrants	−0.068 (0.042)	−91.907*** (33.624)	290
2024 entrants	−0.016 (0.039)	−3.141 (17.214)	293
Incumbents	0.249 (0.184)	−102.199 (648.892)	97
Covered market	0.159 (0.224)	−520.786 (755.750)	86

Notes: Equal-weight averages retain the main specifications and windows, restricting each sample to markets with both service outcomes resolved for every required facility-year, including documented exit zeros. Selection into complete reporting limits generalization. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null.

G.3. Supplementary Follow-Up in 2025

Figure A12 and Table A21 extend the cohort paths in Figure 2 through 2025, reporting every annual coefficient and the extended averages. Available 2025 observations are appended before re-estimating the full annual specification; all earlier observations, the main HRR lists, controls, and baseline adjustments are retained. Missing outcomes affect only the additional year, with annual HRR counts identifying its observed support.

Figure A12. Fixed entry-cohort home dialysis supply through 2025



Notes: Figure 2 is extended through 2025 using the original cohort rosters, HRR lists, and pre-2025 observations. Missing 2025 outcomes remain missing. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A21. Annual entry-cohort home dialysis supply through 2025

Outcome / population	2021	2022	2023	2024	2025	Average
Panel A. 2021 entrants: Home dialysis–providing facilities						
ETC effect	0.087 (0.062)	0.146* (0.082)	0.140* (0.078)	0.127 (0.078)	0.157* (0.079)	0.131* (0.074)
Control mean	0.16	0.18	0.19	0.20	0.19	0.18
HRR-years	290	290	290	290	288	1,448
HRRs	290	290	290	290	288	290
Panel B. 2021 entrants: Medicare home treatments						
ETC effect	55.23 (36.90)	193.86* (101.82)	220.57* (125.72)	262.37 (158.16)	321.14 (189.96)	210.63* (118.80)
Control mean	46.16	134.62	195.34	257.63	285.51	183.65
HRR-years	290	290	290	290	288	1,448
HRRs	290	290	290	290	288	290
Panel C. 2022 entrants: Home dialysis–providing facilities						
ETC effect	—	0.167*** (0.062)	0.118* (0.067)	0.116* (0.067)	0.080 (0.065)	0.120* (0.062)
Control mean	—	0.11	0.17	0.17	0.19	0.16
HRR-years	—	296	296	296	292	1,180
HRRs	—	296	296	296	292	296
Panel D. 2022 entrants: Medicare home treatments						
ETC effect	—	50.73 (40.66)	97.60 (72.52)	110.03 (96.49)	9.01 (116.93)	66.84 (74.29)
Control mean	—	24.15	93.66	143.62	244.66	126.09
HRR-years	—	296	296	296	292	1,180
HRRs	—	296	296	296	292	296
Panel E. 2023 entrants: Home dialysis–providing facilities						
ETC effect	—	—	−0.070* (0.039)	−0.033 (0.053)	−0.047 (0.056)	−0.050 (0.048)
Control mean	—	—	0.13	0.15	0.17	0.15
HRR-years	—	—	293	293	289	875
HRRs	—	—	293	293	289	293
Panel F. 2023 entrants: Medicare home treatments						
ETC effect	—	—	−32.87 (19.28)	−148.60*** (50.54)	−187.63*** (65.46)	−123.04*** (43.25)
Control mean	—	—	37.12	149.42	209.58	131.79
HRR-years	—	—	293	293	289	875
HRRs	—	—	293	293	289	293

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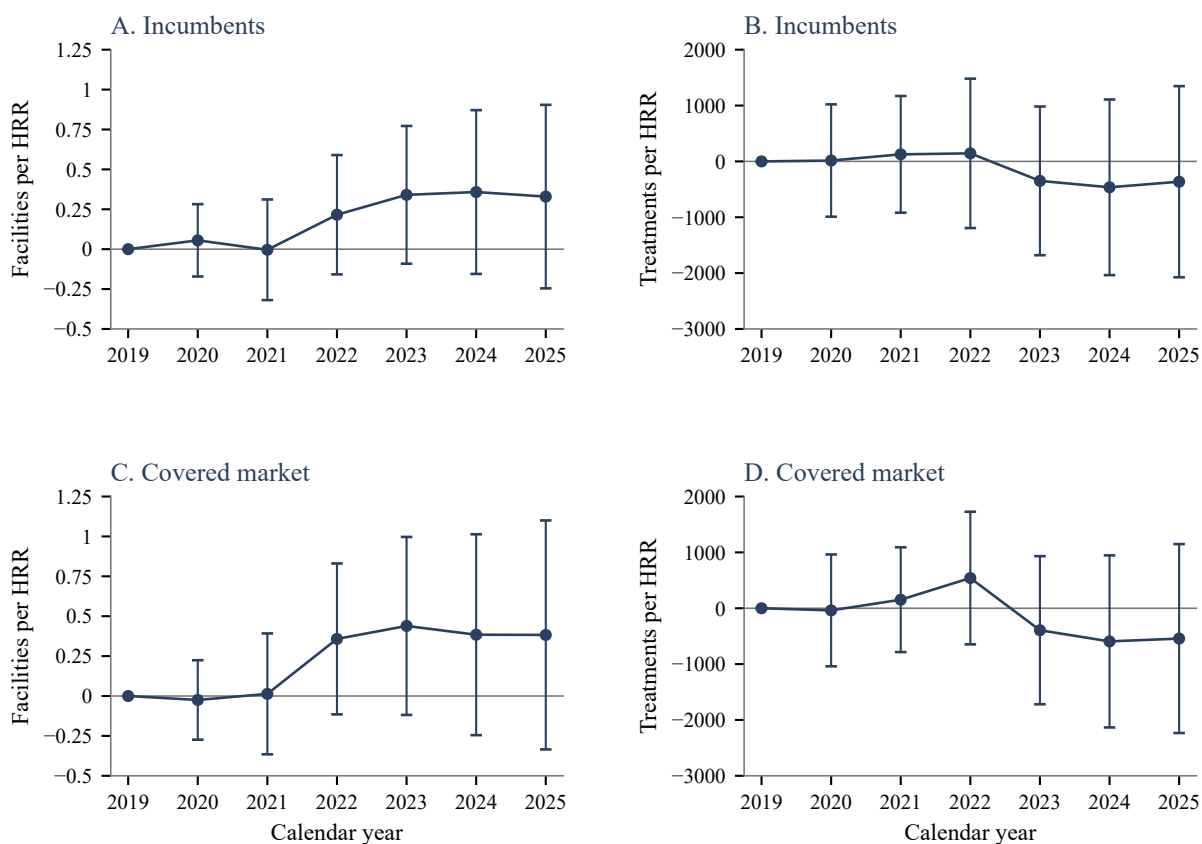
Table A21. Annual entry-cohort home dialysis supply through 2025 (continued)

Outcome / population	2021	2022	2023	2024	2025	Average
Panel G. 2024 entrants: Home dialysis–providing facilities						
ETC effect	—	—	—	−0.015	0.005	−0.005
	—	—	—	(0.039)	(0.039)	(0.038)
Control mean	—	—	—	0.09	0.09	0.09
HRR-years	—	—	—	295	290	585
HRRs	—	—	—	295	290	295
Panel H. 2024 entrants: Medicare home treatments						
ETC effect	—	—	—	−2.71	−0.70	−1.71
	—	—	—	(17.23)	(57.21)	(33.87)
Control mean	—	—	—	32.06	103.56	67.37
HRR-years	—	—	—	295	290	585
HRRs	—	—	—	295	290	295

Notes: Equation (2) appends available 2025 outcomes, retaining original HRR lists and pre-2025 observations. Equal-weight averages run from entry through 2025. Of 290, 296, 293, and 295 main HRRs, respectively, 288, 292, 289, and 290 contribute in 2025. Outcomes are facilities or Medicare home-treatment equivalents per HRR; totals retain partial coverage and missing outcomes remain missing. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes indicate years before entry. Control means pool the available control HRR-years; annual coefficients receive equal weights.

Figure A13 and Table A22 provide the corresponding full-path extension of Figure 4 for incumbents and the covered market. Both retain 301 HRRs; the market extension follows the verified 2024 facility roster. The data gaps are specified in Section 3.2. These displays assess whether the supply patterns continue beyond the main window without allowing incomplete 2025 records to select the pre-2025 sample.

Figure A13. Incumbent and covered-market home dialysis supply through 2025



Notes: Figure 4 is extended through 2025 on the same 301 HRRs, relative to 2019. The 2025 market follows the verified 2024 roster; unreported facility activity remains unknown. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A22. Annual incumbent and covered-market home dialysis supply through 2025

Outcome / population	2020	2021	2022	2023	2024	2025	Average
Panel A. Incumbents: Home dialysis–providing facilities							
ETC effect	0.055 (0.116)	−0.004 (0.161)	0.216 (0.191)	0.340* (0.220)	0.358 (0.262)	0.329 (0.293)	0.248 (0.204)
Control mean	10.29	10.39	10.15	9.88	9.64	9.47	9.90
HRR-years	301	301	301	301	301	301	1,505
HRRs	301	301	301	301	301	301	301
Panel B. Incumbents: Medicare home treatments							
ETC effect	14.95 (513.20)	126.07 (532.98)	143.90 (682.38)	−348.50 (679.08)	−463.96 (802.20)	−363.45 (872.97)	−181.19 (670.60)
Control mean	17,315.19	15,072.57	13,615.65	12,666.93	11,991.46	11,531.44	12,975.61
HRR-years	301	301	301	301	301	301	1,505
HRRs	301	301	301	301	301	301	301

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Table A22. Annual incumbent and covered-market home dialysis supply through 2025 (continued)

Outcome / population	2020	2021	2022	2023	2024	2025	Average
Panel C. Covered market: Home dialysis–providing facilities							
ETC effect	−0.025 (0.127)	0.013 (0.193)	0.358 (0.241)	0.439 (0.285)	0.384 (0.321)	0.383 (0.366)	0.315 (0.258)
Control mean	10.51	10.88	10.86	10.81	10.75	10.56	10.77
HRR-years	301	301	301	301	301	301	1,505
HRRs	301	301	301	301	301	301	301
Panel D. Covered market: Medicare home treatments							
ETC effect	−38.42 (510.93)	152.83 (478.58)	540.61 (605.60)	−393.67 (676.37)	−593.87 (785.97)	−543.01 (862.79)	−167.42 (615.05)
Control mean	17,416.62	15,447.29	14,277.34	13,559.32	13,205.62	13,057.81	13,909.48
HRR-years	301	301	301	301	301	301	1,505
HRRs	301	301	301	301	301	301	301

Notes: Equation (1) is re-estimated through 2025 on the original 301 HRRs, retaining pre-2025 observations and the 2019 reference. Equal-weight averages cover 2021–2025. The 2025 market follows the verified 2024 roster. Outcomes are facilities or Medicare home-treatment equivalents per HRR; totals retain partial coverage and unreported activity remains unknown. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Control means pool the available control HRR-years; annual coefficients receive equal weights.

G.4. Accounting Uncertainty and Denominator Diagnostics

Table A23 compares bootstrap and Delta standard errors for the decomposition in Section 5.2. The same resampled HRRs enter every equation and year, preserving covariance across the calculation, and all 2,000 finite draws enter the uncertainty estimates. Economic admissibility additionally requires positive adjusted cohort counts, nonnegative service quantities, and provider counts bounded by original cohort size in both assignment states and every included year. For the 2023 treatment allocation, 781 draws contain a negative fitted quantity in at least one year, limiting a literal count-and-rate interpretation in those draws.

Table A23. Accounting Uncertainty and Denominator Diagnostics

Cohort / outcome	Count bootstrap SE	Rate bootstrap SE	Count Delta SE	Rate Delta SE	Admissible draws
2021 / providers	0.071	0.050	0.071	0.049	2,000
2021 / treatments	72.891	81.849	71.975	79.137	2,000
2022 / providers	0.045	0.042	0.044	0.040	1,990
2022 / treatments	28.579	54.680	28.124	52.329	1,999
2023 / providers	0.033	0.029	0.032	0.027	1,971
2023 / treatments	16.051	32.322	15.449	30.737	1,219

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Table A23. Accounting Uncertainty and Denominator Diagnostics (continued)

Cohort / outcome	Count bootstrap SE	Rate bootstrap SE	Count Delta SE	Rate Delta SE	Admissible draws
2024 / providers	0.025	0.029	0.024	0.028	1,996
2024 / treatments	9.716	17.183	8.439	16.043	1,981

Notes: Standard errors use all 2,000 finite HRR bootstrap draws. Admissibility requires positive adjusted entrant counts and nonnegative service in both arms, with provider counts bounded by entrant counts in every averaged year. Delta-method calculations retain covariance across equations and years on fixed samples.